

Decision-Aware Denoising

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Modern decision-making in urban planning, climate change, and healthcare leverage large geospatial and panel datasets. These data are often extremely noisy, resulting in low-quality downstream decisions. We propose a new “light touch” framework to adapt techniques from machine learning originally designed for denoising these data to instead guide decision-making. The key to our method is a novel estimator of out-of-sample policy performance that we call the *one-shot Variance Gradient Correction* (one-shot VGC). Using the one-shot VGC, we tune the machine learning methods to minimize downstream costs (instead of minimizing prediction error or maximizing signal recovery). We uniformly bound the relative error of the one-shot VGC as an estimate of downstream costs by an intuitive measure of solution stability for the problem plus a term that vanishes as problem dimension grows. Solution stability depends both on the policy and the structure of the downstream optimization problem. We bound the solution stability for three classes of policies and problems – i) regularized plug-in policies for convex feasible regions ii) (unregularized) plug-in policies for strongly-convex feasible regions and iii) (unregularized) affine plug-in policies for weakly-coupled (potentially non-convex) problems. In all cases, we show the solution stability vanishes (relative to out-of-sample cost) as the dimension of the optimization problem grows. Finally, we present a case study based on real traffic accident data from New York City on deploying speed humps to reduce pedestrian injury. Our “light-touch” decision-aware approach outperforms traditional decision-blind techniques and highlights that the optimal level of smoothing for a denoising algorithm should depend on the downstream decision-problem.

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Predict-then-Optimize Framework.

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1. Introduction

Many decision-making problems involve large-scale systems, such as cities in urban design, road networks in transportation, and ecosystems in wildlife conservation. To analyze and make decisions in these settings, practitioners must use data to construct estimates of their many uncertain parameters. Crucial data are often missing or very imprecise as they must

be collected over time through a limited number of sensors, satellites, and people. As a result, estimates of the key model parameters can be noisy leading to poor decision-making.

A canonical solution to reducing the noise of estimates is denoising. Denoising describes a broad variety of approaches that seek to estimate an underlying signal that has been corrupted by noise. Across domains, denoising has been successfully applied to extract the heart’s electrical signal from electrocardiograms (Chatterjee et al., 2020), separate speech from background noise (Wilson et al., 2008), study the impact of drugs, cancer progression, and cell development at a molecular level (Kavran and Clauset, 2021), and improve weather forecasts (Kim et al., 2020). Importantly, in each of these applications, the goal is either signal recovery or improved prediction; there is no explicit downstream decision task.

By contrast, this paper proposes a new method for denoising when the estimated signal informs a downstream optimization problem by explicitly leveraging the structure of that problem. Such approaches are sometimes called *decision-aware*. By contrast *estimate-then-optimize* or *decision-blind* procedures denoise the parameters using standard statistical and machine learning tools (without knowledge of the downstream optimization) and then plug-in the denoised estimates into the downstream optimization. Recent works have shown that decision-aware methods often out-perform decision-blind methods, particularly in small-data, large-scale settings (Gupta, Huang, and Rusmevichientong, 2022a) and when models are misspecified (Elmachtoub and Grigas, 2022; Elmachtoub et al., 2023; Huang and Gupta, 2024).

More specifically, we study the following optimization problem with linear objective:

$$\mathbf{x}^* \in \arg \min_{\mathbf{x} \in \mathcal{X} \subseteq [-1,1]^n} \boldsymbol{\mu}^\top \mathbf{x}, \quad (1)$$

where $\mathcal{X}$ is a known, potentially non-convex, feasible region, and $\boldsymbol{\mu} \in \mathbb{R}^n$ is an unknown vector of parameters (the signal). We observe a corrupted version of μ_j , i.e., $Z_j = \mu_j + \xi_j$ for each j , and features $\mathbf{W}_j \in \mathbb{R}^p$ that are fixed and known. While we present our formal results for a general distribution for the corruption $\boldsymbol{\xi}$, our method is practically most useful when ξ_j are approximately Gaussian (see Proposition 4.3 and surrounding discussion)

We consider a class of policies that first tries to denoise $\mathbf{Z}$ to recover $\boldsymbol{\mu}$ and then plugs in this denoised estimate $\mathbf{T}(\mathbf{Z})$ for $\boldsymbol{\mu}$ in Problem (1) or a regularized variant. Such policies are called *plug-in* and *regularized plug-in* policies, respectively (c.f. Definition 2.4). Crucially, typical denoising methods leverage components Z_k with $k \neq j$ in addition to Z_j

when constructing the estimate $T_j(\mathbf{Z})$ of μ_j . Indeed, the key intuition in denoising is to combine information from similar observations to smooth out corruptions. The definition of “similar” differs by method but often involves comparing features $\mathbf{W}_k$ and $\mathbf{W}_j$.

Given a class of such policies, our goal is to find the policy $\mathbf{x}(\mathbf{Z}, \mathbf{W})$ in that class with the best out-of-sample performance $\boldsymbol{\mu}^\top \mathbf{x}(\mathbf{Z}, \mathbf{W})$ with respect to the objective in Problem (1).

Since $\boldsymbol{\mu}$ is unknown, even estimating the out-of-sample performance of a given policy $\mathbf{x}(\mathbf{Z}, \mathbf{W})$ is not trivial. The naive in-sample estimator $\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \mathbf{W})$ is often badly biased, i.e., $\mathbb{E}[\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \mathbf{W})] \neq \mathbb{E}[\boldsymbol{\mu}^\top \mathbf{x}(\mathbf{Z}, \mathbf{W})]$ (Gupta, Huang, and Rusmevichientong, 2022a).

Instead, we propose a novel estimator we call the *one-shot Variance Gradient Correction* (one-shot VGC) that debiases the in-sample performance $\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \mathbf{W})$ to obtain an estimate of $\boldsymbol{\mu}^\top \mathbf{x}(\mathbf{Z}, \mathbf{W})$. The one-shot VGC approximately computes the gradient of the objective value with respect to $\mathbf{Z} - \boldsymbol{\mu}$, similar to the method of Gupta, Huang, and Rusmevichientong (2022a), which we refer to as the *multi-shot VGC*. We provide a detailed comparison of the one-shot VGC and multi-shot VGC in Sections 1.2 and 8 below.

We prove uniform bounds on the error of our estimator of out-of-sample performance. The key insight is that this error is bounded (almost surely) by a particular measure of solution stability plus an approximation term that is easy to control (c.f. Theorem 4.2). Thus, to bound the error of our estimator, it is sufficient to study solution stability.

This perspective yields a simple analysis for problems where the plug-in policy is continuous. For example, with regularized plug-in policies and convex $\mathcal{X}$, the plug-in policy is Lipschitz continuous, and we show that the error of the one-shot VGC essentially scales like $\tilde{O}_p(n^{1/4})$ for ℓ_1 -strongly-convex regularizers and like $\tilde{O}_p(n^{3/4})$ for ℓ_2 -strongly-convex regularizers, uniformly over the policy class (c.f. Theorem 5.2 and Corollary 5.3). Similarly, for (unregularized) plug-in policies and strongly-convex $\mathcal{X}$ (c.f. Definition 5.4), the plug-in policy is *almost* Lipschitz continuous, and we prove similar rates of convergence on the uniform error (c.f. Theorem 5.6). In typical applications, the full information optimum of Problem (1) is $\Theta(n)$, and hence, these results imply that the *relative* error of the one-shot VGC as an estimator of out-of-sample performance vanishes as $n \rightarrow \infty$, uniformly over the policy class. This further implies that choosing a policy by optimizing our debiased estimate incurs vanishing relative regret as $n \rightarrow \infty$. The proofs of these results are short and straightforward highlighting the pivotal role of our new notion of solution stability.

The case of non-convex $\mathcal{X}$ and potentially unregularized plug-in policies is more delicate because plug-ins are typically discontinuous. Solution stability in this case depends strongly on the interplay between the plug-in function and the constraints of $\mathcal{X}$. We restrict attention to affine plug-in functions and feasible regions $\mathcal{X}$ that are “weakly-coupled” in the sense of Gupta, Huang, and Rusmevichientong (2022a). As we argue in Section 2.3, many denoising methods can be rewritten as linear smoothers or linear filters (Buja, Hastie, and Tibshirani, 1989; Wahba, 1990) and, hence, give rise to affine plug-in policies. Moreover, linear optimization problems with wide constraint matrices and some two-stage stochastic programs are weakly-coupled, so this class subsumes many applications of interest.

We tailor our one-shot VGC correction to this setting. We prove that the error of our estimator essentially scales like $\tilde{O}_p(n^{3/4})$ (c.f. Theorem 6.12 and discussion surrounding Corollary 6.8), so that in the typical case, the relative error of our debiased estimator of out-of-sample performance vanishes as $n \rightarrow \infty$. Again, this implies that choosing a policy by optimizing our debiased estimate incurs vanishing relative regret as $n \rightarrow \infty$.

The key to our analysis in this setting is a novel proof technique that breaks the dependence structure in weakly-coupled problems by considering a suitably lifted problem. In the lifted space, we can characterize dependence between components of the plug-in policy by the chromatic number of a particular graph constructed from $\mathcal{X}$ and the plug-in function. This analysis allows us to bypass many of the involved duality arguments and approximate strong-convexity arguments used in Gupta, Huang, and Rusmevichientong (2022a) to analyze weakly-coupled programs, and may have independent interest.

Finally, in Section 9, we perform a numerical case study on prioritizing requests for speed humps using motor vehicle accident data from New York City. Our decision-aware denoising approach outperforms decision-blind denoising in identifying locations with the most pedestrian injuries caused by motor vehicle accidents. The improvement arises because decision-blind methods “oversmooth” in this example – when restricting to regions with higher pedestrian injuries, less smoothing is needed than when considering all of New York, and our decision-aware method correctly exploits this feature.

1.1. Our Contributions

1. We propose the one-shot Variance Gradient Correction (one-shot VGC) for debiasing in-sample policy performance to estimate out-of-sample policy performance. The one-shot VGC can be applied to any plug-in or regularized plug-in policy for Problem (1).

2. Under common assumptions on the data-generating process, we bound the variance (Theorem 3.1) and tail behavior (Theorem 3.3) of the one-shot VGC. Furthermore we show the expected error of the one-shot VGC in estimating in-sample bias is bounded by a specific measure of solution stability that depends on both the policy and structure of Problem (1) (Theorem 4.2).

By further bounding this solution stability, we prove that:

3. When Problem (1) is convex, the plug-in function is Lipschitz, and the policy is regularized, we can uniformly bound the estimation error of the one-shot VGC (Theorem 5.2) in terms of properties of the regularizer. In particular, this bound implies that in typical settings the relative error of our procedure vanishes, and one can learn a best-in-class policy as the problem size grows.
4. When $\mathcal{X}$ is strongly-convex, the plug-in function is Lipschitz, and the policy is unregularized, we can uniformly bound the estimation error (Theorem 5.6) in terms of properties of the feasible region and the magnitude of the plug-in function. This bound also implies that, in typical settings, the relative error of our procedure vanishes, and one can learn a best-in-class policy as the problem size grows.
5. When Problem (1) is potentially non-convex but weakly coupled, the plug-in function is affine, and the policy is unregularized, we uniformly bound the estimation error of the one-shot VGC in terms of the chromatic number of a particular graph depending on both $\mathcal{X}$ and the affine policy (Theorem 6.12). We show that even with crude upper bounds on this chromatic number, the relative error of our procedure vanishes, and it is possible to learn a best-in-class policy as the problem size grows in typical settings.

Finally,

6. We present a case study on deploying speed humps to reduce pedestrian injuries in motor vehicle accidents using data from New York City. We compare our decision-aware denoising approach to a decision-blind estimate-then-optimize approach. We find that the ideal amount of smoothing depends on the downstream loss, and, hence, decision-blind methods cannot always perform optimally. By contrast, our decision-aware approach finds a policy with performance comparable to oracle performance.

1.2. Relationship to Prior Work

Denoising has a long history in both signal processing and statistics. Our review is necessarily incomplete. Methods in signal processing include using Fourier analysis to discard

high-frequency noise (Rabiner and Gold, 1975), solving ℓ_1 -penalized optimization problems to deblur edges in images (Candes, Romberg, and Tao, 2006), and, more recently, leveraging neural networks and autoencoders to enhance blurry video (Tian et al., 2020). In statistics, denoising is often cast as a type of “smoothing” (Buja, Hastie, and Tibshirani, 1989; Seeger, 2004; Wahba, 1990). Such smoothing methods are particularly common when analyzing geospatial data (Chiles and Delfiner, 2012). Both streams of literature focus on the quality of signal recovery.

By contrast, our work focuses on using denoising to improve decision-making in a downstream optimization problem. Unlike the two previous streams, we are not concerned with recovering the original signal or the error of our denoised estimate from the original signal.

In this sense, our work contributes to the growing literature of “decision-aware learning,” also called “operational data analysis” or “decision-focused learning” (Liyanaage and Shanthikumar, 2005; Grigas, Qi, et al., 2021; Elmachoub and Grigas, 2022; Wilder, Dilkina, and Tambe, 2019; Hu, Kallus, and Mao, 2022; Feng and Shanthikumar, 2023; Chu et al., 2023). Most decision-aware learning research studies a contextual setting in which we see a sequence of problems depending on uncertain parameters $\mathbf{Y}_j$ and each problem has a context $\mathbf{X}_j$ for $j = 1 \dots, n$. The goal is, given a *new* context $\mathbf{X}_{n+1}$, predict the corresponding $\mathbf{Y}_{n+1}$ so as to induce a high-quality decision for the $(n + 1)^{\text{th}}$ problem.

By contrast, most denoising applications involve a single problem setting, e.g. state-wide drug-overdose data, and seek to recover the signal for that particular setting. We similarly focus on a single, noisily observed, problem and are interested in performance on that particular problem, not some unseen new instance to be drawn in the future. In this sense, we contribute to recent methods tailored to “small-data” settings (Besbes and Mouchtaki, 2023; Besbes, Ma, and Mouchtaki, 2022). Typical denoising applications also involve many “units,” e.g., census tracts in the state, to enable cross-learning, and exhibit low signal-to-noise ratio due to imprecise sensors, missing covariates, or small-scale surveys. Thus, we focus on a small-data, large-scale optimization regime where the precision of our signal is bounded and the number of units is large. This regime exhibits fundamentally different asymptotics than the large-sample regime (Gupta and Rusmevichientong, 2021; Gupta, 2022), distinguishing our work from most prior work on decision-aware learning.

Our one-shot VGC is based on a perturbation argument and Danskin’s Theorem. Similar perturbation arguments are used in Ito, Yabe, and Fujimaki (2018) and Guo, Jordan,

and Zhou (2022) in very different contexts (parameter estimation and causal inference, respectively). Both works offer asymptotic analysis in the large-sample limit. By contrast, we prove explicit (finite n) bounds in a small-data, large-scale regime.

Finally, our work builds on the results of Gupta, Huang, and Rusmevichientong (2022a) and Gupta and Rusmevichientong (2021). All three works share a similar model setup and goal; they seek to learn high-quality plug-in policies for optimization problems with a linear objective corrupted by noise. Gupta and Rusmevichientong, 2021 proposes a correction based on Stein’s Lemma that applies to very general policies. Under fairly general conditions, the relative error of the Stein correction is vanishing (Gupta, Huang, and Rusmevichientong, 2022b). By contrast, Gupta, Huang, and Rusmevichientong (2022a) more strongly exploits optimization structure through Danskin’s Theorem and focuses specifically on a class of unregularized, **separable** affine plug-in policies and weakly-coupled optimization problems. A separable affine plug-in estimates the unknown μ_j by a linear function $T_j(Z_j)$ that does *not* depend on $Z_k \neq j$. They propose (what we call) the *multi-shot VGC* to debias in-sample policy performance and prove performance guarantees when applying it to separable affine policies and weakly-coupled optimization problems.

As noted above, however, typical denoisers rely on information from components Z_k with $k \neq j$ when estimating μ_j , and hence are *not* separable. Thus, it is unclear whether the multi-shot VGC of Gupta, Huang, and Rusmevichientong (2022a) is provably or practically useful in denoising settings. (In fact our results in Section 8 show multi-shot VGC can, in fact, be badly biased for non-separable policies.) By contrast, although the Stein correction *is* applicable to denoising settings, as observed in Gupta, Huang, and Rusmevichientong, 2022a, the correction is typically discontinuous in policy parameters and the data. A host of research, e.g., Bousquet and Elisseeff (2002) and Breiman (1996), suggests improving smoothness and stability can improve empirical performance of learning methods, and Gupta, Huang, and Rusmevichientong, 2022a show empirically that the Stein Correction’s non-smoothness hurts its empirical performance. A continuous, stable alternative would be preferred.

Our work improves upon and complements Gupta, Huang, and Rusmevichientong (2022a) and Gupta and Rusmevichientong (2021) by proposing a novel debiasing scheme (the one-shot VGC) with good theoretical and empirical performance for a broad set of

policies and optimization problems (1). Unlike the Stein correction, our estimator is Lipschitz continuous in the data and policy parameters. Moreover, we prove performance guarantees when applying one-shot VGC with

- i) any regularized plug-in policy with a Lipschitz plug-in for convex $\mathcal{X}$,
- ii) any unregularized plug-in policy with a Lipschitz plug-in for strongly-convex $\mathcal{X}$, or
- iii) any affine plug-in policy (potentially non-separable) for weakly-coupled $\mathcal{X}$.

For clarity, the multi-shot VGC of Gupta, Huang, and Rusmevichientong (2022a) is not well-defined for nonlinear plug-in functions (cases i) and ii) above), and has poor performance for non-separable affine plug-in policies (case iii).

Finally, from a more theoretical perspective, our work introduces two technical tools that facilitate our simpler and more general proofs. First, we use a novel notion of solution stability (c.f. Theorem 4.1) to bound our estimation error with high-probability. By comparison, Gupta, Huang, and Rusmevichientong (2022a) introduced a different notion of stability – average solution stability – in their work. They show small average solution stability is a sufficient condition to bound the variance of their estimator for a fixed policy. However, such pointwise variance bounds are not sufficient for learning, and hence their performance guarantees do not explicitly leverage average solution stability. By contrast, we prove our bounds by directly analyzing our novel solution stability concept leading to more streamlined and intuitive proofs. The second technical tool is our aforementioned “lifting” technique that allows us to bypass the difficult duality and approximate strong-convexity argument of Gupta, Huang, and Rusmevichientong (2022a) used to analyze weakly-coupled problems. Both ideas may be useful in other settings.

2. Model

Let $\boldsymbol{\mu} \in \mathbb{R}^n$ be a fixed but unknown signal. We observe corrupted estimates $\mathbf{Z} = \boldsymbol{\mu} + \boldsymbol{\xi}$ where $\boldsymbol{\xi} \in \mathbb{R}^n$. Throughout the remainder, we assume

Assumption. (Mean-Zero, Independent Corruptions) *The noise $\boldsymbol{\xi}$ is mean-zero, has independent components, and known covariance $\boldsymbol{\Sigma} = \text{diag}(1/\nu_1, \dots, 1/\nu_n)$.*

More explicitly, ν_j is the precision of Z_j . For convenience, let $\nu_{\min} \equiv \min_j \nu_j$. We also observe fixed features $\mathbf{W}_j \in \mathbb{R}^p$ that potentially contain information relating components of $\boldsymbol{\mu}$, and let $\mathbf{W} \in \mathbb{R}^{n \times p}$ be the matrix with j^{th} row $\mathbf{W}_j$.

In traditional signal-processing applications, e.g., with geospatial data, $\mathbf{Z}$ represents sensor readings at n locations. The random corruptions $\boldsymbol{\xi}$ represent irreducible measurement

error, and the features $\mathbf{W}_j$ encode information about the j^{th} location (latitude, longitude, elevation, terrain type, rainfall level, etc.). Often, such applications entail many locations (large n), and improving sensor quality is prohibitively expensive (low precisions ν_j). Such applications are perhaps best studied in the so-called *small-data, large-scale regime*, (Gupta and Rusmevichientong, 2021; Gupta, Huang, and Rusmevichientong, 2022a; Gupta and Kallus, 2022), and our analysis takes this perspective.

Our model can also approximate some machine learning applications with panel data (Ignatiadis and Wager, 2019; Chung et al., 2022; Gupta and Kallus, 2023). Here, j refers to individual “units” such as patients, facilities, or regions, with $\mathbf{W}_j$ encoding the corresponding unit information. The “observations” $\mathbf{Z}$ represent predictions from an upstream, black-box machine learning model with known precisions $\boldsymbol{\nu}$. Because of privacy constraints, missing data and a host of other issues, precisions can be low (see above references for examples). Moreover, typical applications often focus on a specific set of existing units — e.g., a certain set of existing facilities — rather than assessing the performance of a policy on a hypothetical “out-of-sample” unit.

In either interpretation, we might believe two components j and k with “similar” features $\mathbf{W}_j$ and $\mathbf{W}_k$ have similar underlying μ_j and μ_k . Thus, we expect the observation Z_k to be informative of the underlying signal μ_j .

REMARK 1 (CORRELATED NOISE). That ξ_j is independent across j is not without loss of generality, and, in particular, may fail for some machine learning models with panel data. In some instances, however, we can transform a problem with correlated noise into one that matches our assumption. Specifically, suppose $\text{Cov}(\boldsymbol{\xi}) = \boldsymbol{\Omega}$ is full-rank and that we can write $\mathbf{Z} = \boldsymbol{\mu} + \boldsymbol{\Omega}^{1/2}\boldsymbol{\eta}$ where $\boldsymbol{\eta}$ a mean-zero with independent components. (A canonical example is a multivariate Gaussian.) Then, the transformed problem with $\mathbf{Z} \leftarrow \boldsymbol{\Omega}^{-1/2}\mathbf{Z}$ and $\mathcal{X} \leftarrow \{\boldsymbol{\Omega}^{1/2}\mathbf{x} : \mathbf{x} \in \mathcal{X}\}$ has independent noise components with unit precision.

To highlight an application of Problem (1) with such data, consider the following:

Example 2.1 (Speed Hump Planning in NYC) Certain studies such as Elvik et al. (2009) have shown speed humps can reduce the number of crashes by 50%–60%. They are highly requested traffic calming measures used to reduce crashes, injuries, and fatalities. In New York City, residents can request speed humps; however, Kuntzman (2022) recently found that the wait time for such requests can be as long as 12 *years*. Speed studies

that assess the feasibility of deploying a speed hump at a requested location are a salient bottleneck, taking an average two years to complete.

To prioritize speed hump requests, consider the following linear optimization problem that chooses speed humps that would, if built, reduce the most pedestrian injuries:

$$\min_{\mathbf{x}} \quad \sum_{k=1}^K \mu_k x_k, \quad \text{s.t.} \quad \sum_{k=1}^K x_k \leq B, \quad x_k \in \{0, 1\} \quad \forall k = 1, \dots, K. \quad (2)$$

Here $\{1, 2, \dots, K\}$ denotes the set of requests for speed humps, and B is the number of speed humps that can be processed in a year. The binary variable x_k encodes if request k will be processed. We assume that the potential *relative* reduction of pedestrian injuries is the same for all requests. The cost μ_k is the expected number of pedestrian injuries per year at location k . In cities like New York City, estimates Z_k of μ_k can be formed from traffic accident reports, but are often noisy. Features $\mathbf{W}_k$ capture information such as longitude and latitude of the accident, street width, speed limit, and number of bike lanes.

As a second application of Problem (1), we consider seeking a portfolio allocation problem similar to El Balghiti et al., 2023.

Example 2.2 (Portfolio Allocation) Consider a portfolio allocation problem in which we observe noisy returns $R_t \in \mathbb{R}^n$ for $t = 1, \dots, T$ of n distinct assets. We seek to solve

$$\max_{\mathbf{x}} \mathbf{x}^\top \boldsymbol{\mu} \quad \text{s.t.} \quad \mathbf{x}^\top \boldsymbol{\Omega} \mathbf{x} \leq \Gamma,$$

where $\boldsymbol{\mu} \equiv \mathbb{E}[\mathbf{R}]$, $\boldsymbol{\Omega}$ is a (given) covariance matrix for $\mathbf{R}$ and Γ is a risk threshold. Note, Da, Nagel, and Xiu (2024) argue that for typical financial settings, $\boldsymbol{\Omega}$, which is primarily driven by systemic risk, is much more easily estimated than the expected return $\boldsymbol{\mu}$, which also depends on idiosyncratic price premia. We let $\mathbf{Z} = \frac{1}{T} \sum_{t=1}^T R_t$ to be our noisy observation of $\boldsymbol{\mu}$. Features $\mathbf{W}_j$ encode information about each asset (e.g. its industrial sector). This problem is an instance of Problem (1) with a strongly convex feasible region. (See Definition 5.4). While it is reasonable to invoke the central limit theorem to claim $\mathbf{Z}$ is approximately Gaussian, its components may *not* be independent. However, we can appeal to Remark 1.

As mentioned, many denoising methods share information across similar locations to estimate $\boldsymbol{\mu}$. We next introduce two policy classes that exploit this intuition.

2.1. Policy Classes

We study a class of (potentially regularized) plug-in policies. Let $\mathbf{T} : \mathbb{R}^n \times \bar{\Theta} \mapsto \mathbb{R}^n$ be a transformation of the data depending on a user-defined parameter $\bar{\theta} \in \bar{\Theta}$, and let $\phi : \mathcal{X}_i \mapsto \mathbb{R}_+$ be a strongly convex regularizer, where $\mathcal{X}_i$ is the projection of $\mathcal{X}$ onto the i^{th} coordinate. Throughout the remainder, we require that $\mathbf{T}(\cdot)$ is Lipschitz:

Assumption 2.3 (Lipschitz Plug-in) *The function $\mathbf{Z} \mapsto \mathbf{T}(\mathbf{Z}, \bar{\theta})$ is $C_T(\bar{\theta})$ -Lipschitz with respect to the ℓ_1 norm, i.e., $\|\mathbf{T}(\mathbf{Z}, \bar{\theta}) - \mathbf{T}(\mathbf{Y}, \bar{\theta})\|_1 \leq C_T(\bar{\theta})\|\mathbf{Z} - \mathbf{Y}\|_1$, and $C_T = \max_{\bar{\theta} \in \bar{\Theta}} C_T(\bar{\theta})$ is finite.*

When $\mathbf{T}(\mathbf{Z}, \bar{\theta}) = \mathbf{L}(\bar{\theta})\mathbf{Z} + \mathbf{l}(\bar{\theta})$ is affine, $C_T(\bar{\theta}) = \max_j \|\mathbf{L}_j(\bar{\theta})\|_1$, i.e., the maximal ℓ_1 -norm of the columns.

Given $\mathbf{T}(\cdot)$, ϕ and $\bar{\Theta}$, we can define our policy class:

Definition 2.4 (Regularized Plug-in Policy) *A regularized plug-in policy $\mathbf{x}(\mathbf{Z}, (\rho, \bar{\theta}))$ for Problem (1) takes the form¹*

$$\mathbf{x}(\mathbf{Z}, (\rho, \bar{\theta})) \in \arg \min_{\mathbf{x} \in \mathcal{X}} \sum_{j=1}^n (T_j(\mathbf{Z}, \bar{\theta})x_j + \rho\phi(x_j)). \quad (3)$$

The policy is defined by $(\rho, \bar{\theta})$ – i.e., T and ϕ are fixed – hence we sometimes write $\boldsymbol{\theta} = (\rho, \bar{\theta})$ for brevity. Given $\bar{\Theta}$ and an interval $[\rho_{\min}, \rho_{\max}]$, let $\Theta = [\rho_{\min}, \rho_{\max}] \times \bar{\Theta}$ for brevity, and define the policy class $\mathcal{X}_\Theta(\mathbf{Z}) \equiv \{\mathbf{x}(\mathbf{Z}, (\rho, \bar{\theta})) : (\rho, \bar{\theta}) \in \Theta\}$.

We assume throughout that Θ is compact. In order to prove uniform error bounds, however, we also require that Θ is suitably “low-complexity.” We make a standard covering assumption:

Assumption 2.5 (Plug-in and Regularization Covering) *For any $\varepsilon > 0$, there exists a deterministic set $\Theta_0(\varepsilon)$ such that for every $(\bar{\theta}, \rho) \in \Theta$ there exists a $(\bar{\theta}_0, \rho_0) \in \Theta_0(\varepsilon)$ such that $\|\mathbf{T}(\mathbf{Z}, \bar{\theta}) - \mathbf{T}(\mathbf{Z}, \bar{\theta}_0)\|_1 \leq \varepsilon C_\Theta(\|\mathbf{Z}\|_1 + 1)$ and $|\rho - \rho_0| \leq \varepsilon$ for some constant $C_\Theta \geq 1$.*

Affine plug-ins where the matrix varies smoothly in $\boldsymbol{\theta}$ typically satisfy Assumption 2.5 because Θ is compact. Indeed, for some denoisers, Θ is finite, and hence Assumption 2.5 is trivially satisfied. See Section 2.3 for examples.

¹ When Eq. (3) admits multiple optima, we tie-break via some deterministic rule, e.g., taking the lexicographically smallest solution.

Intuitively, regularized plug-in policies proxy the unknown μ by a transformation $T(\mathbf{Z}, \bar{\theta})$ of the data $\mathbf{Z}$, and plug this proxy into a version of Problem (1) with a regularizer. The use of the regularizer $\phi(\cdot)$ in this proxy problem is common (see, e.g. Grigas, Qi, et al. (2021) and Wilder, Dilkina, and Tambe (2019)), because when $\mathcal{X}$ is convex it ensures policies are (almost everywhere) differentiable with respect to $\bar{\theta}$. However, in settings where $\mathcal{X}$ is non-convex – e.g., when Problem (1) encodes a minimum spanning tree problem – we lose this differentiability, and the value of regularization is unclear. Hence, in what follows, we treat i) regularized policies, i.e., $\rho > 0$ and ii) unregularized policies, i.e., $\rho = 0$ separately in Sections 5 and 6, respectively.

We focus on regularized plug-in policies because they are “light touch” in the sense of Chung et al. (2022), i.e., they are only marginally more complex to implement than an estimate-then-optimize procedure. Specifically, for a fixed $\bar{\theta}$, one need not alter the measurement or prediction procedure used to obtain $\mathbf{Z}$. The transformation $T(\mathbf{Z}, \bar{\theta})$ is applied as a post-processing step. Unregularized plug-in policies are also “light touch” with respect to the optimization. If one has a specialized algorithm to solve Problem (1) efficiently, one can apply this algorithm directly to the transformed vector $T(\mathbf{Z}, \bar{\theta})$.

Remark 2.6 (Relation to Separable, Affine Plug-in Policies) Regularized plug-in policies generalize the *separable* affine plug-in policy class studied in Gupta, Huang, and Rusmevichientong (2022a). Namely, separable affine plug-in policies take $\rho = 0$ and $T_j(\mathbf{Z}, \bar{\theta}) = a_j(\bar{\theta})Z_j + b_j(\bar{\theta})$, where the functions a_j and b_j may depend on $\mathbf{W}$. We focus on general plug-ins as they are necessary to exploit our earlier denoising intuition that Z_k may be informative for μ_j when $\mathbf{W}_j$ and $\mathbf{W}_k$ are similar. Because $T_j(\mathbf{Z}, \bar{\theta})$ may depend on Z_k for $k \neq j$, our analysis of general, regularized plug-in policies below must use substantively different techniques than those in Gupta, Huang, and Rusmevichientong (2022a).

2.2. A Decision-Aware Oracle Benchmark

Unlike previous works on denoising, we do *not* assume that $\mu = T(\mathbf{Z}, \bar{\theta}^*)$ for some value of $\bar{\theta}^*$. Instead, we take a best-in-class perspective. We define the oracle parameter

$$\theta^{OR} \in \arg \min_{\theta \in \Theta} \mu^\top x(\mathbf{Z}, \theta), \quad (4)$$

and corresponding oracle policy $x(\mathbf{Z}, \theta^{OR})$. This oracle is “best-in-class” over Θ , but not easily computable because μ is unknown.

In what follows, we consider two tasks:

1. (Policy Evaluation) Given a fixed θ , provide an estimate of $\mu^\top \mathbf{x}(\mathbf{Z}, \theta)$. Our strategy will be to debias the observed performance $\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \theta)$.
2. (Policy Learning) Find a parameter $\hat{\theta}(\mathbf{Z})$ (depending on the data) such that $\mu^\top (\mathbf{x}(\mathbf{Z}, \hat{\theta}(\mathbf{Z})) - \mathbf{x}(\mathbf{Z}, \theta^{OR}))$ is small. Our strategy will be to optimize our debiased estimator from Task 1. Then, a standard result shows that $\mu^\top (\mathbf{x}(\mathbf{Z}, \hat{\theta}(\mathbf{Z})) - \mathbf{x}(\mathbf{Z}, \theta^{OR}))$ is at most twice the uniform error of our estimator over Θ (c.f. Lemma A.2). Hence, if this error is small, our learned policy achieves near oracle performance.

2.3. Examples of Denoising Plug-Ins

Before delving into the details of our approach, we describe popular denoising plug-ins inspired by modern machine learning methods. The following examples are all *affine* transformations of $\mathbf{Z}$. Affine structure is not necessary for the majority of our paper, but is necessary for our results in Section 6. Moreover, for each example, one can explicitly compute $C_{\mathbf{T}}$ as the maximal ℓ_1 norm of the relevant matrix and verify Assumption 2.5 holds when Θ is compact.

Example 2.7 (Kernel Smoothers) Kernel smoothers define similarity via a kernel function $K_{\bar{\theta}}(\mathbf{W}_j, \mathbf{W}_k)$ with parameters $\bar{\theta}$. For example, the box-kernel is defined by $K_{\bar{\theta}}(\mathbf{W}_j, \mathbf{W}_k) = \mathbb{I}\{\|\mathbf{W}_j - \mathbf{W}_k\| \leq \bar{\theta}\}$. Inspired by kernel smoothing, we consider a plug-in policy using Nadaraya-Watson regression as our transformation, i.e.,

$$T_j^{KR}(\mathbf{Z}, \bar{\theta}) = \sum_{i=1}^n \frac{K_{\bar{\theta}}(\mathbf{W}_j, \mathbf{W}_i)}{\sum_{l=1}^n K_{\bar{\theta}}(\mathbf{W}_j, \mathbf{W}_l)} Z_i. \quad (5)$$

This expression is affine in $\mathbf{Z}$. By varying $\bar{\theta} \in \bar{\Theta}$, we obtain a class of plug-in-policies each with a different amount of smoothing.

Example 2.8 (Local Regression) Local linear regression uses a kernel to upweight similar data points when predicting for a given test point. To estimate μ_j , it solves

$$\beta_j^{\text{LR}}(\mathbf{Z}, \bar{\theta}) \in \arg \min_{\beta} \sum_{i=1}^n K_{\bar{\theta}}(\mathbf{W}_j, \mathbf{W}_i) (\mathbf{W}_j \beta - Z_j)^2,$$

and outputs $T_j^{\text{LR}}(\mathbf{Z}, \bar{\theta}) = \mathbf{W}_j^\top \beta_j^{\text{LR}}(\mathbf{Z}, \bar{\theta})$. Defining the Graham matrix $\mathbf{K} = (K_{\bar{\theta}}(\mathbf{W}_j, \mathbf{W}_k))_{ij} \in \mathbb{R}^{n \times n}$, we can rewrite this estimator as

$$T_j^{\text{LR}}(\mathbf{Z}, \bar{\theta}) = \mathbf{W}_j^\top (\mathbf{W}^\top \text{diag}(\mathbf{K}_j) \mathbf{W})^{-1} \mathbf{W}^\top \text{diag}(\mathbf{K}_j) \mathbf{Z},$$

which is again affine in $\mathbf{Z}$. By varying $\bar{\boldsymbol{\theta}}$ (and thus the Graham Matrix), we obtain a class of plug-in policies based on local linear regression with different sized neighborhoods. Polynomial local regression methods can also be used in a similar way.

Example 2.9 (Ridge Regression and Random Features) In high dimensional settings when $p \gg n$, regularized regression is often use to improve stability. For example, ridge regression solves

$$\boldsymbol{\beta}^{\text{RR}}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) \in \arg \min_{\boldsymbol{\beta}} \|\mathbf{W}\boldsymbol{\beta} - \mathbf{Z}\|_2^2 + \bar{\boldsymbol{\theta}}\|\boldsymbol{\beta}\|_2, \quad (6)$$

and then proxies $\boldsymbol{\mu}$ by

$$\mathbf{T}^{\text{RR}}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) = \mathbf{W}\boldsymbol{\beta}^{\text{RR}}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) = \mathbf{W}(\mathbf{W}^\top \mathbf{W} + \bar{\boldsymbol{\theta}}\mathbf{I})^{-1} \mathbf{W}^\top \mathbf{Z},$$

which is affine in $\mathbf{Z}$. By varying $\bar{\boldsymbol{\theta}}$, we can construct a class of plug-in policies corresponding to different degrees of regularization.

If we first apply a non-linear transformation to each feature $\mathbf{W}_j \mapsto \psi(\mathbf{W}_j) \in \mathbb{R}^{\bar{p}}$ and then apply ridge regression, the resulting transformation is still in affine in $\mathbf{Z}$. Thus, plug-in policies based on kernelized ridge regression also fall within our scope. Recently, randomized feature maps (Gallicchio and Scardapane, 2020; Rahimi and Recht, 2008) have been used to approximate kernelized ridge regression. Such methods also give rise to affine policies.

Example 2.10 (Clustering) Consider a clustering method indexed by a hyperparameter $\bar{\boldsymbol{\theta}}$. (For example, take K -means clustering indexed by the number of clusters K .) Applied to the features $\mathbf{W}_j$, $j = 1, \dots, n$, clustering yields a partition $\mathcal{S}(\bar{\boldsymbol{\theta}}) = \{C_1, \dots, C_K\}$ of the set $\{1, \dots, n\}$. We estimate μ_j by the average of its cluster:

$$T_j^{\text{CL}}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) = \sum_{k=1}^K \mathbb{I}\{j \in C_k\} \left(\frac{1}{|C_k|} \sum_{i \in C_k} Z_i \right).$$

This transformation is affine in $\mathbf{Z}$. By varying $\bar{\boldsymbol{\theta}}$ (e.g. varying the number of clusters), we again obtain a class of plug-in policies with different local neighborhoods. For this example, $C_T = 1$ and if we take $\bar{\boldsymbol{\theta}}$ to represent the number of clusters, $\bar{\boldsymbol{\Theta}}$ is finite.

The above list is not exhaustive. Many transformations $\mathbf{T}(\cdot)$ can be cast as linear smoothers (see Buja, Hastie, and Tibshirani (1989)). Such predictors yield affine plug-in

policies, and are often indexed by a low dimensional parameter $\bar{\theta}$. Moreover, some non-affine plug-ins, e.g., arising from neural networks, can be approximated by affine plug-ins. For example, Jacot, Gabriel, and Hongler (2018) show that kernel regression using neural tangent kernels approximates infinitely wide fully-connected neural networks, and Arora et al. (2020) provide evidence that this approximation has comparably strong performance in small-data tasks.

That said, there are common denoising strategies that do not yield affine plug-ins, e.g., those based upon LASSO. Again, only the results in Section 6 require an affine plug-in. Our other results apply both to affine and non-affine plug-ins.

3. One-Shot Variance Gradient Correction for Policy Evaluation

To denoise the in-sample performance, we instead estimate the in-sample optimism since

$$\underbrace{\mu^\top \mathbf{x}(\mathbf{Z}, \theta)}_{\text{Out-of-Sample Performance}} = \underbrace{\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \theta)}_{\text{In-Sample Performance}} - \underbrace{\xi^\top \mathbf{x}(\mathbf{Z}, \theta)}_{\text{In-Sample Optimism}}.$$

Our approach to estimating $\xi^\top \mathbf{x}(\mathbf{Z}, \theta)$ is as follows: Define

$$\hat{\mathbf{x}}(\mathbf{t}, \rho) \in \arg \min_{\mathbf{x} \in \mathcal{X}} \mathbf{t}^\top \mathbf{x} + \rho \sum_{j=1}^n \phi(x_j) \quad \text{and} \quad V(\mathbf{t}, \rho) \equiv \mathbf{t}^\top \hat{\mathbf{x}}(\mathbf{t}, \rho) + \rho \sum_{j=1}^n \phi(\hat{x}_j(\mathbf{t}, \rho)).$$

Let $\mathbf{L}(\mathbf{Z}, \bar{\theta}) \in \mathbb{R}^{n \times n}$ be the Jacobian of $\mathbf{T}(\mathbf{Z}, \bar{\theta})$ with respect to $\mathbf{Z}$. Consider the symmetric matrix $\mathbf{L}(\mathbf{Z}, \bar{\theta})\Sigma + \Sigma\mathbf{L}(\mathbf{Z}, \bar{\theta})^\top$ and let $(\cdot)_+$ denote projection onto the positive semidefinite cone.² Fix some small user-defined constant $h > 0$ and define the random vector

$$\delta_h \sim \mathcal{N}(\mathbf{0}, h(\mathbf{L}(\mathbf{Z}, \bar{\theta})\Sigma + \Sigma\mathbf{L}(\mathbf{Z}, \bar{\theta})^\top)_+ + h^2\Sigma). \quad (7)$$

The *one-shot VGC Correction* is defined as

$$D(\mathbf{Z}, \theta) = D(\mathbf{Z}, (\rho, \bar{\theta})) \equiv \frac{1}{h} \mathbb{E} [V(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + \delta_h, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\theta}), \rho) | \mathbf{Z}]. \quad (8)$$

Computationally, the expectation can be approximated by simulating δ_h and averaging. The one-shot VGC $D(\mathbf{Z}, \theta)$ is our estimate of the in-sample optimism $\xi^\top \mathbf{x}(\mathbf{Z}, \theta)$. Hence our estimate of out-of-sample performance is $\mu^\top \mathbf{x}(\mathbf{Z}, \theta) \approx \mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \theta) - D(\mathbf{Z}, \theta)$.

² For a symmetric matrix $\mathbf{S}$, let $\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ be its eigendecomposition. Then its projection onto the positive semidefinite cone is $\mathbf{S}_+ = \mathbf{Q}\mathbf{\Lambda}_+\mathbf{Q}^\top$ where $\mathbf{\Lambda}_+ = \text{diag}(\max(0, \lambda_i))$.

For concreteness, we define the one-shot VGC Correction with respect to the particular δ_h in Eq. (7). We motivate this choice in Proposition 4.3 after developing our error bound. However, unless otherwise specified, all of our results below hold for an *arbitrary* distribution of δ_h . See Section 4.1 for discussion.

3.1. A Heuristic Derivation

We next motivate the one-shot VGC via a heuristic derivation similar to that of the multishot VGC Gupta, Huang, and Rusmevichientong, 2022a. We consider a fixed policy and drop $(\rho, \bar{\theta})$ from the notation in this derivation.

Consider the function

$$\lambda \mapsto V(\mathbf{T}(\mathbf{Z}) + \lambda \boldsymbol{\xi}). \quad (9)$$

Intuitively, this function *adds* noise to the system $(\lambda \boldsymbol{\xi})$ to see how the plug-in value degrades. Studying systems with increased noise is philosophically similar to the progressive noise corruption/learned denoiser paradigm underlying modern diffusion based generative models (Song and Ermon, 2019; Ho, Jain, and Abbeel, 2020). Here, we are interested in the gradient of this function, i.e., how quickly does the plugin degrade?

By Danskin’s theorem, whenever $\hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}) + \lambda \boldsymbol{\xi})$ is the unique optimizer, the derivative of this function is

$$\left. \frac{\partial}{\partial \lambda} V(\mathbf{T}(\mathbf{Z}) + \lambda \boldsymbol{\xi}) \right|_{\lambda=0} = \boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z})) = \boldsymbol{\xi}^\top \mathbf{x}(\mathbf{Z}). \quad (10)$$

(When $\hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}) + \lambda \boldsymbol{\xi})$ is not the unique optimizer, then $\boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}))$ is a subgradient.) This is exactly the optimism. Said differently, the derivative measures how the plugin degrades as we *add* noise to the system. If we take a gradient step in the opposite direction, we effectively *remove* noise, since

$$\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}) - \left. \frac{\partial}{\partial \lambda} V(\mathbf{T}(\mathbf{Z}) + \lambda \boldsymbol{\xi}) \right|_{\lambda=0} = \mathbf{Z}^\top \mathbf{x}(\mathbf{Z}) - \boldsymbol{\xi}^\top \mathbf{x}(\mathbf{Z}) = \boldsymbol{\mu}^\top \mathbf{x}(\mathbf{Z}).$$

Since estimating optimism (the rightside of Eq. (10)) is difficult, we attempt to estimate the leftside. We approximate the leftside with a first-order, forward finite-difference approximation (LeVeque, 2007, Chapter 1). For sufficiently small $h > 0$, we expect that

$$\left. \frac{\partial}{\partial \lambda} V(\mathbf{T}(\mathbf{Z}) + \lambda \boldsymbol{\xi}) \right|_{\lambda=0} = \frac{1}{h} \left(V(\mathbf{T}(\mathbf{Z}) + h \boldsymbol{\xi}) - V(\mathbf{T}(\mathbf{Z})) \right) + o_p(h) \quad \text{as } h \rightarrow 0. \quad (11)$$

Finally, since ξ is unknown, we approximate $\mathbf{T}(\mathbf{Z}) + h\xi$ by $\mathbf{T}(\mathbf{Z}) + \delta_h$. The resulting estimator $\frac{1}{h}(V(\mathbf{T}(\mathbf{Z}) + \delta_h) - V(\mathbf{T}(\mathbf{Z})))$ depends on the random variable δ_h . To reduce variance, we derandomize our estimator by taking a conditional expectation over δ_h conditional on $\mathbf{Z}$ (a.k.a. Rao-Blackwellization), and yielding Eq. (8).

The above motivation is intentionally heuristic in order to be intuitive. Before presenting a rigorous analysis of our estimator, we must establish some basic properties.

3.2. Properties of the One-Shot VGC

We apply the Efron-Stein Inequality to bound the variance of the one-shot VGC:

Theorem 3.1 (Variance of the VGC) *Under Assumption 2.3, the variance of the one-shot VGC satisfies*

$$\text{Var}\left(D(\mu + \xi, (\bar{\theta}, \rho))\right) \leq \frac{4C_T^2(\bar{\theta})n}{h^2\nu_{\min}}.$$

We can strengthen this result to a high probability tail bound assuming ξ has light tails:

Assumption 3.2 (Independent Sub-Gaussian Corruptions) *For all $j = 1, \dots, n$, ξ_j is sub-Gaussian with variance proxy at most κ^2 .*

Theorem 3.3 (Concentration of the VGC) *Suppose Assumptions 2.3 and 3.2 hold. For any $0 < \epsilon < \frac{1}{2}$ the following holds with probability at least $1 - \epsilon$,*

$$|D(\mathbf{Z}, (\bar{\theta}, \rho)) - \mathbb{E}[D(\mathbf{Z}, (\bar{\theta}, \rho))]| \leq 2 \cdot C_T(\bar{\theta}) \frac{\kappa}{h} \sqrt{n \log\left(\frac{2}{\epsilon}\right)}.$$

Both Theorems 3.1 and 3.3 suggest that if $C_T(\bar{\theta})$ is $O_p(1)$, the stochastic fluctuations of the one-shot VGC are $O_p(n^{1/2}/h)$. Thus, in typical cases where the full information solution to Problem (1) is $O(n)$, the stochastic contributions are asymptotically negligible if $h\sqrt{n} \rightarrow \infty$ as $n \rightarrow \infty$. It may help the reader to keep this scaling in mind for the remainder. It provides a guideline for the size of h to achieve good performance.

4. Bounding the Estimation Error by Solution Stability

In this section, we bound the estimation error of our method by a more manageable term related to the stability of our policies:

Definition 4.1 *For any $\theta \in \Theta$ and $h > 0$, we define the solution stability of a policy by*

$$\text{SS}(\xi, h, \theta) = |\xi^\top (\hat{x}(\mathbf{T}(\mathbf{Z}, \bar{\theta}), \rho) - \hat{x}(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h\xi, \rho))|.$$

Intuitively, solution-stability measures how much the solution changes given a small perturbation of the data $\mathbf{Z}$ *in the direction of the noise* $\boldsymbol{\xi}$. In particular, solution-stability is random, and not directly computable (because it depends on $\boldsymbol{\xi}$). Below, we first show that the error of our estimator is bounded by solution stability (plus a term that is easy to control) and then discuss how to further bound solution-stability.

To that end, write

$$\begin{aligned} & \sup_{\boldsymbol{\theta} \in \Theta} \underbrace{\left| \mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta}) - \boldsymbol{\mu}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) \right|}_{\text{Estimation Error}} \\ & \leq \sup_{\boldsymbol{\theta} \in \Theta} \underbrace{\left| \boldsymbol{\xi}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - \frac{1}{h} (V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), \rho)) \right|}_{\text{Finite Difference Error}} \end{aligned} \quad (12a)$$

$$+ \sup_{\boldsymbol{\theta} \in \Theta} \underbrace{\left| \frac{1}{h} (V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), \rho)) - D(\mathbf{Z}, \boldsymbol{\theta}) \right|}_{\text{Replication Error}}. \quad (12b)$$

Intuitively, the Finite Difference Error measures the error incurred by approximating the derivative by a forward finite step approximation. The Replication Error measures the error introduced by replacing $\boldsymbol{\xi}$ by $\boldsymbol{\delta}$.

We can further bound Finite Difference Error by invoking the concavity of $f(\lambda) = V(\mathbf{T}(\mathbf{Z}, \boldsymbol{\theta}) + \lambda\boldsymbol{\xi}, \rho)$.³ By concavity,

$$\begin{aligned} f(h) & \leq f(0) + hf'(0) \implies \frac{V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), \rho)}{h} \leq \boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), \rho), \\ f(0) & \leq f(h) - hf'(h) \implies \frac{V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), \rho)}{h} \geq \boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho), \end{aligned}$$

where we have again used Danskin's theorem to compute the derivatives. Combining shows

$$\text{Eq. (12a)} \leq \sup_{\boldsymbol{\theta} \in \Theta} \left| \boldsymbol{\xi}^\top (\hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), \rho) - \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho)) \right| = \sup_{\boldsymbol{\theta} \in \Theta} \text{SS}(\boldsymbol{\xi}, h, \boldsymbol{\theta}). \quad (13)$$

As we will argue in Section C below, Solution Stability depends strongly on the structure of Problem (1) and the choice of policy class.

By contrast, we can bound the Replication Error without any additional assumptions, yielding the main workhorse result of our paper:

³ This map is concave because it is the minimum of linear functions of λ .

Theorem 4.2 (Bounding Estimation Error by Stability) *Suppose Assumptions 2.3, 2.5 and 3.2 hold. Then, there exists a constant C (depending on $\phi_{\max}$, $\nu_{\min}$, $\|\boldsymbol{\mu}\|_{\infty}$) such that the following holds with probability at least $1 - 2\epsilon$,*

$$\begin{aligned} \sup_{\boldsymbol{\theta} \in \Theta} \underbrace{\left| \mathbf{Z}^{\top} \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta}, h) - \boldsymbol{\mu}^{\top} \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) \right|}_{\text{Estimation Error}} &\leq \underbrace{C \frac{(C_{\mathbf{T}} + h)\kappa}{h} \sqrt{n \cdot \log n \cdot \log |\Theta_0(n^{-1/2})| \cdot \log \left(\frac{2}{\epsilon} \right)}}_{\text{Deviation of Replication Error}} \\ &\quad + \underbrace{\frac{1}{h} \sup_{\boldsymbol{\theta} \in \Theta} \mathbb{W}_1(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \boldsymbol{\delta}_h)}_{\text{Expected Replication Error}} \\ &\quad + \sup_{\boldsymbol{\theta} \in \Theta} \text{SS}(\boldsymbol{\xi}, h, \boldsymbol{\theta}) \end{aligned}$$

where $\mathbb{W}_1$ the Wasserstein 1-distance metric.

Recall that in a typical setting, we expect the full-information solution to Problem (1) to be $O(n)$. Hence, if the error of our estimate is of strictly lower order than $O(n)$, the relative error is asymptotically vanishing. For the first term, the order of the error depends on the choice of policy class. For many of the policies presented earlier in Section 2.3, e.g., Examples 2.7 and 2.10, $|\Theta_0(n^{-1/2})|$ is polynomial in n and, hence, the first term in the bound is $\tilde{O}(\sqrt{n}/h)$ or less. Hence, again, if $h\sqrt{n} \rightarrow \infty$, its relative error vanishes.

The second term of the bound depends on how well we replicate the (unknown) perturbation $h\boldsymbol{\xi}$ and is discussed in Section 4.1. Therein we argue it can be close to zero.

Finally, the third term — solution stability — depends intimately on the problem structure and policy class. We argue such dependence is fundamental and unavoidable in Appendix C. In Sections 5 and 6 we give precise, sufficient conditions under which solution stability grows slower than n to ensure the relative error of our procedure is vanishing.

4.1. The distribution of $\boldsymbol{\delta}_h$

The one-shot VGC can, in principle, be implemented with *any* choice of distribution for $\boldsymbol{\delta}_h$, and Theorem 4.2 still holds as written. Nonetheless, we advocate for Eq. (7) because, under assumptions common in the denoising literature, it leads to zero replication error.

Proposition 4.3 (Exact Distribution Replication with Multivariate Gaussians)

Suppose $\boldsymbol{\xi}$ has Gaussian distribution, that $\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) = \mathbf{L}(\bar{\boldsymbol{\theta}})\mathbf{Z} + \mathbf{l}(\bar{\boldsymbol{\theta}})$ is affine, and $\mathbf{L}(\bar{\boldsymbol{\theta}})\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{L}(\bar{\boldsymbol{\theta}})^{\top}$ is positive semidefinite. Then, if $\boldsymbol{\delta}_h$ follows Eq. (7), $\mathbb{W}_1(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \boldsymbol{\delta}_h) = 0$.

Many fundamental techniques for denoising are based on Gaussian assumptions (Johnstone, 2002) and, as seen in Section 2.3, many denoisers are affine. In these settings, Eq. (7) ensures the replication error is zero.

That said, given other apriori knowledge on $\mathbf{Z}$, one might construct an alternate distribution for δ_h that more closely matches $h\xi$. Indeed, techniques from the “data fission” and “data thinning” (Leiner et al., 2025; Dharamshi et al., 2025; Neufeld et al., 2024) — wherein we seek to split a single data sample into multiple samples with similar properties to the original — might be used to tailor a distribution for δ_h in these settings, but this is somewhat beyond our scope.

In sum, to be both completely rigorous and general, we prove results for an arbitrary distribution of δ_h and any mean-zero, independent noise ξ . Thus, should a user elect a different distribution for δ_h , our results still describe valid bounds on the error. Practically speaking, though, we expect Eq. (7) to yield good results with affine plug-ins whenever data are approximately Gaussian, since, in that case, the replication error should be approximately zero. We adopt this perspective in our case-study in Section 9 where we use Eq. (7) with non-Gaussian (i.e. Binomial) data.

Remark 4.4 (Misspecified Precisions) Algorithmically, the precisions ν_j are only leveraged in the specification of δ_h via Eq. (7). Hence, if they are not known, but estimated, we run the risk of “misspecifying” δ_h , which, by Theorem 4.2, may increase the replication error. If the misspecification is small, the resulting Wasserstein distance should be similar to well-specified setting.

The correlated case (c.f. Remark 1) is more subtle. If the matrix Ω is misspecified, not only do we potentially increase the replication error, but we may be left with residual dependence between components of $\mathbf{Z}$. In our proofs, this independence is primarily used to justify concentration to the mean. Insofar as mildly dependent random variables still concentrate, we again intuit our approach to be robust to mild misspecifications.

5. Bounds for Continuous Plug-in Policies

5.1. Regularized Policies

We first present bounds for regularized plug-in policies when the feasible set $\mathcal{X}$ is convex. The key insight is that when $\mathbf{t} \mapsto V(\mathbf{t}, \rho)$ is strongly-convex, plug-in policies are Lipschitz

in their argument. Other authors have utilized this Lipschitz continuity in a variety of contexts. We use it to derive a simple upper bound on solution stability. For any norm $\|\cdot\|$, let $\|\cdot\|_*$ denote the dual norm.

Lemma 5.1 (Solution Stability of Regularized Policies) *If $\mathcal{X}$ is convex, $\rho > 0$, and $\phi(\cdot)$ is 1-strongly convex with respect to the norm $\|\cdot\|$,*

$$|\text{SS}(\boldsymbol{\xi}, h, \boldsymbol{\theta})| \leq \frac{h}{\rho} \|\boldsymbol{\xi}\|_*^2, \text{ a.s..}$$

The bound only depends on the regularizer through the dual norm, but does not otherwise depend on the plug-in $\mathbf{T}(\mathbf{Z}, \boldsymbol{\theta})$. In this sense, it is general purpose.

Plugging Lemma 5.1 into Theorem 4.2 bounds estimation error uniformly.

Theorem 5.2 (Estimation Error for Regularized Policies) *Suppose Assumptions 2.3, 2.5 and 3.2 hold, $\mathcal{X}$ is convex, and $\Theta = [\rho_{\min}, \rho_{\max}] \times \bar{\Theta}$ for $0 < \rho_{\min} < \rho_{\max}$. If $\phi(\cdot)$ is 1-strongly convex with respect to $\|\cdot\|$, then there exists a constant C (depending on $\phi_{\max}$, $\nu_{\min}$, $\|\boldsymbol{\mu}\|_{\infty}$) such that with probability at least $1 - \epsilon$*

$$\begin{aligned} \sup_{\boldsymbol{\theta} \in \Theta} |\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta}, h) - \boldsymbol{\mu}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta})| &\leq \frac{h}{\rho_{\min}} \|\boldsymbol{\xi}\|_*^2 \\ &+ C \frac{(C_T + h)\kappa}{h} \sqrt{n \log |\Theta_0| (n^{-1/2}) \log \left(\frac{1}{\epsilon} \right)} + \frac{1}{h} \sup_{\boldsymbol{\theta} \in \bar{\Theta}} \mathbb{W}_1(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \boldsymbol{\delta}_h). \end{aligned}$$

Common regularizers include $\phi(x) = \frac{1}{2}x^2$, which is 1-strongly convex with respect to the ℓ_2 norm (so $\|\cdot\|_*$ is the ℓ_2 norm), and $\phi(x) = x \log x$, which is 1-strongly convex with respect to the ℓ_1 norm (so $\|\cdot\|_*$ is the ℓ_{∞} norm) on the unit simplex.

Under assumptions common in denoising, we can both compute such a bound and optimize the choice of h to obtain explicit rates:

Corollary 5.3 (Regularized Policies with Exact Replication) *Suppose the assumptions of Theorem 5.2 and Proposition 4.3 hold and $|\Theta_0(n^{-1/2})|$ is polynomial in n .*

i) *If $\phi(\cdot)$ is 1-strongly convex with respect to $\|\cdot\|_1$, letting $h = \rho_{\min}^{1/2} n^{1/4}$ shows*

$$\sup_{\boldsymbol{\theta} \in \Theta} |\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta}, h) - \boldsymbol{\mu}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta})| = \tilde{O}_p \left(n^{1/4} \rho_{\min}^{-1/2} \right),$$

ii) *If $\phi(\cdot)$ is 1-strongly convex with respect to $\|\cdot\|_2$, letting $h = \rho_{\min}^{1/2} n^{-1/4}$ shows*

$$\sup_{\boldsymbol{\theta} \in \Theta} |\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta}, h) - \boldsymbol{\mu}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta})| = \tilde{O}_p \left(n^{3/4} \rho_{\min}^{-1/2} \right).$$

Recall that in the typical case, the full-information optimal value of Problem (1) is $O(n)$. Thus, as long as $\rho_{\min}$ does not tend to 0 too fast as n grows, Corollary 5.3 shows that the relative error of our approach vanishes as n gets large.

5.2. Strongly Convex Feasible Regions

We next study unregularized policies in the special case where $\mathcal{X}$ is strongly convex by adapting techniques from El Balghiti et al., 2023. We first summarize key results from that work and then show how they can be used to bound solution stability and, hence, estimation error, in our setting.

For a given norm $\|\cdot\|$, let $B(\mathbf{x}_0, r) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\| \leq r\}$ be the ball of radius r centered at $\mathbf{x}_0$. Recall the following classical definition:

Definition 5.4 (Strongly Convex Feasible Region) *We say the feasible region $\mathcal{X}$ is α -strongly convex with respect to $\|\cdot\|$ if for any $\mathbf{x}, \mathbf{y} \in \mathcal{X}$, and any $\lambda \in [0, 1]$,*

$$B(\lambda\mathbf{x} + (1-\lambda)\mathbf{y}, \frac{\alpha}{2}\lambda(1-\lambda)\|\mathbf{x} - \mathbf{y}\|^2) \subseteq \mathcal{X}$$

In words, strong convexity of the feasible region ensures that $\mathcal{X}$ is “round.” El Balghiti et al. (2023, Theorem 7, Theorem 3a) establishes that when $\mathcal{X}$ is α -strongly convex, the plug-policies are *almost* Lipschitz continuous, namely⁴

$$\|\hat{\mathbf{x}}(\mathbf{t}, 0) - \hat{\mathbf{x}}(\bar{\mathbf{t}}, 0)\| \leq \frac{2\|\mathbf{t} - \bar{\mathbf{t}}\|_*}{\alpha(\|\mathbf{t}\|_* + \|\bar{\mathbf{t}}\|_*)}, \quad \forall \mathbf{t}, \bar{\mathbf{t}} \in \mathbb{R}^n.$$

These results provide a simple bound on solution-stability:

Lemma 5.5 (Solution Stability for Strongly-Convex Feasible Regions) *Suppose $\mathcal{X}$ is α -strongly convex with respect to $\|\cdot\|$. Then,*

$$|\text{SS}(\boldsymbol{\xi}, h, \boldsymbol{\theta})| \leq \frac{2h\|\boldsymbol{\xi}\|_*^2}{\alpha\|\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}})\|_*}.$$

Unlike Lemma 5.1, our bound depends not only on the strong-convexity parameter, but also the magnitude of $\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}})$. Intuitively, if $\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}})$ is close to $\mathbf{0}$, then a small perturbation might cause the plug-in policy to change wildly.

We can combine this bound with our previous results to provide an estimation bound for unregularized policies and strongly-convex feasible regions:

⁴El Balghiti et al. (2023, Theorem 3a) uses a slightly worse constant in which they bound $\frac{1}{2}(\|\mathbf{t}\|_* + \|\bar{\mathbf{t}}\|_*) \geq \min(\|\mathbf{t}\|_*, \|\bar{\mathbf{t}}\|_*)$. We use the tighter constant above.

Theorem 5.6 (Estimation Error for Strongly-Convex Feasible Regions)

Suppose Assumptions 2.3, 2.5 and 3.2 hold, $\mathcal{X}$ is α -strongly convex with respect to $\|\cdot\|$ and $\Theta = [0] \times \bar{\Theta}$ (i.e. $\rho = 0$) and $\inf_{\bar{\theta} \in \bar{\Theta}} \|\mathbf{T}(\mathbf{Z}, \bar{\theta})\|_* \geq T_{\min}$. Then there exists a constant C (depending on $\phi_{\max}, \nu_{\min}, \|\boldsymbol{\mu}\|_{\infty}$) such that with probability at least $1 - \epsilon$,

$$\begin{aligned} \sup_{\boldsymbol{\theta} \in \Theta} |\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta}, h) - \boldsymbol{\mu}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta})| &\leq \frac{Ch \|\boldsymbol{\xi}\|_*^2}{\alpha T_{\min}} \\ &+ C \frac{(C_T + h)\kappa}{h} \sqrt{n \log |\Theta_0| (n^{-1/2}) \log \left(\frac{1}{\epsilon} \right)} + \frac{1}{h} \sup_{\bar{\theta} \in \bar{\Theta}} \mathbb{W}_1(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h\boldsymbol{\xi}, \mathbf{T}(\mathbf{Z}, \bar{\theta}) + \boldsymbol{\delta}_h). \end{aligned}$$

Under the similar assumptions to Corollary 5.3, we can optimize the choice of h . The results are identical to Corollary 5.3 with $\rho_{\min}$ replaced by $\alpha T_{\min}$ and hence omitted for brevity.

6. Bounds for Unregularized Plug-in Policies in Weakly-Coupled Problems

For combinatorial optimization problems with non-convex feasible sets, the plug-in policy is typically *not* continuous with respect to its parameter. Hence, bounding solution stability requires us to exploit the structure of the underlying problem more heavily.

In this section, we prove estimation error bounds for non-regularized plug-in policies, i.e. when $\rho = 0$ and $\Theta = \{0\} \times \bar{\Theta}$. Since $\rho = 0$ throughout this section, we write $\hat{\mathbf{x}}(\mathbf{t})$ and drop ρ from the notation. We focus on a general class of linear optimization problems known as *weakly-coupled* linear optimization problems, also studied in Gupta, Huang, and Rusmevichientong (2022a).

To describe such problems, let $S_0, \dots, S_K$ be a disjoint partition of $\{1, \dots, n\}$ and let $\mathbf{x}^k = (x_j : j \in S_k)$ denote vectors of decision variables for $k = 0, \dots, K$. Without loss of generality, reorder the indices so that the S_k occur “in order,” i.e., $(j \in S_0), \dots, (j \in S_K)$ is a consecutive sequence. Weakly-coupled optimization problems have the following form:

$$\min_{\mathbf{x}^0, \dots, \mathbf{x}^K} \quad (\boldsymbol{\mu}^0)^\top \mathbf{x}^0 + \sum_{k=1}^K (\boldsymbol{\mu}^k)^\top \mathbf{x}^k \quad (14a)$$

$$\text{s.t.} \quad \sum_{k=0}^K \sum_{j \in S_k} \mathbf{A}_j^0(\mathbf{x}^0) x_j \leq \mathbf{b}^0(\mathbf{x}^0) \quad (14b)$$

$$\mathbf{x}^0 \in \mathcal{X}^0 \quad (14c)$$

$$\mathbf{x}^k \in \mathcal{X}^k(\mathbf{x}^0), \quad \forall k = 1, \dots, K, \quad (14d)$$

where for each fixed $\mathbf{x}^0$, $\mathbf{A}_j^0(\mathbf{x}^0) \in \mathbb{R}^m$ for all j , $\mathbf{b}^0(\mathbf{x}^0) \in \mathbb{R}^m$, and $\mathcal{X}^k(\mathbf{x}^0)$ for $k = 1, \dots, K$ are bounded, non-empty polyhedra. The coupling set $\mathcal{X}^0$ is possibly non-convex. We let $\mathcal{X}^{WC}$ denote the feasible set that satisfies Eqs. (14b) to (14d) and assume $\mathcal{X}^{WC} \subseteq [-1, 1]^n$. For convenience, define $S_{\max} \equiv \max_k |S_k|$ to be the size of the largest subproblem.

The optimization problem is “weakly-coupled” in the sense that by removing the linear constraints (Eq. (14b)) and fixing a choice of $\mathbf{x}^0 \in \mathcal{X}^0$ (Eq. (14c)), the optimization problem decouples into K separate linear optimization subproblems of the form:

$$\min_{\mathbf{x}^k} (\boldsymbol{\mu}^k)^\top \mathbf{x}^k, \quad \text{s.t. } \mathbf{x}^k \in \mathcal{X}^k(\mathbf{x}^0). \quad (15)$$

Many applications of linear optimization have the form of Problem (14). For example, two-stage stochastic optimization problems are often weakly coupled by variables. In these problems, $\mathbf{x}^0$ are first-stage decision variables (which are sometimes binary), and each $\mathbf{x}^k$ represents second-stage decisions for the K different scenarios. Gupta, Huang, and Rusmevichientong (2022a) consider such an instance to model drone assisted emergency medical response. In that application, $m = 0$ and $|\mathcal{X}^0| = O(1)$ as $K \rightarrow \infty$. Other applications are weakly-coupled by constraints, where the binding constraints (Eq. (14b)) model resource budgets, e.g., on time or labor. Example 2.1 is such an example with $m = 1$, where the single binding constraint Eq. (14b) limits the number of speed humps that can be built, and $\mathcal{X}^0 = \{0\}$.

In general, we believe that in typical applications, the number of subproblems $K = O(n)$, the number of coupling constraints $m = O(1)$, and $\log |\mathcal{X}^0| = o(n)$. Intuitively, this corresponds to a problem where there are many small subproblems, a relatively small number of coupling constraints, and a relatively small number of key, potentially discrete, decisions that affect the global structure. Both Example 2.1 and the drone-assisted medical response of Gupta, Huang, and Rusmevichientong, 2022a satisfy these conditions. Hence, although we present results for general instances, the reader may want to focus on these scalings as they interpret our results.

Additional Assumptions

Proving strong estimation error bounds for Problem (14) with general policies appears challenging. We instead restrict attention to a class of well-behaved affine policies. Let $\sigma_j(\bar{\boldsymbol{\theta}})$ be the j^{th} largest singular value of $\mathbf{L}(\bar{\boldsymbol{\theta}})$.

Definition 6.1 (Affine Plug-in Policy) *An affine plug-in policy $\mathbf{x}(\mathbf{Z}, (0, \bar{\boldsymbol{\theta}}))$ is a plug-in policy where $\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}}) = \mathbf{L}(\bar{\boldsymbol{\theta}})\mathbf{z} + \mathbf{l}(\bar{\boldsymbol{\theta}})$ and $\mathbf{L}(\cdot), \mathbf{l}(\cdot)$ may implicitly depend $\mathbf{W}$.*

Assumption 6.2 (Plug-ins are Affine) *The policy class $\Theta = \{0\} \times \bar{\Theta}$ consists of unregularized ($\rho = 0$) affine plug-in policies. Furthermore, $\|\mathbf{L}(\bar{\boldsymbol{\theta}})\|_2 = \sigma_1(\bar{\boldsymbol{\theta}}) \leq 1$ for all $\bar{\boldsymbol{\theta}} \in \bar{\Theta}$.*

As highlighted in Section 2.3, many machine learning methods give rise to unregularized affine plug-in policies. Furthermore, given any policy $\mathbf{x}(\mathbf{Z}, \boldsymbol{\theta})$, we can always rescale $\mathbf{L}(\boldsymbol{\theta})$ and $\mathbf{l}(\boldsymbol{\theta})$ to ensure the singular values of $\mathbf{L}(\bar{\boldsymbol{\theta}})$ are at most 1 without changing $\mathbf{x}(\mathbf{Z}, \boldsymbol{\theta})$. In this sense, the second part of the assumption amounts to a reparameterization of the policy class, and, in this sense, holds without loss of generality. As an aside, the separable affine plug-in policy class studied in Gupta, Huang, and Rusmevichientong (2022a) corresponds to the special case where $\mathbf{L}(\bar{\boldsymbol{\theta}})$ is diagonal for all $\bar{\boldsymbol{\theta}} \in \bar{\Theta}$.

Finally, we require two somewhat technical, regularity assumptions:

Assumption 6.3 (Invertibility of Submatrices) *Let $\mathbf{L}^k(\bar{\boldsymbol{\theta}}) \in \mathbb{R}^{|S_k| \times |S_k|}$ be the submatrix of $\mathbf{L}(\bar{\boldsymbol{\theta}})$ induced by the set S_k . We assume there exists a constant $\sigma_{\min}$ such that*

$$0 < \sigma_{\min} \leq \inf_{\bar{\boldsymbol{\theta}} \in \bar{\Theta}} \sigma_{\min}(\mathbf{L}^k(\bar{\boldsymbol{\theta}})), \quad \forall k = 1, \dots, K,$$

where $\sigma_{\min}(\cdot)$ denotes the smallest singular value, and h is the step size in the one-shot VGC (cf. Eq. (8)).

Intuitively, $\sigma_{\min} > 0$ bounds the above matrices “away from singularity.” When these matrices are singular, $\mathbf{T}(\mathbf{Z})$ is a lossy-transformation, and it is not possible to recover $\mathbf{Z}$ from $\mathbf{T}(\mathbf{Z})$, complicating the analysis. Often, we can satisfy Assumption 6.3 by adding a small perturbation $\gamma \mathbf{I}$ to $\mathbf{L}(\bar{\boldsymbol{\theta}})$.

Assumption 6.4 (Smoothly Varying Densities) *The random vector $\boldsymbol{\xi}$ admits a density $p(\cdot)$ and there exists a constant $C_{\boldsymbol{\xi}}$ such that $\mathbb{E}[\|\nabla \log p(\boldsymbol{\xi})\|^2] \leq C_{\boldsymbol{\xi}} \epsilon$.*

The expectation in the assumption is the Fisher Information of $\boldsymbol{\xi}$. Intuitively, it ensures the log-density does not fluctuate too sharply as we vary $\boldsymbol{\xi}$ in high-density regions. Such an assumption holds, e.g., when the components of $\boldsymbol{\xi}$ are Gaussian, Laplace, Student-t, or logistic – essentially any location family with a smooth density. In this sense we consider it to be a mild regularity requirement.

Assumptions 6.3 and 6.4 are more stringent than Assumption 6.2. In our numerical experiments, we study a setting where these two assumptions do not strictly hold and our method still demonstrates strong performance.

6.1. Block Decoupled Problems

To build intuition, we first analyze the special case of block decoupled problems, i.e., when $m = 0$ and $|\mathcal{X}^0| = 0$. For block-decoupled problems, the fundamental challenge is that $T_j(\mathbf{Z})$ may depend on $T_\ell(\mathbf{Z})$ for ℓ in a different block; thus the policy solutions $\mathbf{x}^k(\mathbf{Z}, \boldsymbol{\theta})$ for each block are not independent across k . We need to bound this dependence to prove concentration.

Our approach leverages the chromatic number of a graph built from the policy class to bound the dependence between subproblems. Specifically, define the graph $G(V, E)$ where vertex $V = \{0, \dots, K\}$ corresponds to our disjoint partition $\{S_0, \dots, S_K\}$. An edge $e(k, \bar{k})$ exists between vertices k and $\bar{k}$ if there exists $i \in S_k$, $j \in S_{\bar{k}}$, $l \in \{1, \dots, n\}$, and $\bar{\boldsymbol{\theta}} \in \bar{\Theta}$ such that $L_{il}(\bar{\boldsymbol{\theta}}) \neq 0$ and $L_{jl}(\bar{\boldsymbol{\theta}}) \neq 0$. In words, this means that $T_i(\mathbf{Z}, \bar{\boldsymbol{\theta}})$ and $T_j(\mathbf{Z}, \bar{\boldsymbol{\theta}})$ both depend on ξ_l , despite i and j belonging to different subproblems.

Let $\chi(G)$ be the chromatic number of $G(V, E)$ and fix a vertex coloring for the graph.

Lemma 6.5 (Colors Encode Independent Subproblems) *If $k \neq \bar{k}$ have the same color, then $\boldsymbol{\xi}^{k^\top} \hat{\mathbf{x}}^k(\mathbf{T}(\mathbf{Z}, \boldsymbol{\theta}))$ and $\boldsymbol{\xi}^{\bar{k}^\top} \hat{\mathbf{x}}^{\bar{k}}(\mathbf{T}(\mathbf{Z}, \boldsymbol{\theta}))$ must be independent.*

For simplicity of exposition, we present both the bound based on the chromatic number and a weaker using Brook's Theorem to upperbound this chromatic number.

Lemma 6.6 (Solution Stability of Block-Decoupled Problems) *Consider a decoupled instance of Problem (14) ($m = 0$, and $|\mathcal{X}^0| = 0$). Suppose Assumptions 2.3, 3.2 and 6.2 to 6.4 hold and $0 < h < \min(S_{\max}^{-1}, \sigma_{\min}/2)$. Then, there exists a constant C (depending on $\nu_{\min}, \nu_{\max}, \sigma_{\min}, \kappa, C_\xi$) such that, for any $0 < \epsilon < \frac{1}{2}$, with probability at least $1 - \epsilon$,*

$$\begin{aligned} \sup_{\boldsymbol{\theta} \in \Theta} \text{SS}(\boldsymbol{\xi}, h, \boldsymbol{\theta}) &\leq CS_{\max} nh \sqrt{\log\left(\frac{1}{h}\right)} + C\kappa S_{\max} \sqrt{\chi(G)n \log(\mathcal{X}_{\max}) \log(\chi(G)) \log(1/\epsilon)} \\ &\leq CS_{\max} nh \sqrt{\log\left(\frac{1}{h}\right)} + CS_{\max} \sqrt{n S_{\max} T_{\max} \log \mathcal{X}_{\max}} \cdot \log\left(\frac{S_{\max} T_{\max}}{\epsilon}\right) \end{aligned}$$

where $T_{\max} = \max_j \sum_{i=1}^n \max_{\bar{\theta} \in \bar{\Theta}} \mathbb{I}\{\mathbf{L}_{ij}(\bar{\theta}) \neq 0\}$, and $\mathcal{X}_{\max} = \max_{\mathbf{z} \in \mathbb{R}^n} |\mathcal{X}_{\Theta}(\mathbf{z})| = \max_{\mathbf{z} \in \mathbb{R}^n} |\{\mathbf{x}(\mathbf{Z}, (0, \bar{\theta})) : \bar{\theta} \in \bar{\Theta}\}|$.

The first term above bounds the expected solution stability $\mathbb{E}[\text{SS}(\boldsymbol{\xi}, h, \boldsymbol{\theta})]$. Even though the plug-in policies themselves can “jump” from vertex to vertex as $\mathbf{Z}$ varies, Assumption 6.4 ensures enough smoothness in the density that in expectation, solution stability still scales roughly with h . The second term measures stochastic fluctuations and depends on two key constants, $T_{\max}$ and $S_{\max}$. In words, $T_{\max}$ upper bounds the number of non-zero elements in the j^{th} column of $\mathbf{L}(\bar{\theta})$ for any $\bar{\theta}$ and j , i.e., it bounds how many components are affected by a change in ξ_j . Since there are at most $S_{\max}$ components in a subproblem, the product $T_{\max}S_{\max}$ essentially bounds the degree of dependence between subproblems (although the $\chi(G)$ provides a tighter bound). On the other hand, since each subproblem is a linear optimization problem, by Assumption 6.2 the plug-in policy must return a vertex. Thus, the constant $\mathcal{X}_{\max}$ bounds the number of plug-in policies for any realization $\mathbf{Z}$ of the data. It is necessarily finite (since each $\mathcal{X}^k(\mathbf{x}^0)$ is a bounded polyhedron), and prior works have shown how to bound its size in terms of the complexity of the plug-in function class. (See, e.g., El Balghiti et al., 2023; Hu, Kallus, and Mao, 2022; Gupta, Huang, and Rusmevichientong, 2022a).

That said, in denoising settings, we can often analyze $T_{\max}$ and $\mathcal{X}_{\max}$ more directly. Loosely, when $\mathbf{L}(\bar{\theta})$ is sparse, e.g. when denoising is based on “local” neighborhoods, $T_{\max} = O(1)$. Furthermore, if $|\bar{\Theta}|$ is small, we expect $\mathcal{X}_{\max}$ is also small. We formalize these intuitions in Section F.1 by revisiting our examples from Section 2.3.

We can now combine Theorem 4.2 and Lemma 6.6 to obtain a bound on the estimation error of the one-shot VGC. Let $L_{\max} = \sup_{\bar{\theta} \in \bar{\Theta}} \sup_{i,j} |\mathbf{L}_{ij}(\bar{\theta})|$.

Theorem 6.7 (Estimation Error for Block Decoupled Problems) *Suppose Assumptions 2.3, 2.5, 3.2 and 6.2 to 6.4 hold. Consider Problem (14) with $m = 0$, and $|\mathcal{X}^0| = 0$. Finally assume $h < \min(S_{\max}^{-1}, \sigma_{\min}/2, C_T)$. Then, there exists a constant C (depending on $\nu_{\min}, \nu_{\max}, \sigma_{\min}, \|\boldsymbol{\mu}\|_{\infty}, L_{\max}$) such that with probability at least $1 - \epsilon$,*

$$\begin{aligned} & \sup_{\boldsymbol{\theta} \in \bar{\Theta}} |\mathbf{Z}^{\top} \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta}, h) - \boldsymbol{\mu}^{\top} \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta})| \\ & \leq \underbrace{C \left(S_{\max} n^{1/2} h + \frac{T_{\max}}{h} \right)}_{\text{Step-Size Trade-Off}} \sqrt{n \log n \cdot \log \left(\frac{1}{h} \right) \cdot \log |\Theta_0(n^{-1/2})| \cdot \log \left(\frac{2}{\epsilon} \right)} \end{aligned}$$

$$+ CS_{\max}^{3/2} \sqrt{nT_{\max} \log \mathcal{X}_{\max}} \log \left(\frac{S_{\max} T_{\max}}{\epsilon} \right) + \underbrace{\frac{1}{h} \sup_{\bar{\theta} \in \bar{\Theta}} \mathbb{W}_1(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h\xi, \mathbf{T}(\mathbf{Z}, \bar{\theta}) + \delta_h)}_{\text{Replication Error}}.$$

The same terms from earlier results — $S_{\max} T_{\max}$ as a measure of dependence between subproblems, $\sqrt{\log \mathcal{X}_{\max}}$ to measure complexity of the policy class, and Replication Error — appear with the same interpretations.

Neglecting the replication error, we can approximately optimize the choice of h minimize the rate.

Corollary 6.8 (Optimized Step-Size) *Suppose $\mathbf{Z}$ is multivariate Gaussian and $\mathbf{L}(\bar{\theta})^\top \Sigma + \Sigma \mathbf{L}(\bar{\theta})$ is positive semidefinite for all $\bar{\theta} \in \bar{\Theta}$. Then, under the assumptions of Theorem 6.7, if $h = O\left(n^{-1/4} T_{\max}^{1/2} S_{\max}^{-1/2}\right)$ there exists a constant C (depending on $\nu_{\min}, \nu_{\max}, \sigma_{\min}, \kappa, \|\boldsymbol{\mu}\|_{\infty}, L_{\max}$) such that with probability at least $1 - \epsilon$,*

$$\begin{aligned} & \sup_{\boldsymbol{\theta} \in \Theta} |\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta}, h) - \boldsymbol{\mu}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta})| \\ & \leq CS_{\max}^{3/4} n^{3/4} \log n \sqrt{T_{\max} \log \mathcal{X}_{\max} \log |\Theta_0(n^{-1/2})|} \cdot \log \left(\frac{S_{\max} T_{\max}}{\epsilon} \right). \end{aligned}$$

The corollary helps clarify in what settings our proposal should perform well. Recall that in typical settings, the full-information solution to Problem (1) is $O(n)$, so if each term above is strict smaller, the relative error of our procedure vanishes. Moreover, when denoising depends only on “local” neighborhoods, we expect $T_{\max} = O(1)$, and if Problem (14) consists of “small” subproblems, $S_{\max} = O(1)$. Finally, if, as in the examples from Section 2.3, $\Theta_0(n^{-1/2})$ grows at most polynomially in n , the above bound is $\tilde{O}(n^{3/4})$. Thus, if there is zero replication error, the relative error of our estimator vanishes. If there is non-zero replication error, then since $h = O(n^{-1/4})$, if data is sufficiently close to Gaussian that the replication error is strictly less than $\tilde{O}(n^{3/4})$ — then the relative error of the procedure will still vanish. It is precisely these settings — local denoisers, small subproblems, simple policy classes, and nearly Gaussian data — where we expect the method to perform well.

6.2. A Modified VGC for Weakly-Coupled Problems

Our strategy for the weakly-coupled Problem (14) will be to i) “fix” values of the coupling variables Eq. (14c) and Lagrangian duals for the coupling constraints Eq. (14b) ii) apply our previous results to the resulting decoupled problem iii) take a union bound over all possible values of the “fixed” quantities. To execute this strategy we define a lifted policy class indexed by the “fixed” quantities and then define a modified VGC for this lifted class.

Lifted Affine Policy Class

We first introduce dual notation. Recall the general affine plug-in for the weakly-coupled optimization problem Problem (14) solves

$$\mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) \in \arg \min_{\mathbf{x}} \sum_{j=1}^n T_j(\mathbf{Z}, \bar{\boldsymbol{\theta}}) x_j \quad (16a)$$

$$\text{s.t.} \quad \sum_{k=0}^K \sum_{j \in S_k} \mathbf{A}_j^0(\mathbf{x}^0) x_j \leq \mathbf{b}^0(\mathbf{x}^0) \quad (16b)$$

$$\mathbf{x}^0 \in \mathcal{X}^0 \quad (16c)$$

$$\mathbf{x}^k \in \mathcal{X}^k(\mathbf{x}^0), \quad \forall k = 1, \dots, K. \quad (16d)$$

where $\mathbf{A}_j^0(\mathbf{x}^0) \in \mathbb{R}^m$ for $j = 1, \dots, n$ and $\mathbf{b}^0(\mathbf{x}^0) \in \mathbb{R}^m$ are vectors that may depend on $\mathbf{x}^0$.

Fix some $\mathbf{x}^0 \in \mathcal{X}^0$ and consider relaxing Eq. (16b) with dual variables $\boldsymbol{\lambda} \in \mathbb{R}_+^m$ to obtain the block decoupled optimization problem

$$\min_{\mathbf{x}} \sum_{k=1}^K \langle \mathbf{T}^k(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \mathbf{A}_{S_k}^0(\mathbf{x}^0)^\top \boldsymbol{\lambda}, \mathbf{x}^k \rangle, \quad \text{s.t.} \quad \mathbf{x}^k \in \mathcal{X}^k(\mathbf{x}^0), \quad \forall k = 1, \dots, K \quad (17)$$

where $\mathbf{A}_{S_k}^0(\mathbf{x}^0)$ is formed by taking columns $j \in S_k$. Let $\tilde{\mathbf{x}}^k(\mathbf{Z}, \boldsymbol{\lambda}, \mathbf{x}^0, \bar{\boldsymbol{\theta}})$ denote the solution of the k^{th} subproblem and $\tilde{\mathbf{x}}(\mathbf{Z}, \boldsymbol{\lambda}, \mathbf{x}^0, \bar{\boldsymbol{\theta}}) = (\mathbf{x}^0, \tilde{\mathbf{x}}^1(\mathbf{Z}, \boldsymbol{\lambda}, \mathbf{x}^0, \bar{\boldsymbol{\theta}})^\top, \dots, \tilde{\mathbf{x}}^K(\mathbf{Z}, \boldsymbol{\lambda}, \mathbf{x}^0, \bar{\boldsymbol{\theta}})^\top)^\top$. Although the notation is onerous, the intuition is to view Eq. (17) as defining the plug-in policy $\tilde{\mathbf{x}}(\mathbf{Z}, \boldsymbol{\lambda}, \mathbf{x}^0, \bar{\boldsymbol{\theta}})$ for the target optimization problem

$$\min_{\mathbf{x}} \sum_{k=1}^K \langle \boldsymbol{\mu}^k, \mathbf{x}^k \rangle, \quad \text{s.t.} \quad \mathbf{x}^k \in \mathcal{X}^k(\mathbf{x}^0), \quad \forall k = 1, \dots, K,$$

where $(\mathbf{x}^0, \boldsymbol{\lambda}, \bar{\boldsymbol{\theta}})$ are the parameters of the plug-in function. Our strategy will be to use Theorem 6.7 to debias the in-sample performance of this policy, and then relate it back to the original plug-in policy for our weakly coupled problem. To that end, define the lifted affine policy class

$$\tilde{\mathcal{X}}(\mathbf{Z}) = \{ \tilde{\mathbf{x}}(\mathbf{Z}, \boldsymbol{\lambda}, \mathbf{x}^0, \bar{\boldsymbol{\theta}}) : \boldsymbol{\lambda} \in \mathbb{R}_+^m, \mathbf{x}^0 \in \mathcal{X}^0, \bar{\boldsymbol{\theta}} \in \bar{\Theta} \}$$

and the corresponding plug-in function

$$\tilde{V}(\mathbf{t}, \mathbf{x}^0) = \min \{ \mathbf{t}^\top \mathbf{x} : \mathbf{x}^k \in \mathcal{X}^k(\mathbf{x}^0), \quad \forall k = 1, \dots, K. \}$$

Our one-shot VGC applied to this lifted, block-decoupled problem is

$$\tilde{D}(\mathbf{Z}, \boldsymbol{\lambda}, \mathbf{x}^0, \bar{\boldsymbol{\theta}}, h) \equiv \frac{1}{h} \mathbb{E} \left[\tilde{V}(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + (\mathbf{A}^0)^\top \boldsymbol{\lambda} + \boldsymbol{\delta}_h, \mathbf{x}^0) - \tilde{V}(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + (\mathbf{A}^0)^\top \boldsymbol{\lambda}, \mathbf{x}^0) \middle| \mathbf{Z} \right].$$

Of course Eq. (17) is not the same as Problem (16). However, one might intuit that if we set $(\boldsymbol{\lambda}, \mathbf{x}^0)$ to their optimal values, Eq. (17) might closely approximate Problem (16) and the corresponding VGC might approximately debias Problem (16). To that end, for a fixed $\mathbf{x}^0$, we define the optimal dual variables

$$\boldsymbol{\lambda}(\mathbf{Z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0) \in \arg \max_{\boldsymbol{\lambda} \geq \mathbf{0}} - \langle \mathbf{b}^0(\mathbf{x}^0), \boldsymbol{\lambda} \rangle + \sum_{k=0}^K \langle \mathbf{T}^k(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \mathbf{A}_{S_k}^0(\mathbf{x}^0)^\top \boldsymbol{\lambda}, \tilde{\mathbf{x}}^k(\mathbf{Z}, \boldsymbol{\lambda}, \mathbf{x}^0, \bar{\boldsymbol{\theta}}) \rangle \quad (18)$$

and optimal coupling variables $\mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta})$ as the portion of the optimal solution to Problem (16) corresponding to block S_0 . Our modified VGC correction is then

$$D^{WC}(\mathbf{Z}, \boldsymbol{\theta}, h) \equiv \tilde{D}(\mathbf{Z}, \boldsymbol{\lambda}(\mathbf{Z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta})), \mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta}), \bar{\boldsymbol{\theta}}, h). \quad (19)$$

Remark 6.9 (Computing the Modified VGC) Computing the modified VGC is not much harder than solving Problem (16) since one can extract $\mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta})$ and then solve a linear optimization problem to obtain $\boldsymbol{\lambda}(\mathbf{Z}, \boldsymbol{\theta}, \mathbf{x}^0)$ and $\tilde{\mathbf{x}}(\mathbf{Z}, \boldsymbol{\lambda}(\mathbf{Z}, \boldsymbol{\theta}, \mathbf{x}^0), \mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta}), \boldsymbol{\theta})$. In practice, however, we adopt a simpler approach. Let $V(\mathbf{t})$ denote the optimal objective value of Problem (16) when the cost coefficients are $\mathbf{t} \in \mathbb{R}^n$. We approximate $D^{WC}(\mathbf{Z}, \boldsymbol{\theta}, h) \approx \frac{1}{h} \mathbb{E} [V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \boldsymbol{\delta}_h, 0) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), 0) \middle| \mathbf{Z}]$. One might intuit that these quantities are asymptotically identical. We treat Eq. (19) in our analysis for simplicity.

6.3. Estimation Error of Weakly-Coupled One-Shot VGC

The key insight to leveraging Theorem 6.7 to analyzing D^{WC} is relating solutions to Problem (16) to our lifted affine policy class. In particular, we will show that

$$\mathbf{x}^k(\mathbf{Z}, \boldsymbol{\theta}) = \tilde{\mathbf{x}}^k(\mathbf{Z}, \boldsymbol{\lambda}(\mathbf{Z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta})), \mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta}), \bar{\boldsymbol{\theta}})$$

whenever $\tilde{\mathbf{x}}^k(\mathbf{Z}, \boldsymbol{\lambda}(\mathbf{Z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0), \mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta}), \bar{\boldsymbol{\theta}})$ is the unique solution to its defining optimization problem. We next introduce a technical assumption that allows us to bound the number of subproblems with multiple solutions.

Let $\mathcal{D}^k(\mathbf{x}^0)$ be the set of unit vectors $\mathbf{d}$ such that $\min_{\mathbf{x} \in \mathcal{X}^k(\mathbf{x}^0)} \mathbf{d}^\top \mathbf{x}$ has multiple optima and define the set of vectors

$$\mathcal{F}(\mathbf{Z}, \mathbf{x}^0, \bar{\boldsymbol{\theta}}) \equiv \bigcup_{k=1}^K \left\{ \begin{pmatrix} T_j(\mathbf{Z}, \bar{\boldsymbol{\theta}}) \\ \mathbf{A}_j^0 \\ -d_j \end{pmatrix} : j \in S_k, \mathbf{d} \in \mathcal{D}^k(\mathbf{x}^0) \right\} \subseteq \mathbb{R}^{m+2}.$$

We assume these vectors are in general position.⁵

Assumption 6.10 (Induced Cost Vectors in General Position) *We have*

$$\mathbb{P}(\mathcal{F}(\mathbf{Z}, \mathbf{x}^0, \bar{\boldsymbol{\theta}}) \text{ are in general position for all } \bar{\boldsymbol{\theta}} \in \bar{\Theta} \text{ and } \mathbf{x}^0 \in \mathcal{X}^0) = 1.$$

In Lemma G.1, we use this assumption to prove that the number of subproblems of our lifted affine policy with multiple optima is at most m .

Additionally, we require assumptions on the boundedness of the optimal dual values $\lambda^0(\mathbf{Z}, \boldsymbol{\theta}, \mathbf{x}^0)$ to cover the policy class. We adopt a simple assumption similar to Gupta, Huang, and Rusmevichientong (2022a) and Gupta and Rusmevichientong (2021).

Assumption 6.11 ($\bar{s}$ -Strict Feasibility of Coupling Constraints) *For each $\mathbf{x}^0 \in \mathcal{X}^0$, there exists an $\bar{s} > 0$ and $\bar{\mathbf{x}} \in \mathcal{X}^{WC}$ such that $\sum_{j=1}^n \mathbf{A}_j^0(\mathbf{x}^0) \bar{x}_j + \bar{s} \mathbf{e} \leq \mathbf{b}^0(\mathbf{x}^0)$.*

In words, $\bar{s}_0$ -strict feasibility generalizes feasibility. Instead of requiring a point where all inequalities are satisfied, we require one where all inequalities are satisfied with slack at least s_0 . Lemma G.2 bounds $\|\lambda(\mathbf{Z}, \boldsymbol{\theta})\|_1$ under this assumption.

We now can present our uniform bound on the estimation error of the weakly-coupled one-shot VGC. Let $\|\mathbf{A}^0\|_\infty$ be the entry-wise infinity norm for matrices. For simplicity, we've chosen h to optimize the rate of the bound (excluding the replication error).

Theorem 6.12 (Estimation Error of the Weakly-Coupled VGC) *Suppose Assumptions 2.3, 2.5, 3.2, 6.2 to 6.4, 6.10 and 6.11 hold. Choose $h = O\left(n^{-1/4} T_{\max}^{1/2} S_{\max}^{-1/2}\right)$ and δ_h as in Proposition 4.3. Consider Problem (14). For $h < \min(S^{-1}_{\max}, \sigma_{\min}/2, C_T$,*

⁵ Recall, any finite set of vectors not in general position can be made to be in general position by randomly perturbing them by an arbitrarily small amount.

there exists a constant C (depending on $\nu_{\min}$, $\nu_{\max}$, $\sigma_{\min}$, $\|\boldsymbol{\mu}\|_{\infty}$, $L_{\max}$, $\log \bar{s}^{-1}$, $\|\mathbf{A}^0\|_{\infty}$, κ) such that the following holds with probability at least $1 - \epsilon - \exp(-n)$,

$$\begin{aligned}
& \sup_{\boldsymbol{\theta} \in \Theta} \underbrace{\left| \mathbf{Z}^{\top} \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D^{WC}(\mathbf{Z}, \boldsymbol{\theta}, h) - \boldsymbol{\mu}^{\top} \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) \right|}_{\text{Estimation Error}} \\
& \leq \underbrace{C_4 S_{\max}^{3/4} n^{3/4} \log^{3/2} n \sqrt{m T_{\max} \log \mathcal{X}_{\max} \log |\Theta_0| (n^{-1/2}/2)} \cdot \log \left(\frac{S_{\max} T_{\max} |\mathcal{X}^0|}{\epsilon} \right)}_{\text{Block Decoupled Bound}}, \\
& + \underbrace{\frac{n^{1/4} S_{\max}}{T_{\max}} \sup_{\boldsymbol{\theta} \in \Theta} \mathbb{W}_1(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h \boldsymbol{\xi}, \mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \boldsymbol{\delta}_h)}_{\text{Replication Error}} + \underbrace{C m S_{\max} \sqrt{\log n \cdot \log \left(\frac{1}{\epsilon} \right)}}_{\text{Lifted Policy Estimation Error}}.
\end{aligned}$$

Intuitively, we are applying Theorem 6.7 to the lifted problem which is decoupled. The only additional term that arises is the “lifted policy approximation” error, which bounds the difference between the in-sample optimism of the lifted policy and the original policy. In typical settings where $m = O(1)$ (c.f. Example 2.1), this term is $\tilde{O}(1)$.

When data are Gaussian, Gupta, Huang, and Rusmevichientong (2022a) bounds the relative error of the multi-shot VGC for problems that are “weakly-coupled by constraints” that essentially scales like $\tilde{O}(C_{\text{PI}} n^{-1/4})$, where C_{PI} is a problem dependent constant measuring the complexity of the policy class. For comparison, when applying the one-shot VGC, Theorem 6.12 implies the relative error scales like $\tilde{O}(\bar{C}_{\text{PI}} n^{-1/4})$, where $\bar{C}_{\text{PI}}$ is a different constant (since it treats a different class of policies). Thus, both bounds agree, although our applies in a more general setting.

As an aside, our weakly-coupled VGC applies to the lifted decoupled problem, it depends on the complexity of the lifted policy class $\tilde{\mathcal{X}}_{\Theta}(\mathbf{Z})$, not just $\mathcal{X}_{\Theta}(\mathbf{Z})$. Since the lifted class simply shifts the plug-in by an affine amount, it can often be bounded using the same techniques as in Section F.1

7. Using Lepski’s Method to Select h

Corollaries 5.3 and 6.8 and Theorem 6.12 provide some guidance in choosing the step-size h to maximize the rate of convergence, but require a careful analysis of solution stability. We next describe a more computational approach to selecting h based on Lepski’s Method.

Lepski’s method (Lepskii, 1992) is a classical technique in adaptive estimation. Our treatment follows Tucker and Lee (2021). Fix some $\boldsymbol{\theta} \in \Theta$ and fix a finite grid of candidate step-sizes $0 < h_1 < h_2 \dots < h_M$. Let $\text{Bias}_j(\boldsymbol{\theta}) \equiv \left| \mathbb{E} [\boldsymbol{\xi}^{\top} \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta}, h_j)] \right|$ and

$\text{CNF}_j(\boldsymbol{\theta}) \equiv 2C_T(\bar{\boldsymbol{\theta}})_{h_j}^{\frac{\kappa}{h_j}} \sqrt{n \log\left(\frac{2}{\epsilon}\right)}$, from Theorem 3.3. Then, the union bound implies that with probability at least $1 - M\epsilon$,

$$\begin{aligned} & |Z^\top x(Z, \boldsymbol{\theta}) - D(Z, \boldsymbol{\theta}, h_j) - \mu^\top x(Z, \boldsymbol{\theta})| \\ &= |\xi^\top x(Z, \boldsymbol{\theta}) - D(Z, \boldsymbol{\theta}, h_j)| \\ &\leq \underbrace{|\xi^\top x(Z, \boldsymbol{\theta}) - \mathbb{E}[\xi^\top x(Z, \boldsymbol{\theta})]|}_{\text{Does not depend on } h_j} + \text{Bias}_j(\boldsymbol{\theta}) + \text{CNF}_j(\boldsymbol{\theta}) \end{aligned}$$

for all j simultaneously. $\text{Bias}_j(\boldsymbol{\theta})$ includes both finite difference and replication error. When replication error is small, $\text{Bias}_j(\boldsymbol{\theta})$ is driven by finite difference error which is increasing in h . Hence, $\max_{i \leq j} \text{Bias}_i(\boldsymbol{\theta})$ is likely not a bad upper-bound on this term. $\text{CNF}_j(\boldsymbol{\theta})$ is decreasing in h by inspection.

Since the first term does not depend on h_j , a reasonable choice of h might minimize this bound, i.e., choose h_{j^*} where $j^* \in \arg \min_{j=1, \dots, M} \{\max_{i \leq j} \text{Bias}_i(\boldsymbol{\theta}) + \text{CNF}_j(\boldsymbol{\theta})\}$. Of course, computing h_{j^*} requires analyzing $\text{Bias}_i(\boldsymbol{\theta})$ which is in general difficult.

Lepski’s method instead prescribes the data-driven choice

$$\hat{j} = \max \left\{ j : |D(Z, \boldsymbol{\theta}, h_j) - D(Z, \boldsymbol{\theta}, h_i)| \leq \text{CNF}(j) + (\sqrt{6} - 1)\text{CNF}(i), \forall i \leq j \right\}. \quad (20)$$

Notice, identifying $\hat{j}$ does not require computing $\text{Bias}_j(\boldsymbol{\theta})$.

Tucker and Lee, 2021 prove

$$|D(Z, \boldsymbol{\theta}, h_{\hat{j}}) - \mathbb{E}[\xi^\top x(Z, \boldsymbol{\theta})]| \leq (\sqrt{6} + 3) \min_j \left\{ \max_{i \leq j} \text{Bias}_i(\boldsymbol{\theta}) + \text{CNF}_j(\boldsymbol{\theta}) \right\}.$$

In other words, $h_{\hat{j}}$ is no more than a constant times worse than h_{j^*} . In this sense, Eq. (20) provides a theoretically justified choice of h for a given $\boldsymbol{\theta}$ that is readily computed.

When policy learning over a finite class, we can choose h by Lepski’s method and then compare the performance of each policy and choose the best. Since our earlier proven bounds are uniform in $\boldsymbol{\theta}$, if M is at most polynomial in n , a straightforward union bound yields an error bound. For infinite policy classes, Lepski’s method requires discretization.

We apply Lepski’s method in our numerical experiments and find the Lepski choice of h is often comparable to the “rate-optimized” h in both synthetic (Fig. EC.2) and real-world settings (Fig. 3). We discuss some implementation details for tuning performance in Section H.2.

8. Comparing the One-Shot and Multi-Shot VGC

In this section, we provide some further theoretical and empirical comparisons to the multi-shot VGC of Gupta, Huang, and Rusmevichientong (2022a). To facilitate a transparent “apples-to-apples” comparison, we restrict attention to affine plug-in policies and assume all diagonal elements of $\mathbf{L}$ are non-zero and the precisions $\nu_j = 1$ for all j .

The key intuition behind the multishot VGC is that when $\mathbf{L}$ is diagonal, by Danskin’s Theorem, $\left. \frac{\partial}{\partial \lambda} V(\mathbf{T}(\mathbf{Z} + \lambda \xi_j \mathbf{e}_j)) \right|_{\lambda=0} = L_{jj} \xi_j \hat{x}_j(\mathbf{Z})$. Dividing both sides by L_{jj} and summing shows

$$\sum_{j=1}^n \frac{1}{L_{jj}} \frac{\partial}{\partial \lambda} V(\mathbf{T}(\mathbf{Z} + \lambda \xi_j \mathbf{e}_j)) \Big|_{\lambda=0} = \boldsymbol{\xi}^\top \mathbf{x}(\mathbf{Z}). \quad (21)$$

The multishot VGC is an approximation of the left side using finite differences and $\delta_j \sim \mathcal{N}(0, \bar{h}^2 + 2\bar{h})$ to approximate ξ_j . Thus, it estimates the optimism $\boldsymbol{\xi}^\top \mathbf{x}(\mathbf{Z})$ by

$$D^{MS}(\mathbf{Z}) = \sum_{j=1}^n \frac{1}{L_{jj} \bar{h}} \mathbb{E}[V(\mathbf{T}(\mathbf{Z} + \delta_j \mathbf{e}_j)) - V(\mathbf{T}(\mathbf{Z})) \mid \mathbf{Z}]. \quad (22)$$

Gupta, Huang, and Rusmevichientong (2022a) proves that, under some assumptions, $D^{MS}(\mathbf{Z})$ equals the left side of Eq. (21) plus an $o_p(n)$ remainder. Since the optimism is typically $O_p(n)$, this result implies the relative error is vanishing for separable affine policies. It also suggests that the left side of Eq. (21) governs the asymptotic behavior of the multi-shot VGC.

8.1. Bias for General Affine Policies

Eq. (21) already suggests the multi-shot VGC can be asymptotically biased for general affine policies. Evaluating the left side of Eq. (21) for a non-diagonal matrix $\mathbf{L}$ with Danskin’s Theorem yields

$$\sum_{j=1}^n \frac{1}{L_{jj}} \xi_j \mathbf{L}_j^\top \mathbf{x}(\mathbf{Z}) = \boldsymbol{\xi}^\top \text{diag}(\mathbf{L})^{-1} \mathbf{L}^\top \mathbf{x}(\mathbf{Z}).$$

The error in estimating the optimism is thus $\boldsymbol{\xi}^\top (\text{diag}(\mathbf{L})^{-1} \mathbf{L}^\top - \mathbf{I}) \mathbf{x}(\mathbf{Z})$. When $\mathbf{L}$ is not diagonal (i.e. the policy is not separable), the middle matrix does not vanish. Hence, whenever there exists components j such that $x_j(\mathbf{Z})$ depends on component k and ξ_k appears in the sum $\boldsymbol{\xi}^\top \text{diag}(\mathbf{L})^{-1} \mathbf{L}^\top \mathbf{e}_j$, we expect the multishot VGC to be biased. As we next show empirically, this bias can be of order n , causing the relative error to be non-vanishing.

8.2. An Example where Multi-shot VGC is Biased

We illustrate our previous analysis with a block-decoupled optimization problem and Gaussian data. Multi-shot VGC has provably good performance in this setting for separable, affine policies. We instead consider a *mildly* non-separable plug-in $T_j(\mathbf{Z}) = 0.5Z_j + 0.25(Z_{j-1} + Z_{j+1})$, which is 1-nearest-neighbor smoothing.

Our full information optimization problem is

$$\max_{\mathbf{x} \in \mathcal{X}} \sum_{k=1}^K \sum_{j=1}^B \mu_{kj} x_{kj}, \quad \text{where } \mathcal{X} = \left\{ \mathbf{x} : \sum_{j=1}^B x_{kj} \leq 1 \forall k; x_{kj} \in \{0, 1\} \forall k, j \right\}$$

The parameter B controls how many components each $x_{kj}(\cdot)$ depends upon.

We observe $W_j = j$, and $Z_{kj} \sim \mathcal{N}(\mu_{kj}, 1/\nu_{kj})$ with

$$(\mu_{kj}, \nu_{kj}) = \begin{cases} (0, 2) & \text{w.p. } \frac{1}{2}, \\ (0.5, 10) & \text{w.p. } \frac{1}{4}, \\ (1, 6) & \text{w.p. } \frac{1}{4}. \end{cases}$$

The “ground-truth” values of $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ are generated once and fixed throughout the experiment. The data $\mathbf{Z}$ are generated each simulation run.

As B grows, our previous intuition suggests the amount of bias grows, because $x_{kj}(\cdot)$ depends on all ξ_{kl} for all $l = 1, \dots, B$. Figure 1 confirms this intuition. Panel (a) demonstrates the intuition numerically by showing the *relative* size of the bias increases as B grows. However, panel (b) highlights the fundamental difference between one-shot and multi-shot VGC. We see that the multi-shot VGC bias is non-vanishing even as h goes to 0. By contrast, bias (scaled by full information) of the one-shot VGC vanishes, closely corresponding to the predicted rate of $O(Bh)$ from Theorem 6.7.

8.3. What about separable affine policies?

When restricting to separable, affine policies, the theoretical results of Gupta, Huang, and Rusmevichientong (2022a) show the method has a vanishing bias of $o(n)$. However, as we next argue, the multi-shot VGC can have worse variance than our one-shot VGC. To build intuition, we study the separable plug-in $\mathbf{T}(\mathbf{Z}) = \mathbf{Z}$.

Define the random vectors $\boldsymbol{\delta} \sim \mathcal{N}(0, (\bar{h}^2 + 2\bar{h})\mathbf{I})$ and $\bar{\boldsymbol{\delta}} \in \mathbb{R}^n$ by $\bar{\boldsymbol{\delta}} \sim \delta_j \mathbf{e}_j$ with probability $\frac{1}{n}$. Finally, let $h = \bar{h}/n$. Then, the multi-shot VGC Eq. (22) can be rewritten as

$$D^{MS}(\mathbf{Z}) = \frac{1}{h} \mathbb{E} [V(\mathbf{Z} + \bar{\boldsymbol{\delta}}) - V(\mathbf{Z}) \mid \mathbf{Z}],$$

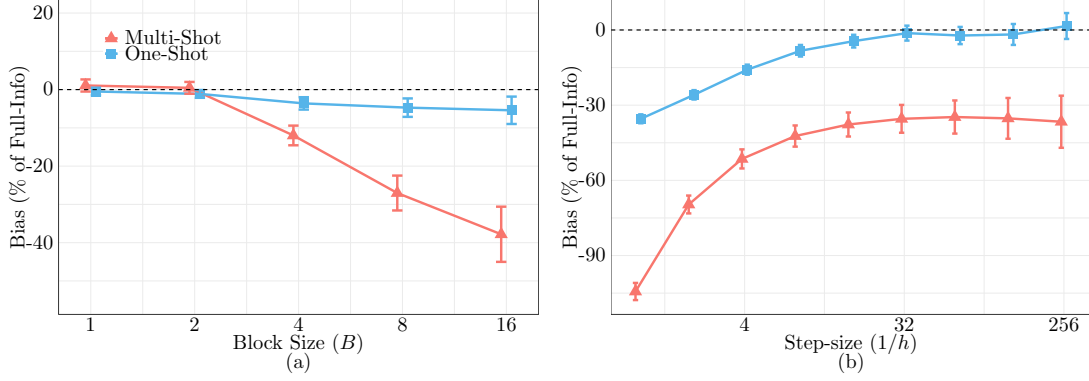


Figure 1 Multi-Shot vs. One-Shot: Non-Separable Policy. Estimates of the bias of the one-shot VGC and multi-shot VGC (Eq. (22)) for one-nearest neighbors, a non-separable plug-in policy. Figure (a) increases the size of the blocks (B) for fixed $n = KB = 160$ and $h = 2^{-5}$. Figure (b) decreases h for a fixed $B = 20$ and $n = 320$. Error bars are 95% confidence intervals computed using 200 trials.

i.e., it of the same form as the one-shot VGC but uses a different random step $\bar{\delta}$. Recall, the one-shot VGC uses the step δ_h defined by $\delta_h \sim \mathcal{N}(\mathbf{0}, (2h + h^2)\mathbf{I})$.

Both our analysis and that in Gupta, Huang, and Rusmevichientong, 2022a suggest h must go to zero fairly slowly as n grows to balance bias and variance. When $h > 1/n$, multi-shot VGC takes a step of size $O_p(nh)$, while the one-shot VGC takes a smaller step of size $O_p(\sqrt{nh})$. In both cases, the multi-shot VGC only steps in coordinate directions, while the one-shot VGC steps in a uniformly random direction.

From a theoretical point of view, the different structure of one-shot and multi-shot VGC give rise to different analyses. Because it restricts to coordinate directions, bounding the bias of the multi-shot VGC is straightforward (Gupta, Huang, and Rusmevichientong, 2022a, Theorem 3.2), but analyzing its stochastic behavior is quite difficult and involves a subtle duality argument and a number of assumptions on the policy and optimization. See Gupta, Huang, and Rusmevichientong (2022a, Theorem 3.5). By contrast, by fully randomizing the step direction, the one-shot VGC admits a very simple analysis of its stochastic behavior Theorems 3.1 and 3.3, but analyzing its bias is now more difficult.

Intuitively, the larger step sizes in the multi-shot VGC produce a higher-variance estimator than the one-shot version, since larger perturbations to the objective coefficients lead to greater variation in the perturbed optimization problem’s optimal solution. This additional variation is harder to quantify theoretically because it depends on the structure of the underlying optimization problem. In the next section, we examine this intuition empirically.

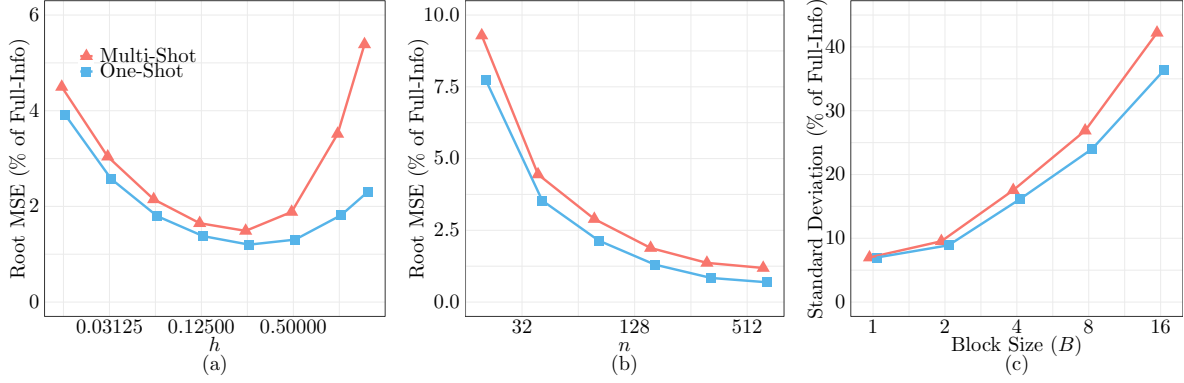


Figure 2 Multi-Shot vs. One-Shot: Separable Policy. The first two panels show one-shot VGC achieves uniformly better RMSE than the multi-shot VGC (estimated over 200 trials). In panel (a) we vary the step size h and fix $n = 160$. For panel (b), we vary n for a fixed $h = 0.5$. Figure (c) plots the estimated standard deviation of the estimators over block sizes for a fixed $n = 160$ and step-size $h = 2^{-5}$. All quantities in the figure are computed using 200 trials.

8.4. Variability of the Multi-Shot Correction

We revisit the synthetic experiment of Section 8.2, using a separable policy $T(\mathbf{Z}) = \mathbf{Z}$ and $B = 4$. The first two panels of Figure 2 show that multi-shot VGC has uniformly worse mean-square error than our one-shot variant across different step-sizes and problem sizes. The third panel shows the variability of the VGC estimators increases with the subproblem size. The bias (not pictured) remains approximately constant as the subproblem size grows, further supporting the earlier “step-size” intuition that the multi-shot VGC estimator has uniformly higher variance.

9. Computational Case Study: Prioritizing Speed Hump Requests

Our case study further develops Example 2.1, i.e., prioritizing speed hump requests to improve traffic safety. We use motor vehicle accident data from the New York Police Department from 2012-2023 (New York, 2023). Our goal is to identify census tracts that have the highest incidence of pedestrian injury due to motor vehicle accidents subject to a budget constraint on the number of tracts that can be serviced in a given year (B).

Notably, our data lack covariate information such as foot and vehicle traffic that are (intuitively) highly predictive of pedestrian injury rates. Hence, obtaining precise estimates is difficult or impossible. While one might argue for collecting and integrating these (and other) useful covariates into the dataset, such integration can be costly, time-consuming, and may conflict with data-privacy regulations. Our approach instead seeks to identify the highest quality decisions possible using only the data at hand.

All code and data for these experiments are available on Github⁶.

Data and Setup. We consider selecting the top 5% of the 2,157 total census tracts in New York city to prioritize speed hump requests, i.e., $B = 107$ and $n = 2,157$.

We let μ_j represent the expected number of crashes per year with pedestrian, cyclist, or motorist injuries in census tract j . We model $\mu_j = \lambda_j p_j$ where λ_j and p_j are the respective crash rate per year and probability a pedestrian is injured given an accident for census tract j . As ground truth, we take p_j to be the observed pedestrian injury probability for each census tract across the *entire* time period covered by the dataset (2012-2023). These ground-truth values range from 0 to 0.1576 across census tracts. We treat these values as *unknown*. Similarly, we take the ground-truth λ_j to be the observed average number of crashes per year over the time frame. These values range from 1.4 to 448 across census tracts with an average of 52 crashes per year. However, we treat these values as *known* and fixed.

We set N_j to be the number of accidents observed in 2018. To generate our observations, we let $I_j \sim \text{Binomial}(\gamma N_j, p_j)$ for $\gamma \in \{1, 2, 3, 4\}$. Intuitively, the I_j represents the number of pedestrian injuries based on γ years of historical data. The predicted number of crashes with injuries is then $Z_j = \lambda_j \hat{p}_j$ where $\hat{p}_j = I_j / (\gamma N_j)$. This setup allows us to generate multiple simulation paths Z_j and also vary the amount of historical data available in forming these estimates.

Policy Class. We consider a non-regularized kernel smoother plug-in policy class (Example 2.7):

$$T_j(\hat{\mathbf{p}}, \mathbf{W}, \theta) = \lambda_j \sum_{i=1}^n \hat{p}_i \frac{\mathbb{I}\{\|\mathbf{W}_j - \mathbf{W}_i\| \leq \theta\}}{\sum_{l=1}^n \mathbb{I}\{\|\mathbf{W}_j - \mathbf{W}_l\| \leq \theta\}},$$

where we let the bandwidth $\theta \geq 0$ vary. Here $\mathbf{W}_j$ are the longitude and latitude coordinates of the centroid of census tract j . Our goal is to select θ that maximizes the out-of-sample performance $\sum_{j=1}^n \lambda_j p_j x_j(\hat{\mathbf{p}}, \mathbf{W}, \theta)$.

Experiments. We compare decision-aware methods to decision-blind methods for selecting θ as we vary the amount of crash data observed (γ).

For decision-aware methods, we compare the multi-shot VGC implementation discussed in Section 8 and the Stein correction from Gupta and Rusmevichientong (2021). All three

⁶ <https://github.com/mh3166/Decision-Aware-Denoising>

approaches estimate the out-of-sample performance of different choices of θ . We set $h = n^{-1/4}$ for the one-shot VGC and simulate δ_h as multivariate Gaussians similar to Proposition 4.3. The Stein correction and multi-shot VGC also require a choice of h which we set as $h = n^{-1/6}$ as recommended. For full implementation details, see Appendix H.1.

For our decision-blind policy, we use a predict-then-optimize policy that chooses the θ that minimizes (oracle) mean squared error (MSE) of the predicted pedestrians injured. Using oracle MSE favors the decision-blind by avoiding any noise from misestimating MSE. Formally, the MSE policy selects

$$\theta^{MSE} \in \arg \min_{\theta \geq 0} \frac{1}{n} \sum_{j=1}^n (T_j(\hat{\mathbf{p}}, \mathbf{W}, \theta) - \lambda_j p_j)^2$$

Finally, we also benchmark against the oracle θ that optimizes the true out-of-sample performance for each realization of our data $\hat{\mathbf{p}}$. This oracle represents an upperbound on achievable performance.

For ease of comparison, we present performance relative to the full information optimal performance $\max_{\mathbf{x} \in \mathcal{X}} \sum_{j=1}^n \lambda_j p_j x_j$.

9.1. Case Study Results

Figure 3 plots the estimates of the expected out-of-sample performance of the all the different methods as we increase the amount of data. We highlight several key observations.

First, we show that decision-aware and decision-blind approaches can both achieve performance comparable to the oracle policy as the data increases. However, our proposed decision-aware one-shot VGC approach converges faster, showing that it can also perform well in small-data settings. We also stress that we are using an *oracle* version of MSE, so the MSE performance here optimisitic.

Second, we see that among decision-aware approaches, the one-shot VGC also converges faster to oracle performance. We hypothesize the one-shot VGC converges faster than the Stein correction due to leveraging the optimization structure more directly. The same observation was also made in Gupta, Huang, and Rusmevichientong (2022a) with the multi-shot VGC. For the multi-shot VGC, we see the bias and variance issues discussed in Section 8 directly translate into learning worse policies.

Lifted Optimization Problems. Inspired by Theorem 6.12, we next consider a modifications of the one-shot VGC and multi-shot VGC in which we apply these corrections

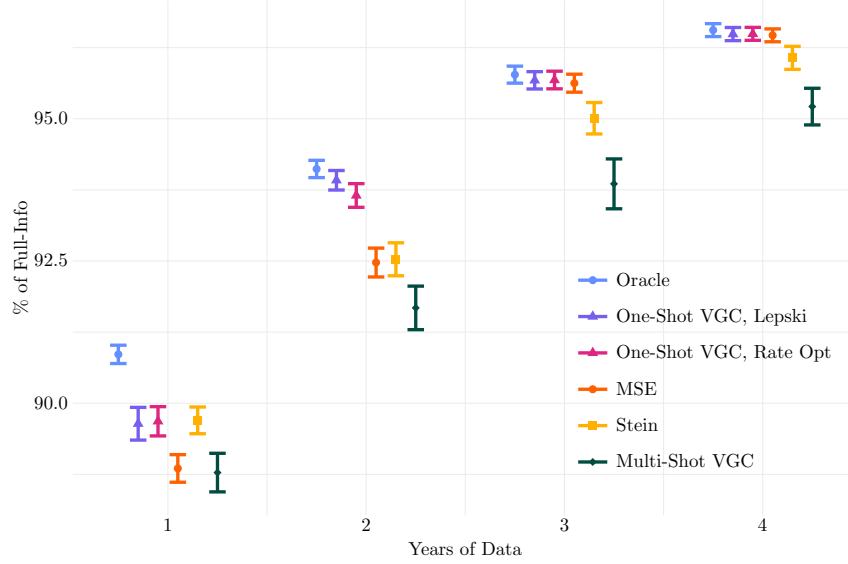


Figure 3 Performance Results. We compare the estimated expected out-of-sample performance of our method to various benchmarks and various over 100 trials. The error bars are 95% confidence intervals. The experiments vary the amount of data available but keep the number of decisions fixed.

to the lifted policy class introduced in Section 6.2. Fig. EC.1 in the appendix provides a larger plot with all methods for ease of comparison.

As predicted by Theorem 6.12, the “lifted” one-shot VGC performs similarly to the non-lifted variant. However, we see that the “lifted” multi-shot VGC outperforms its non-lifted version, achieving performance similar to the Stein correction. This empirical result combined with Proposition B.2 suggest some of the bias of multi-shot VGC from ?? might be heuristically addressed by “lifting” coupling constraints. While not pursued here, we conjecture that techniques from Theorem 6.12 and Gupta, Huang, and Rusmevichientong (2022b) might be used to formalize this heuristic.

Advantage of Decision-Aware Approach. In Figure 4, we map the selected census tracts of the decision-aware one-shot VGC policy and the decision-blind MSE policy. The map also plots contours of pedestrian injury rates generated from observed crashes. Many of the census tracts selected by the MSE policy lie on regions where there are multiple contour lines indicating a changing density of crashes with pedestrian injury. By contrast, the one-shot VGC policy tends to select census tracts in the center of regions with a high pedestrian injury rate.

To better understand this phenomenon, we plot the distribution of θ^{MSE} , θ^{OR} and θ^{VGC} (the θ that optimizes our one-shot VGC estimator) in Fig. 5. Note that θ^{OR} and θ^{MSE}

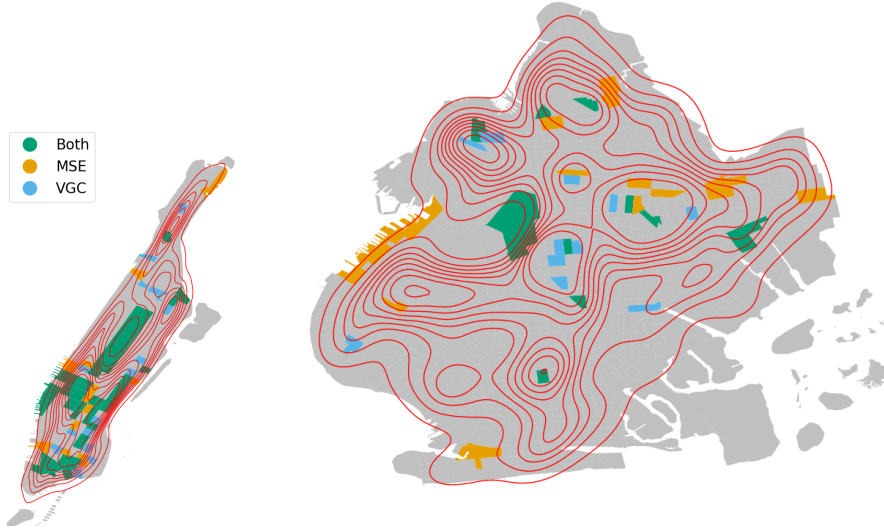


Figure 4 Map of Census Tracts Selected. We plot the census tracts selected by the MSE policy and the one-shot VGC policy for the Manhattan and Brooklyn boroughs of New York City. The contour lines plot the density of crashes with pedestrian injuries.

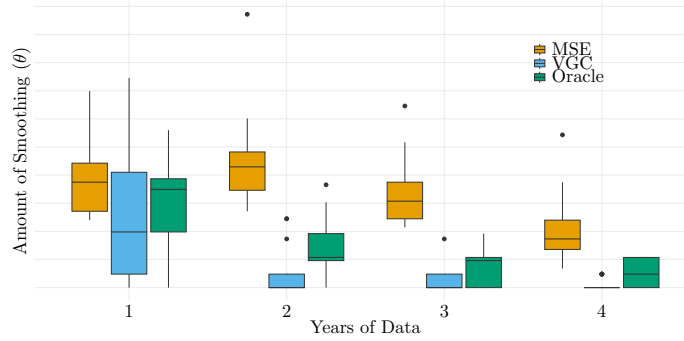


Figure 5 Amount of Smoothing by Method The figure compares the values of θ chosen by each method. Across realizations, θ^{MSE} tends to oversmooth relative to θ^{OR} , particularly as the amount of historical data grows. By contrast, the one-shot VGC performs less smoothing.

are quite different. By construction, θ^{MSE} is the best possible θ to minimize MSE, not downstream decision loss. In this example, it systematically over-smooths the data. This partially explains why it chooses census tracts that cross contour lines – the larger θ effectively smooths away the differences across the contour. By contrast, the one-shot VGC selects a θ^{VGC} much closer to the oracle value. We see this as strong evidence that i) Decision-aware approaches are necessary for finding the “right” level of smoothing in these types of applications, and ii) our one-shot VGC, despite being light-touch, is able to identify a good level of smoothing.

10. Conclusion

Our paper highlights the benefits of adapting traditional denoising techniques to decision-making settings. We provide rigorous guarantees of our approach for a wide class of regularized plug-in and non-regularized plug-in policies. Specifically, we provide a general bound of the error of the approach in terms of the stability of the underlying policies. For regularized plug-in policies and convex optimization, this approach readily yields strong guarantees that show the relative error of the method is vanishing as the problem size increases. We also develop a customized VGC estimator for general affine plug-in policies for weakly-coupled, potentially non-convex optimization problems. Under somewhat stronger assumptions, we also show the relative error in these settings vanishes as the problem size grows.

The fundamental role of stability in these results complements the role of stability in more traditional generalization guarantees in machine learning. Exploring other aspects of stability in data-driven optimization is an open area.

Finally, while our results primarily focus on debiasing policy performance for optimization problems with linear objectives, developing similar debiasing techniques for other structured optimization classes remains an exciting direction for future work.

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Online Appendix: Decision-Aware Denoising

Appendix A: Background Results

Recall, a mean-zero random variable X is subgaussian with variance proxy κ^2 if $\mathbb{E}[\exp(\lambda X)] \leq \exp\left(\frac{\lambda^2 \kappa^2}{2}\right)$. Let $f : \mathbb{R}^n \mapsto \mathbb{R}$ be a function. We say that f is 1-Lipschitz with respect to the $\mathbf{c}$ -weighted ℓ_1 norm if

$$|f(x_1, \dots, x_n) - f(y_1, \dots, y_n)| \leq \sum_{i=1}^n c_i |x_i - y_i|,$$

for some $\mathbf{c} \in \mathbb{R}_+^n$. Kontorovich (2014) proved that such functions applied to subGaussian random variables concentrate.

Theorem A.1 (Kontorovich 2014) *Let $\mathbf{X} = (X_1, \dots, X_n)$ be a vector of independent random variables such that $X_i - \mathbb{E}[X_i]$ is subGaussian with variance proxy κ_i^2 . Suppose $f(\cdot)$ is 1-Lipschitz with respect to the $\mathbf{c}$ -weighted ℓ_1 norm. Then, there exists a universal constant C such that for any $0 < \epsilon < \frac{1}{2}$, with probability at least $1 - \epsilon$,*

$$|f(\mathbf{X}) - \mathbb{E}[f(\mathbf{X})]| \leq 2 \sqrt{\sum_{i=1}^n c_i^2 \kappa_i^2} \cdot \sqrt{\log(2/\epsilon)}.$$

REMARK 2. The original result in Kontorovich, 2014 is stated with respect the sub-Gaussian diameters of the X_i . Note however that if $X_i - \mathbb{E}[X_i]$ is sub-Gaussian with variance proxy κ_i^2 , then X_i has sub-Gaussian diameter at most a $\sqrt{2}\kappa_i$.

Lemma A.2 (Uniform Error Bounds Suboptimality) *Suppose $\hat{\boldsymbol{\theta}} \in \arg \min_{\boldsymbol{\theta} \in \Theta} \mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta})$ and $\boldsymbol{\theta}^{OR} \in \arg \min_{\boldsymbol{\theta} \in \Theta} \boldsymbol{\mu}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta})$. Then,*

$$0 = \boldsymbol{\mu}^\top (\mathbf{x}(\mathbf{Z}, \hat{\boldsymbol{\theta}}) - \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}^{OR})) \leq 2 \sup_{\boldsymbol{\theta} \in \Theta} |\boldsymbol{\xi}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta})|.$$

Thus, the suboptimality of policy (w.r.t. an oracle policy) is at most twice the uniform error in our estimator.

Proof. The first inequality follows from the optimality of $\boldsymbol{\theta}^{OR}$. For the second, expand

$$\begin{aligned} \boldsymbol{\mu}^\top (\mathbf{x}(\mathbf{Z}, \hat{\boldsymbol{\theta}}) - \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}^{OR})) &\leq (\boldsymbol{\mu} - \mathbf{Z})^\top \mathbf{x}(\mathbf{Z}, \hat{\boldsymbol{\theta}}) + D(\mathbf{Z}, \hat{\boldsymbol{\theta}}) \\ &\quad + \mathbf{Z}^\top (\mathbf{x}(\mathbf{Z}, \hat{\boldsymbol{\theta}}) - \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}^{OR})) - D(\mathbf{Z}, \hat{\boldsymbol{\theta}}) + D(\mathbf{Z}, \boldsymbol{\theta}^{OR}) \\ &\quad + (\mathbf{Z} - \boldsymbol{\mu})^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}^{OR}) - D(\mathbf{Z}, \boldsymbol{\theta}^{OR}). \end{aligned}$$

The middle line is non-positive by the optimality of $\hat{\boldsymbol{\theta}}$. The other two lines can be bounded by $\sup_{\boldsymbol{\theta} \in \Theta} |(\mathbf{Z} - \boldsymbol{\mu})^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta})|$. Combining and using $\mathbf{Z} = \boldsymbol{\mu} + \boldsymbol{\xi}$ proves the lemma. $\square$

Lemma A.3 (A Pathwise Bound for Total Variation) *Let $\{q_t\}_{t \in [0, h]}$ be a family of probability densities on $\mathbb{R}^d$ such that $t \mapsto q_t$ is absolutely continuous as a map into $L^1(\mathbb{R}^d)$ and $\partial_t q_t \in L^1(\mathbb{R}^d)$ for all t . Then*

$$\text{TV}(q_0, q_h) \leq \frac{1}{2} \int_0^h \|\partial_t q_t\|_{L^1} dt.$$

Proof. By the fundamental theorem of calculus,

$$q_h - q_0 = \int_0^h \partial_t q_t dt \quad \text{in } L^1.$$

Taking norms and applying Minkowski's integral inequality,

$$\|q_h - q_0\|_{L^1} \leq \int_0^h \|\partial_t q_t\|_{L^1} dt.$$

By definition, $\text{TV}(q_0, q_h) = \frac{1}{2} \|q_h - q_0\|_{L^1}$.

Appendix B: Section 3 Proofs

B.1. Auxilliary Lemmas

Many of our results leverage that the function $V(\mathbf{z}, \rho) = \min_{\mathbf{x} \in \mathcal{X}} \mathbf{z}^\top \mathbf{x} + \rho \sum_{j=1}^n \phi(x_j)$ is Lipschitz.

Lemma B.1 (Lipschitz Bounds on V) *Let $V(\mathbf{z}, \rho) = \min_{\mathbf{x} \in \mathcal{X}} \mathbf{z}^\top \mathbf{x} + \rho \sum_{j=1}^n \phi(x_j)$ and $\mathcal{X} \subseteq [-1, 1]^n$. Then, the following holds,*

- i) $|V(\mathbf{z}, \rho) - V(\mathbf{y}, \rho)| \leq \|\mathbf{z} - \mathbf{y}\|_1$.
- ii) $|V(\mathbf{z}, \rho) - V(\mathbf{z}, \rho')| \leq n\phi_{\max} |\rho - \rho'|$, where $\phi_{\max} = \sup_{x \in [-1, 1]} |\phi(x)|$

Proof of Lemma B.1:

Part i) We first upper-bound the difference. We see,

$$\begin{aligned} V(\mathbf{z}, \rho) - V(\mathbf{y}, \rho) &= \underbrace{\mathbf{z}^\top \hat{\mathbf{x}}(\mathbf{z}, \rho) + \rho \sum_{j=1}^n \phi(\hat{x}_j(\mathbf{z}, \rho)) - \mathbf{z}^\top \hat{\mathbf{x}}(\mathbf{y}, \rho) - \rho \sum_{j=1}^n \phi(\hat{x}_j(\mathbf{y}, \rho))}_{\leq 0, \text{ by optimality of } \hat{\mathbf{x}}(\mathbf{z}, \rho)} + (\mathbf{z} - \mathbf{y})^\top \hat{\mathbf{x}}(\mathbf{y}, \rho) \\ &\leq \|\mathbf{z} - \mathbf{y}\|_1 \|\hat{\mathbf{x}}(\mathbf{y}, \rho)\|_\infty \leq \|\mathbf{z} - \mathbf{y}\|_1 \end{aligned}$$

where the first inequality holds by Holder's inequality and the second inequality holds because $\|\hat{\mathbf{x}}(\mathbf{y}, \rho)\|_\infty \leq 1$. By symmetry, this completes the proof of Part i).

Part ii) We apply a similar argument. Letting $\phi(\mathbf{x}) = \sum_{j=1}^n \phi(x_j)$,

$$\begin{aligned} V(\mathbf{z}, \rho) - V(\mathbf{z}, \rho') &= \underbrace{\mathbf{z}^\top \hat{\mathbf{x}}(\mathbf{z}, \rho) + \rho \phi(\hat{\mathbf{x}}(\mathbf{z}, \rho)) - \mathbf{z}^\top \hat{\mathbf{x}}(\mathbf{z}, \rho') - \rho' \phi(\hat{\mathbf{x}}(\mathbf{z}, \rho'))}_{\leq 0, \text{ by optimality of } \hat{\mathbf{x}}(\mathbf{z}, \rho)} + (\rho - \rho') \phi(\hat{\mathbf{x}}(\mathbf{z}, \rho')) \\ &\leq |\rho - \rho'| \sum_{j=1}^n \sup_{x \in [-1, 1]} |\phi(x)| = n\phi_{\max} |\rho - \rho'|. \end{aligned}$$

By symmetry, we can also bound $V(\mathbf{z}, \rho') - V(\mathbf{z}, \rho)$, completing the proof. $\square$

Since $V(\cdot)$ is 1-Lipschitz relative to $\|\cdot\|_1$, we can bound the expected error introduced by proxying ξ by δ_h in our proof:

□

B.2. Proof for Theorem 3.1

We fix θ throughout and drop it from the notation. Recall $\mathbf{Z} = \xi + \mu$. Let $\bar{\xi}$ be an i.i.d. copy of ξ . Let ξ^k denote the vector ξ with element k replaced with $\bar{\xi}_k$. By the Efron-Stein Inequality,

$$\text{Var}(D(\mathbf{Z})) \leq \frac{1}{2} \sum_{k=1}^n \mathbb{E} \left[\left(D(\xi + \mu) - D(\xi^k + \mu) \right)^2 \right]$$

Focusing on the term inside the square, we see

$$\begin{aligned} |D(\xi + \mu) - D(\xi^k + \mu)| &\leq \frac{1}{h} \left| \mathbb{E} [V(\mathbf{T}(\xi + \mu) + \delta_h) - V(\mathbf{T}(\xi^k + \mu) + \delta_h)] \right| \\ &\quad + \frac{1}{h} |V(\mathbf{T}(\xi + \mu)) - V(\mathbf{T}(\xi^k + \mu))| \\ &\leq \frac{2}{h} \|\mathbf{T}(\xi + \mu) - \mathbf{T}(\xi^k + \mu)\|_1 \\ &\leq \frac{2}{h} C_T \|\xi - \xi^k\|_1 \\ &\leq \frac{2}{h} C_T |\xi_k - \bar{\xi}_k|. \end{aligned}$$

The second inequality holds by Lemma B.1 and the third holds by Assumption 2.3.

Plugging in the upper-bound,

$$\mathbb{E} \left[\left(D(\xi + \mu) - D(\xi^k + \mu) \right)^2 \right] \leq \mathbb{E} \left[\frac{4}{h^2} |\xi_k - \bar{\xi}_k|^2 C_T^2 \right] \leq \frac{8C_T^2}{\nu_k h^2}$$

Letting $\nu_{\min} = \min_k \nu_k$, we see,

$$\text{Var}(D(\mathbf{Z})) \leq \frac{4nC_T^2}{\nu_{\min} h^2}$$

This completes the proof.

B.3. Proof for Theorem 3.3

We leverage Theorem A.1 on the function $D(\mathbf{Z})$. Let $\mathbf{Z}^k$ be identical to $\mathbf{Z}$ except possibly in the k^{th} component. Following the same steps of the proof of Theorem 3.1, we have

$$|D(\mathbf{Z}) - D(\mathbf{Z}^k)| \leq \frac{2}{h} |Z_k - \bar{Z}_k| C_T,$$

We take $c_i = \frac{2}{h} C_T$ and $\kappa(Z_i) \leq \kappa$.

Applying Theorem A.1 yields the result.

B.4. The One Shot and Multi-Shot VGC under Decoupling

In Section 8, we showed that in special cases where both corrections apply, the multi-shot VGC can be seen as a special case of the one-shot VGC. We next show that when the optimization problem fully decouples, so that $\hat{x}_j(\mathbf{t}, 0)$ only depends on t_j but not t_k for $k \neq j$, for all j , then the one-shot VGC is equivalent to the multi-shot VGC.

Proposition B.2 (One-Shot vs. Multi-Shot VGC for Decoupled Problems) *Suppose that $T(\mathbf{Z}, \bar{\boldsymbol{\theta}}) = \mathbf{Z} + \ell(\bar{\boldsymbol{\theta}})$ for some $\ell(\cdot)$, $\rho = 0$, and $Z_j \sim \mathcal{N}(\mu_j, 1)$ for each j (not necessarily independent). Let $\boldsymbol{\delta}$ be any random variable such that $\delta_j \sim \mathcal{N}(0, h^2 + 2h)$. Finally, assume that $\hat{x}_j(\mathbf{t}, 0)$ only depends on t_j for each j . Then, the one-shot VGC with stepsize h and perturbation $\boldsymbol{\delta}$ is equivalent to the multi-shot VGC with step size h .*

Proof. We make two observations necessary for the proof: First, since $\hat{x}_j(\mathbf{t}, 0)$ only depends on t_j , $\hat{x}_j(\mathbf{t}, 0) = \hat{x}(t_j \mathbf{e}_j, 0)$ for each j . Second, $V(\mathbf{t} + \delta_j \mathbf{e}_j) - V(\mathbf{t}) = (t_j + \delta_j)\hat{x}((t_j + \delta_j)\mathbf{e}_j) - t_j\hat{x}(t_j\mathbf{e}_j)$ since all the terms $k \neq j$ in the two sums are identical and drop out.

Then, starting with the one-shot VGC, and suppressing $\boldsymbol{\theta}$ in the notation,

$$\begin{aligned} D(\mathbf{Z}) &\equiv \frac{1}{h} \mathbb{E}[V(\mathbf{Z} + \boldsymbol{\ell} + \boldsymbol{\delta}) - V(\mathbf{Z} + \boldsymbol{\ell}) \mid \mathbf{Z}] \\ &= \frac{1}{h} \sum_{j=1}^n \mathbb{E}[(Z_j + \ell_j + \delta_j)\hat{x}_j(\mathbf{Z} + \boldsymbol{\ell} + \boldsymbol{\delta}) - (Z_j + \ell_j)\hat{x}_j(\mathbf{Z} + \boldsymbol{\ell}) \mid \mathbf{Z}] \\ &= \frac{1}{h} \sum_{j=1}^n \mathbb{E}[(Z_j + \ell_j + \delta_j)\hat{x}_j(\mathbf{Z} + \boldsymbol{\ell} + \delta_j \mathbf{e}_j) - (Z_j + \ell_j)\hat{x}_j(\mathbf{Z} + \boldsymbol{\ell}) \mid \mathbf{Z}] \\ &= \frac{1}{h} \sum_{j=1}^n \mathbb{E}[V(\mathbf{Z} + \boldsymbol{\ell} + \delta_j \mathbf{e}_j) - V(\mathbf{Z} + \boldsymbol{\ell}) \mid \mathbf{Z}], \end{aligned}$$

where the second-to-last and last equalities use our two observations from the beginning of the proof. Recognizing the definition of the multi-shot VGC completes the proof. $\square$

Appendix C: Is Dependence on Solution Stability Unavoidable?

We build on our heuristic derivation from Section 3.1 to develop some intuition for why solution stability arises in our analysis and why we believe it is a fundamental quantity. Other authors (Yu, 2013; Shalev-Shwartz et al., 2010) have similarly observed the importance of stability in learning applications. We fix $\boldsymbol{\theta}$ and drop it from the notation throughout this subsection.

For a twice differentiable function, a classical Taylor series argument shows that the error in approximating the first derivative by a first-order finite step difference scales like the second derivative, i.e.,

$$\left| \frac{f(h) - f(0)}{h} - f'(0) \right| = \frac{h}{2} |f''(0)| + o(h).$$

Hence, for the function $\lambda \mapsto V(\mathbf{T}(\mathbf{Z}) + \lambda \boldsymbol{\xi})$ from Eq. (9), we might expect that

$$\left| \frac{V(\mathbf{T}(\mathbf{Z}) + h\boldsymbol{\xi}) - V(\mathbf{T}(\mathbf{Z}))}{h} - \boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z})) \right| \approx \frac{h}{2} \left| \frac{\partial^2}{\partial \lambda^2} V(\mathbf{T}(\mathbf{Z}) + \lambda \boldsymbol{\xi}) \Big|_{\lambda=0} \right|,$$

and

$$\frac{\partial^2}{\partial \lambda^2} V(\mathbf{T}(\mathbf{Z}) + \lambda \boldsymbol{\xi}) \Big|_{\lambda=0} = \frac{\partial}{\partial \lambda} \boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}) + \lambda \boldsymbol{\xi}) \Big|_{\lambda=0} = \boldsymbol{\xi}^\top \mathbf{J}(\mathbf{T}(\mathbf{Z})) \boldsymbol{\xi},$$

where $\mathbf{J}(\mathbf{T}(\mathbf{Z})) \in \mathbb{R}^{n \times n}$ is the Jacobian of $\lambda \mapsto \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}) + \lambda \boldsymbol{\xi})$. This heuristic analysis suggests that the magnitude of changes in the solution vector $\hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}))$ to perturbations of its input drives the error in our method. More specifically, if we apply a first order finite difference to $\frac{\partial}{\partial \lambda} \boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}) + \lambda \boldsymbol{\xi})$ at $\lambda = 0$, we recover $\boldsymbol{\xi}^\top (\hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}) + h\boldsymbol{\xi}) - \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z})))$, the solution stability of Eq. (13).

This perspective also highlights why solution stability depends on the particular structure of Problem (1) and the policy class. Specifically, the (i, j) th element of $\mathbf{J}(\mathbf{T}(\mathbf{Z}))$ is $\frac{\partial}{\partial Z_j} \hat{x}_i(\mathbf{T}(\mathbf{Z}))$, measuring how much the i th component of the solution changes with small perturbations of the j th component of the input. The magnitude of this change depends strongly on the constraints in the feasible set $\mathcal{X}$ of Problem (1) and how $\mathbf{T}(\cdot)$ magnifies/attenuates the perturbation. Hence, unlike bounding the replication error, bounding the finite difference error seemingly requires stronger assumptions on Problem (1).

Appendix D: Section 4 Proofs

D.1. Proof for Theorem 4.2

Recall from Eq. (12), the estimation error can be decomposed into the following two components:

$$\begin{aligned} & \sup_{\boldsymbol{\theta} \in \Theta} \underbrace{\left| \mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D(\mathbf{Z}, \boldsymbol{\theta}) - \boldsymbol{\mu}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) \right|}_{\text{Estimation Error}} \\ & \leq \sup_{\boldsymbol{\theta} \in \Theta} \underbrace{\left| \boldsymbol{\xi}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - \frac{1}{h} (V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), \rho)) \right|}_{\text{Finite Difference Error}_{\boldsymbol{\theta}}} \\ & \quad + \sup_{\boldsymbol{\theta} \in \Theta} \underbrace{\left| \frac{1}{h} (V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), \rho)) - D(\mathbf{Z}, \boldsymbol{\theta}) \right|}_{\text{Replication Error}_{\boldsymbol{\theta}}}. \end{aligned}$$

We bound the finite difference error by solution stability in the main text (c.f. Eq. (13)). Hence, we focus on on bounding the replication error.

To bound replication error, first note that by definition,

$$\left| \frac{V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), \rho)}{h} - D(\mathbf{Z}, \boldsymbol{\theta}, h) \right| = \left| \frac{V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - \mathbb{E}[V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \boldsymbol{\delta}_h, \rho) | \mathbf{Z}]}{h} \right|.$$

Thus,

$$\begin{aligned} \sup_{\boldsymbol{\theta} \in \Theta} \text{Replication Error}_{\boldsymbol{\theta}} &\leq \underbrace{\sup_{\boldsymbol{\theta} \in \Theta} \left| \frac{V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - \mathbb{E}[V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho)]}{h} \right|}_{(i)} \\ &\quad + \underbrace{\sup_{\boldsymbol{\theta} \in \Theta} \left| \frac{\mathbb{E}[V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \boldsymbol{\delta}_h, \rho) | \mathbf{Z}] - \mathbb{E}[V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \boldsymbol{\delta}_h, \rho)]}{h} \right|}_{(ii)} \\ &\quad + \underbrace{\sup_{\boldsymbol{\theta} \in \Theta} \left| \frac{\mathbb{E}[V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \boldsymbol{\delta}_h, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho)]}{h} \right|}_{(iii)} \end{aligned}$$

We next bound each of these components.

Bounding Component (i). Our strategy will be to first show a pointwise bound for a fixed $\boldsymbol{\theta}$ using Theorem A.1, and then extend this to all of Θ via a covering argument using Assumption 2.5.

Fix any $\boldsymbol{\theta} = (\bar{\boldsymbol{\theta}}, \rho) \in \Theta$. Define $\boldsymbol{\xi}^k = (\xi_1, \dots, \bar{\xi}_k, \dots, \xi_n)^\top$ and $\mathbf{Z}^k = (Z_1, \dots, \bar{Z}_k, \dots, Z_n)^\top$ where $\bar{\xi}_k$ is an i.i.d. copy of ξ_k and $\bar{Z}_k \equiv \mu_k + \bar{\xi}_k$. Then,

$$\begin{aligned} |V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}^k, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}^k, \rho)| &\leq \|\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi} - \mathbf{T}(\mathbf{Z}^k, \bar{\boldsymbol{\theta}}) - h\boldsymbol{\xi}^k\|_1 \\ &\leq (C_{\mathbf{T}}(\bar{\boldsymbol{\theta}}) + h)\|\boldsymbol{\xi} - \boldsymbol{\xi}^k\|_1 \\ &\leq (C_{\mathbf{T}} + h)|\xi_k - \bar{\xi}_k|, \end{aligned}$$

where $C_{\mathbf{T}} = \max_{\bar{\boldsymbol{\theta}}} C_{\mathbf{T}}(\bar{\boldsymbol{\theta}})$. The first inequality follows from Lemma B.1 and the second inequality follows from Assumption 2.3. Since ξ_k are sub-Gaussian, Theorem A.1 with $c_k = C_{\mathbf{T}} + h$ implies that with probability at least $1 - \epsilon$,

$$|V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - \mathbb{E}[V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho)]| \leq C \cdot (C_{\mathbf{T}} + h) \kappa \sqrt{n \log \left(\frac{2}{\epsilon} \right)} \quad (23)$$

for a fixed $\boldsymbol{\theta}$ and universal constant C .

Now consider an arbitrary $\boldsymbol{\theta} \in \Theta$ and let $\boldsymbol{\theta}_0$ be the element in $\Theta_0(n^{-1/2})$ closest to $\boldsymbol{\theta}$. Then,

$$\begin{aligned} |V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - \mathbb{E}[V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho)]| &\leq \sup_{\boldsymbol{\theta} \in \Theta_0(n^{-1/2})} |V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - \mathbb{E}[V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho)]| \quad (24a) \\ &\quad + |V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}_0) + h\boldsymbol{\xi}, \rho_0)| \quad (24b) \\ &\quad + |\mathbb{E}[V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}_0) + h\boldsymbol{\xi}, \rho_0)]|. \quad (24c) \end{aligned}$$

Applying Eq. (23) at $\boldsymbol{\theta}_0$ and taking the union bound over the covering shows that with probability at least $1 - \epsilon$

$$\text{Eq. (24a)} \leq C(C_{\mathbf{T}} + h) \cdot \kappa \sqrt{n \log(|\Theta_0(n^{-1/2})|) \log \left(\frac{2}{\epsilon} \right)}.$$

To bound Eq. (24b), recall that $\boldsymbol{\theta} = (\bar{\boldsymbol{\theta}}, \rho)$ and $\boldsymbol{\theta}_0 = (\bar{\boldsymbol{\theta}}_0, \rho_0)$,

$$\begin{aligned} |V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}_0) + h\boldsymbol{\xi}, \rho_0)| &\leq |V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho_0)| \\ &\quad + |V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}_0) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}_0) + h\boldsymbol{\xi}, \rho_0)| \\ &\stackrel{(a)}{\leq} \|\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) - \mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}_0)\|_1 + n\phi_{\max} |\rho - \rho_0| \\ &\stackrel{(b)}{\leq} n^{-1/2} C (\|\mathbf{Z}\|_1 + 1) + n^{-1/2} \cdot n\phi_{\max}, \end{aligned}$$

where $\phi_{\max} = \max_{x \in [-1, 1]} \phi(x)$. For clarity, (a) holds by the Lipschitz results of V in Lemma B.1, (b) holds by applying Assumption 2.5 with $\epsilon = n^{-1/2}$ and universal constant $C \geq 1$. We can bound $\|\mathbf{Z}\|_1$ with high probability. Specifically, write

$$\|\mathbf{Z}\|_1 \leq \|\boldsymbol{\xi}\|_1 + \|\boldsymbol{\mu}\|_1 \leq (\|\boldsymbol{\xi}\|_1 - \mathbb{E}\|\boldsymbol{\xi}\|_1) + \mathbb{E}\|\boldsymbol{\xi}\|_1 + n\|\boldsymbol{\mu}\|_\infty \leq \kappa \sqrt{n \log \left(\frac{1}{\epsilon} \right)} + n\nu_{\min}^{1/2} + n\|\boldsymbol{\mu}\|_\infty,$$

where the last inequality applies Theorem A.1 to $\|\boldsymbol{\xi}\|_1 - \mathbb{E}\|\boldsymbol{\xi}\|_1$ with $c_j = 1$ for all j and that $\mathbb{E}|\xi_j| \leq \mathbb{E} \left| \sqrt{\xi_j^2} \right| \leq \sqrt{\mathbb{E}\xi_j^2} \leq \nu_{\min}^{-1/2}$.

Substituting above and collecting constants shows that

$$\begin{aligned} |V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}_0) + h\boldsymbol{\xi}, \rho_0)| &\stackrel{(c)}{\leq} \frac{C}{\sqrt{n}} \left(\kappa \sqrt{n \log \left(\frac{1}{\epsilon} \right)} + n \left(\nu_{\min}^{-1/2} + \|\boldsymbol{\mu}\|_\infty + \phi_{\max} \right) + 1 \right) \\ &\stackrel{(d)}{\leq} C_1 \kappa \sqrt{n \log \left(\frac{1}{\epsilon} \right)}, \end{aligned}$$

where $C_1 = 4C \left(\nu_{\min}^{1/2} + \|\boldsymbol{\mu}\|_\infty + \phi_{\max} + 1 \right)$. For clarity, the final inequality holds for $\epsilon < 1/e$ by noting $a + b \leq 2ab$ for $a \geq 1, b \geq 1$. Taking the supremum over $\boldsymbol{\theta} \in \Theta$ of both sides bounds Eq. (24b).

Finally, to bound Eq. (24c), note from our previous step and Jensen's inequality

$$\begin{aligned} |\mathbb{E} [V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}_0) + h\boldsymbol{\xi}, \rho_0)]| &\leq \mathbb{E} |V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}_0) + h\boldsymbol{\xi}, \rho_0)| \\ &\leq \mathbb{E} [n^{-1/2} C_\Theta (\|\mathbf{Z}\|_1 + 1) + n^{-1/2} \cdot n\phi_{\max}] \\ &\leq C_\Theta n^{-1/2} \left(n\nu_{\min}^{1/2} + n\|\boldsymbol{\mu}\|_\infty + n\phi_{\max} + 1 \right) \\ &\leq C_1 \sqrt{n}. \end{aligned}$$

Combining Eq. (24a), Eq. (24b), and Eq. (24c), collecting terms and applying a supremum over $\boldsymbol{\theta} \in \Theta$, we have for some universal constant C ,

$$(i) \leq C \frac{C_1(C_T + h)\kappa}{h} \sqrt{n \cdot \log(|\Theta_0| n^{-1/2}) \cdot \log \left(\frac{2}{\epsilon} \right)}$$

Bounding Component (ii). We follow the same strategy as with component (i). We first use Theorem A.1 to establish a pointwise bound. We then use Assumption 2.5 to extend this with a covering argument.

Fix any $\theta = (\bar{\theta}, \rho) \in \Theta$. Define $\xi^k = (\xi_1, \dots, \bar{\xi}_k, \dots, \xi_n)^\top$ and $\mathbf{Z}^k = (Z_1, \dots, \bar{Z}_k, \dots, Z_n)^\top$ where $\bar{\xi}_k$ is an i.i.d. copy of ξ_k and $\bar{Z}_k \equiv \mu_k + \bar{\xi}_k$. Then, by Jensen's inequality and Lemma B.1,

$$\begin{aligned} & |\mathbb{E} [V(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + \delta_h, \rho) | \mathbf{Z}] - \mathbb{E} [V(\mathbf{T}(\mathbf{Z}^k, \bar{\theta}) + \delta_h, \rho) | \mathbf{Z}^k]| \\ & \leq \mathbb{E} [|V(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + \delta_h, \rho) - V(\mathbf{T}(\mathbf{Z}^k, \bar{\theta}) + \delta_h, \rho)| | \mathbf{Z}, \mathbf{Z}^k] \\ & \leq \|\mathbb{E} [\mathbf{T}(\mathbf{Z}, \bar{\theta}) + \delta_h - \mathbf{T}(\mathbf{Z}^k, \bar{\theta}) - \delta_h | \mathbf{Z}, \bar{Z}_k]\|_1 \\ & \leq C_{\mathbf{T}}(\bar{\theta}) \|\xi - \xi^k\|_1 \\ & \leq C_{\mathbf{T}} |\xi_k - \bar{\xi}_k|, \end{aligned}$$

Applying Theorem A.1 with $c_k = C_{\mathbf{T}}$ shows for a fixed $\theta \in \Theta$,

$$|\mathbb{E} [V(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + \delta_h, \rho) | \mathbf{Z}] - \mathbb{E} [V(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + \delta_h, \rho)]| \leq C \cdot C_{\mathbf{T}} \kappa \sqrt{n \log \left(\frac{2}{\epsilon} \right)}.$$

We extend this pointwise bound across the covering in exactly the same manner as done for component (i) and hence omit the details. This procedure yields

$$(ii) \leq C \frac{C_1 C_{\mathbf{T}} \kappa}{h} \sqrt{n \cdot \log(|\Theta_0(n^{-1/2})|) \cdot \log \left(\frac{2}{\epsilon} \right)}$$

Bounding Component (iii). Recall that $V(\cdot)$ is 1-Lipschitz by Lemma B.1. Hence, using the dual-representation of the Wasserstein metric, (Wainwright, 2019, pg. 76)

$$\frac{1}{h} \sup_{\theta \in \Theta} |\mathbb{E} [V(\mathbf{T}(\mathbf{Z}, \theta) + \delta_h, \rho)] - \mathbb{E} [V(\mathbf{T}(\mathbf{Z}, \theta) + h\xi, \rho)]| \leq \frac{1}{h} \mathbb{W}(\mathbf{T}(\mathbf{Z}, \theta) + \delta_h, \mathbf{T}(\mathbf{Z}, \theta) + h\xi).$$

Combining (i), (ii), and (iii), we prove the result.

D.2. Proof for Proposition 4.3.

Canceling identical terms on both sides, we must show that $\mathbb{E} [V(\mathbf{T}(\mathbf{Z}) + h\xi)] = \mathbb{E} [V(\mathbf{T}(\mathbf{Z}) + \delta_h)]$. Thus, it suffices to show that $\mathbf{T}(\mathbf{Z}) + h\xi \sim_d \mathbf{T}(\mathbf{Z}) + \delta_h$.

To that end, first observe that since $h > 0$ and $\mathbf{L}(\bar{\theta})\Sigma + \Sigma\mathbf{L}(\bar{\theta})^\top$ is positive semidefinite, the matrix $h(\mathbf{L}(\bar{\theta})\Sigma + \Sigma\mathbf{L}(\bar{\theta})^\top) + h^2\Sigma$ is also positive semidefinite and δ_h is well-defined.

Since the sum of two multivariate Gaussians is also a multivariate Gaussian, we see both $\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h\xi$ and $\mathbf{T}(\mathbf{Z}, \bar{\theta}) + \delta_h$ are distributed as $\mathcal{N}(\mu, (\mathbf{L} + h\mathbf{I})\Sigma(\mathbf{L} + h\mathbf{I})^\top)$. This completes the proof.

Appendix E: Continuous Policy Proofs

E.1. Proof for Lemma 5.1

We first establish the following well-known result:

Lemma E.1 (Regularized Plug-in Policies are ρ -Lipschitz) *When $\mathcal{X}$ is convex and $\rho > 0$, $\|\hat{\mathbf{x}}(\mathbf{t}, \rho) - \hat{\mathbf{x}}(\mathbf{t}', \rho)\| \leq \frac{1}{\rho} \|\mathbf{t} - \mathbf{t}'\|_*$, where $\|\cdot\|_*$ is the dual norm of $\|\cdot\|$.*

Proof. Let $h_\rho(\mathbf{x}, \mathbf{t}) \equiv \sum_{j=1}^n t_j x_j + \rho \phi(x_j)$. Since $h_\rho(\mathbf{x}, \mathbf{t})$ is strongly convex over $\mathcal{X}$ and by the optimality of $\hat{\mathbf{x}}(\mathbf{t}, \rho)$ and $\hat{\mathbf{x}}(\mathbf{y}, \rho)$, we see

$$\begin{aligned} h_\rho(\hat{\mathbf{x}}(\mathbf{t}, \rho), \mathbf{y}) - h_\rho(\hat{\mathbf{x}}(\mathbf{y}, \rho), \mathbf{y}) &\geq \frac{\rho}{2} \|\hat{\mathbf{x}}(\mathbf{t}, \rho) - \hat{\mathbf{x}}(\mathbf{y}, \rho)\|^2 \\ h_\rho(\hat{\mathbf{x}}(\mathbf{y}, \rho), \mathbf{t}) - h_\rho(\hat{\mathbf{x}}(\mathbf{t}, \rho), \mathbf{t}) &\geq \frac{\rho}{2} \|\hat{\mathbf{x}}(\mathbf{t}, \rho) - \hat{\mathbf{x}}(\mathbf{y}, \rho)\|^2 \end{aligned}$$

Adding the two inequalities, we see

$$\begin{aligned} \rho \|\hat{\mathbf{x}}(\mathbf{t}, \rho) - \hat{\mathbf{x}}(\mathbf{y}, \rho)\|^2 &\leq h_\rho(\hat{\mathbf{x}}(\mathbf{t}, \rho), \mathbf{y}) - h_\rho(\hat{\mathbf{x}}(\mathbf{y}, \rho), \mathbf{y}) + h_\rho(\hat{\mathbf{x}}(\mathbf{y}, \rho), \mathbf{t}) - h_\rho(\hat{\mathbf{x}}(\mathbf{t}, \rho), \mathbf{t}) \\ &= \sum_{j=1}^n y_j (\hat{x}_j(\mathbf{t}, \rho) - \hat{x}_j(\mathbf{y}, \rho)) + t_j (\hat{x}_j(\mathbf{y}, \rho) - \hat{x}_j(\mathbf{t}, \rho)) \\ &= \sum_{j=1}^n (y_j - t_j) (\hat{x}_j(\mathbf{t}, \rho) - \hat{x}_j(\mathbf{y}, \rho)) \\ &\leq \|\mathbf{y} - \mathbf{t}\|_* \|\hat{\mathbf{x}}(\mathbf{t}, \rho) - \hat{\mathbf{x}}(\mathbf{y}, \rho)\|, \end{aligned}$$

where the last line holds by Holder's inequality. Rearranging, we get our intended result.

We can now simply prove Lemma 5.1.

Proof of Lemma 5.1. The result follows directly from Cauchy-Schwarz since

$$\text{SS}(\boldsymbol{\xi}, h, \boldsymbol{\theta}) \leq \|\boldsymbol{\xi}\|_* \|\hat{\mathbf{x}}(T(\mathbf{Z}, \bar{\boldsymbol{\theta}}), \rho) - \hat{\mathbf{x}}(T(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, \rho)\| \leq \frac{h}{\rho} \|\boldsymbol{\xi}\|_*^2.$$

E.2. Proof for Theorem 5.2

The proof is immediate from combining Theorem 4.2 and Lemma 5.1 and collecting constants.

E.2.1. Proof Corollary 5.3.

Proof. First consider the case where $\phi(\cdot)$ is 1-strongly convex with respect to $\|\cdot\|_1$. To form a more explicit bound on the solution stability, we analyze $\|\boldsymbol{\xi}\|_*^2 = \|\boldsymbol{\xi}\|_\infty^2$. Since $\boldsymbol{\xi}$ is mean-zero, $\|\boldsymbol{\xi}\|_\infty = \|\boldsymbol{\xi} - \mathbb{E}[\boldsymbol{\xi}]\|_\infty$. Furthermore, since $\xi_j - \mathbb{E}[\xi_j]$ is sub-Gaussian with variance proxy κ^2 , we have that with probability at least $1 - \epsilon$, there exists an absolute constant C_1 such that

$$|\xi_j - \mathbb{E}[\xi_j]| \leq C_1 \kappa \sqrt{\log(n/\epsilon)} \leq C_1 \kappa \sqrt{\log n \log(1/\epsilon)},$$

simultaneously for all j . Hence, with probability at least $1 - \epsilon$,

$$\|\boldsymbol{\xi}\|_\infty^2 \leq C_1^2 \kappa^2 \log n \log(1/\epsilon).$$

Next, we substitute this bound into Theorem 5.2 and optimize the choice of h . Recall, by assumption the Wasserstein contribution is zero. Hence, the terms depending on h are (neglecting constants)

$$\frac{h}{\rho_{\min}} \log n + \frac{C_T}{h} \sqrt{n \log n}.$$

Choosing $h = \rho_{\min}^{1/2} n^{1/4}$ optimizes this quantity (up to logarithmic factors). Substituting back into the bound and collecting constants proves the first part of the corollary.

We next consider the case that $\phi(\cdot)$ is 1-strongly convex with respect to $\|\cdot\|_2$. To bound solution stability more explicitly, we bound $\|\boldsymbol{\xi}\|_*^2 = \|\boldsymbol{\xi}\|_2^2 = \sum_{j=1}^n \xi_j^2$. Since ξ_j is sub-Gaussian with parameter κ , Vershynin, 2018, Lemma 2.7.6 shows that there exists an absolute constant C_1 such that that $\|\xi_j^2\|_{\Psi_1} \leq C_1 \kappa^2$. (Recall, for any random variable Y , $\|Y\|_{\Psi_1} \equiv \inf\{t > 0 : \mathbb{E}[\exp(|Y|/t)] \leq 2\}$, c.f., Vershynin (2018, pg. 31).) Hence, by the triangle inequality, $\|\sum_{j=1}^n \xi_j^2\|_{\Psi_1} \leq C_1 n \kappa^2$. Finally, by Markov's Inequality, with probability at least $1 - \epsilon$, $\sum_{j=1}^n \xi_j^2 \leq C_2 n \kappa^2 \log(1/\epsilon)$, for some absolute constant C_2 .

Next we plug this upper bound into Theorem 5.2 and optimize the choice of h . Again, the Wasserstein contribution is zero. Neglecting constants, the remaining terms are

$$\frac{h}{\rho_{\min}} n + \frac{C_T}{h} \sqrt{n \log n}.$$

Choosing $h = \rho_{\min}^{1/2} n^{-1/4}$ optimizes the bound up to logarithmic factors. Substituting in and collecting terms completes the proof. $\square$

E.3. Proof of Theorem 5.6.

Proof of Lemma 5.5. By Cauchy-Schwarz,

$$\begin{aligned} |\text{SS}(\boldsymbol{\xi}, h, \boldsymbol{\theta})| &\leq \|\boldsymbol{\xi}\|_* \|\hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), 0) - \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}, 0)\|_* \\ &\leq \frac{2h\|\boldsymbol{\xi}\|_*^2}{\alpha (\|\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}})\|_* + \|\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}\|_*)} \\ &\leq \frac{2h\|\boldsymbol{\xi}\|_*^2}{\alpha \|\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}})\|_*} \end{aligned}$$

This completes the proof. $\square$

Proof of Theorem 5.6. The proof follows from substituting Lemma 5.5 into Theorem 4.2 and collecting terms. $\square$

Appendix F: Proofs and Additional Materials for Block Decoupled Problems

F.1. What are typical values of $T_{\max}$ and $\mathcal{X}_{\max}$?

In this subsection, we revisit some examples from Section 2.3 to provide explicit bounds on the constants $T_{\max}$ and $\mathcal{X}_{\max}$ appearing in Theorem 6.7.

Example F.1 (Clustering Revisited) Recall our clustering based policy class from Example 2.10. Let $\mathbf{x}^{CL}(\mathbf{Z}, \bar{\theta})$ be the resulting policy when clustering into $\bar{\theta}$ clusters. We form a policy class by allowing the number of clusters to vary between $\bar{\theta}_{\min}$ and $\bar{\theta}_{\max}$.

Then, $T_{\max}$ corresponds to the size of the largest cluster. Hence, if $\bar{\theta}_{\min} = O(n)$, i.e., we only want to average over “local” neighborhoods, then $T_{\max} = O(1)$. Furthermore, the number of possible policies $\mathcal{X}_{\max}$ is bounded by the number of possible clusterings $\bar{\theta}_{\max} - \bar{\theta}_{\min}$ which is in turn at most n . Hence, in the typical case where $S_{\max} = 1$ and we constrain $\bar{\theta}_{\min} = O(n)$, Corollary 6.8 bounds the error by a term of size $\tilde{O}_p(n^{3/4})$.

In the previous example, Θ is finite, since we can have at most n clusters. We next provide an example where Θ is not finite, but the induced set of plug-in policies is finite.

Example F.2 (Kernel Smoothers Revisited) Consider the kernel regression policy from Example 2.7 with the box-kernel with bandwidth $\bar{\theta}$, i.e., $K_{\bar{\theta}}(\mathbf{W}_j, \mathbf{W}_k) = \mathbb{I}\{\|\mathbf{W}_j - \mathbf{W}_k\| \leq \bar{\theta}\}$. Let $\mathbf{x}^{KR}(\mathbf{Z}, \bar{\theta})$ be the corresponding plug-in policy. Consider the policy class $\mathcal{X}^{KR}(\mathbf{Z}) = \{\mathbf{x}^{KR}(\mathbf{Z}, \bar{\theta}) : 0 \leq \bar{\theta} \leq \bar{\theta}_{\max}\}$.

Then,

$$T_{\max} = \max_j \sum_{i=1}^n \max_{\bar{\theta} \in [0, \bar{\theta}_{\max}]} \mathbb{I}\{K_{\bar{\theta}}(\mathbf{W}_j, \mathbf{W}_i) \neq 0\} = \max_j \sum_{i=1}^n \mathbb{I}\{\|\mathbf{W}_j - \mathbf{W}_i\| \leq \bar{\theta}_{\max}\},$$

which decreases with $\bar{\theta}_{\max}$. In other words, for small enough $\bar{\theta}_{\max}$, i.e., “local”, neighborhoods, $T_{\max}$ can be made $O(1)$.

That said, $[0, \bar{\theta}]$ is an infinite set. However, the set of kernel weights induced by $\theta \in [0, \bar{\theta}]$ is finite. Namely, partition the real number line at the $O(n^2)$ points $\|\mathbf{W}_i - \mathbf{W}_j\|$. Then, for any θ_1, θ_2 in the same interval, $K_{\theta_1}(\mathbf{W}_i, \mathbf{W}_j) = K_{\theta_2}(\mathbf{W}_i, \mathbf{W}_j)$. Letting $\mathcal{T}_{\Theta}^{KR}(\mathbf{Z}) = \{\mathbf{T}^{KR}(\mathbf{Z}, \theta) : \theta \in \Theta\}$, we see $|\mathcal{T}_{\Theta}^{KR}(\mathbf{Z})| \leq n(n+1)/2 + 1$ by counting the number of intervals. We can then bound $\mathcal{X}_{\max}$ as follows,

$$|\mathcal{X}_{\Theta}^{KR}(\mathbf{Z})| = |\{\hat{\mathbf{x}}(\mathbf{t}) : \mathbf{t} \in \mathcal{T}_{\Theta}^{KR}(\mathbf{Z})\}| \leq |\mathcal{T}_{\Theta}^{KR}(\mathbf{Z})| \leq \frac{n(n+1)}{2} + 1.$$

In settings where neither Θ or $\mathcal{T}_{\Theta}(\mathbf{Z})$ are finite, Gupta, Huang, and Rusmevichientong (2022a) describe how to bound $\mathcal{X}_{\max}$ under a general assumption on the plug-in class. Roughly, they show that $\|\mathcal{X}_{\Theta}(\mathbf{Z})\| = O(n^{\dim(\Theta)})$ for separable affine plug-in classes, or in words, they show $\mathcal{X}_{\max}$ is polynomial in n if the dimension of $\boldsymbol{\theta} \in \Theta$ is fixed. Similar techniques can be applied to bound the affine plug-ins considered in our paper. We omit the details.

F.2. Proof of Lemma 6.6 — Bounding Solution Stability

We bound the solution stability by decomposing it into the following components,

$$\begin{aligned}
& \sup_{\theta \in \Theta} \left| \xi^\top \left(\hat{x}(T(Z, \bar{\theta})) - \hat{x}(T(Z, \bar{\theta}) + h\xi) \right) \right| \\
& \leq \sup_{\bar{\theta} \in \Theta} \left| \mathbb{E} \left[\xi^\top \left(\hat{x}(T(Z, \bar{\theta})) - \hat{x}(T(Z, \bar{\theta}) + h\xi) \right) \right] \right| \\
& \quad + \sup_{\theta \in \Theta} \left| \xi^\top \hat{x}(T(Z, \bar{\theta})) - \mathbb{E} [\xi^\top x(T(Z, \bar{\theta}))] \right| \\
& \quad + \sup_{\theta \in \Theta} \left| \xi^\top \hat{x}(T(Z, \bar{\theta}) + h\xi) - \mathbb{E} [\xi^\top \hat{x}(T(Z, \bar{\theta}) + h\xi)] \right| \\
& \leq \sup_{\bar{\theta} \in \Theta} \left| \mathbb{E} \left[\xi^\top \left(\hat{x}(T(Z, \bar{\theta})) - \hat{x}(T(Z, \bar{\theta}) + h\xi) \right) \right] \right| \tag{25a}
\end{aligned}$$

$$\begin{aligned}
& + 2 \sup_{\substack{\theta \in \Theta \\ h' \in \{0, h\}}} \left| \xi^\top \hat{x}(T(Z, \bar{\theta}) + h'\xi) - \mathbb{E} [\xi^\top \hat{x}(T(Z, \bar{\theta}) + h'\xi)] \right| \tag{25b}
\end{aligned}$$

We bound Eq. (25a) in Lemma F.4 and bound Eq. (25b) by invoking Lemma F.5. Combining the two bounds gives us our bound on the solution stability given in the lemma.

Before executing this strategy, we prove a short lemma bounding the total variation distance between two random variables of interest. When we invoke the lemma, the role of $\mathbf{L}$ will be played by $\mathbf{L}^k$, ξ by ξ^k , and d by $|S_k|$. We phrase the lemma without this k notation for simplicity.

Lemma F.3 (Bounding TV under Linear Shifts) *Let $\mathbf{L} \in \mathbb{R}^{d \times d}$ be invertible with $\sigma_{\min}(\mathbf{L}) > 0$. Let ξ be a random vector in $\mathbb{R}^d$ with density p such that $\mathbb{E} [\|\xi\|^2] < \infty$ and $\mathbb{E} [\|\nabla \log p(\xi)\|^2] < \infty$. Then, for $0 < h < \sigma_{\min}(\mathbf{L})/2$,*

$$\text{TV}(\mathbf{L}\xi, (\mathbf{L} + h\mathbf{I})\xi) \leq \frac{d + \sqrt{\mathbb{E} [\|\xi\|^2] \mathbb{E} [\|\nabla \log p(\xi)\|^2]}}{\sigma_{\min}(\mathbf{L})} h.$$

Proof. Set $\mathbf{A}_t := \mathbf{L} + t\mathbf{I}$ and let q_t denote the density of $\mathbf{A}_t\xi$, that is, $q_t(y) = p(\mathbf{A}_t^{-1}y)/|\det \mathbf{A}_t|$. Our goal is to apply Lemma A.3 to $\{q_t\}_{t \in [0, h]}$. To that end, we differentiate $q_t(y)$ with respect to t .

Standard matrix calculus yields

$$\begin{aligned}
\frac{d}{dt} \mathbf{A}_t^{-1} &= -\mathbf{A}_t^{-1} \left(\frac{d\mathbf{A}_t}{dt} \right) \mathbf{A}_t^{-1} = -\mathbf{A}_t^{-2}, \\
\frac{d}{dt} \log |\det \mathbf{A}_t| &= \text{trace}(\mathbf{A}_t^{-1}).
\end{aligned}$$

Using these identities, the product and chain rules gives

$$\partial_t q_t(y) = \frac{1}{|\det \mathbf{A}_t|} \left[-\nabla p(\mathbf{A}_t^{-1}y) \cdot (\mathbf{A}_t^{-2}y) - p(\mathbf{A}_t^{-1}y) \text{trace}(\mathbf{A}_t^{-1}) \right].$$

Taking the L^1 -norm and changing variables $y = \mathbf{A}_t z$ (so $dy = |\det \mathbf{A}_t| dz$), we obtain

$$\begin{aligned}
\|\partial_t q_t\|_{L^1} &\leq \int \|\nabla p(z)\| \|\mathbf{A}_t^{-1}z\| dz + |\text{trace}(\mathbf{A}_t^{-1})| \\
&\leq \|\mathbf{A}_t^{-1}\|_2 C_p + d \|\mathbf{A}_t^{-1}\|_2,
\end{aligned}$$

where we used $|\text{trace}(\mathbf{B})| \leq d\|\mathbf{B}\|_2$ and defined $C_p := \int \|\nabla p(z)\| \|z\| dz$.

Applying Lemma A.3, we have

$$\text{TV}(\mathbf{L}\boldsymbol{\xi}, (\mathbf{L} + h\mathbf{I})\boldsymbol{\xi}) \leq \frac{1}{2} \int_0^h (C_p + d) \|\mathbf{A}_t^{-1}\|_2 dt.$$

To control $\|\mathbf{A}_t^{-1}\|_2$, we use Weyl's inequality in perturbation form for singular values: for each $t \geq 0$, $|\sigma_{\min}(\mathbf{L} + t\mathbf{I}) - \sigma_{\min}(\mathbf{L})| \leq \|t\mathbf{I}\|_2 = t$. This implies

$$\sigma_{\min}(\mathbf{A}_t) = \sigma_{\min}(\mathbf{L} + t\mathbf{I}) \geq \sigma_{\min}(\mathbf{L}) - t,$$

and hence

$$\|\mathbf{A}_t^{-1}\|_2 = \frac{1}{\sigma_{\min}(\mathbf{A}_t)} \leq \frac{1}{\sigma_{\min}(\mathbf{L}) - t}.$$

Therefore,

$$\text{TV}(\mathbf{L}\boldsymbol{\xi}, (\mathbf{L} + h\mathbf{I})\boldsymbol{\xi}) \leq \frac{1}{2} (C_p + d) \int_0^h \frac{dt}{\sigma_{\min}(\mathbf{L}) - t}.$$

For $0 < h < \sigma_{\min}(\mathbf{L})/2$,

$$\begin{aligned} \int_0^h \frac{dt}{\sigma_{\min}(\mathbf{L}) - t} &= \log\left(\frac{\sigma_{\min}(\mathbf{L})}{\sigma_{\min}(\mathbf{L}) - h}\right) \\ &\leq \frac{h}{\sigma_{\min}(\mathbf{L}) - h} \leq \frac{2h}{\sigma_{\min}(\mathbf{L})}. \end{aligned}$$

Finally, we bound C_p explicitly by Cauchy–Schwarz:

$$\begin{aligned} C_p &= \int \|\nabla p(z)\| \|z\| dz = \int p(z) \|\nabla \log p(z)\| \|z\| dz \\ &\leq \left(\int p(z) \|\nabla \log p(z)\|^2 dz \right)^{1/2} \left(\int p(z) \|z\|^2 dz \right)^{1/2} \quad (\text{Cauchy–Schwarz}) \\ &= \sqrt{\mathbb{E}[\|\nabla \log p(\boldsymbol{\xi})\|^2]} \sqrt{\mathbb{E}[\|\boldsymbol{\xi}\|^2]}. \end{aligned}$$

Combining the inequalities and the bound for C_p , we obtain

$$\text{TV}(\mathbf{L}\boldsymbol{\xi}, (\mathbf{L} + h\mathbf{I})\boldsymbol{\xi}) \leq \frac{C_p + d}{\sigma_{\min}(\mathbf{L})} h \leq \frac{d + \sqrt{\mathbb{E}[\|\boldsymbol{\xi}\|^2]} \sqrt{\mathbb{E}[\|\nabla \log p(\boldsymbol{\xi})\|^2]}}{\sigma_{\min}(\mathbf{L})} h.$$

This concludes the proof.

F.3. Bound on Eq. (25a)

Lemma F.4 (Expected Solution Stability) *Suppose Assumptions 3.2 and 6.2 to 6.4 hold. Then, for any $0 \leq h \leq \min(\sigma_{\min}/2, S_{\max}^{-1})$, there exists a constant C (depending on $\nu_{\min}, \nu_{\max}, \sigma_{\min}, \kappa, C_{\xi}$) such that,*

$$\sup_{\bar{\boldsymbol{\theta}} \in \bar{\Theta}} \left| \mathbb{E} \left[\boldsymbol{\xi}^\top \left(\hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}})) - \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h\boldsymbol{\xi}) \right) \right] \right| \leq CS_{\max} nh \sqrt{\log \left(\frac{1}{h} \right)}$$

Proof for Lemma F.4. Fix $\bar{\boldsymbol{\theta}}$ and suppress it in the notation. Rewrite the argument of the supremum as a sum over blocks S_k :

$$\begin{aligned} & \left| \sum_{k=1}^K \mathbb{E} \left[(\boldsymbol{\xi}^k)^\top \left(\hat{\mathbf{x}}^k(\mathbf{T}(\mathbf{Z})) - \hat{\mathbf{x}}^k(\mathbf{T}(\mathbf{Z}) + h\boldsymbol{\xi}^k) \right) \right] \right| \\ &= \left| \sum_{k=1}^K \mathbb{E} \left[(\boldsymbol{\xi}^k)^\top \left(\hat{\mathbf{x}}^k(\mathbf{T}^k(\mathbf{Z})) - \hat{\mathbf{x}}^k(\mathbf{T}^k(\mathbf{Z}) + h\boldsymbol{\xi}^k) \right) \right] \right| \\ &= \left| \sum_{k=1}^K \mathbb{E} \left[\mathbb{E} \left[(\boldsymbol{\xi}^k)^\top \left(\hat{\mathbf{x}}^k(\mathbf{T}^k(\mathbf{Z})) - \hat{\mathbf{x}}^k(\mathbf{T}^k(\mathbf{Z}) + h\boldsymbol{\xi}^k) \right) \middle| \bar{\mathbf{Z}}^k \right] \right] \right| \end{aligned}$$

where $\bar{\mathbf{Z}}^k \in \mathbb{R}^{n-|S_k|}$ correspond to the components of $\mathbf{Z}$ not in S_k . The first equality holds since the k^{th} block only depends on the objective cost components in the k^{th} block and the second by law of iterated expectations.

We now focus on bounding the conditional expectation

$$\mathbb{E} \left[(\boldsymbol{\xi}^k)^\top \left(\hat{\mathbf{x}}^k(\mathbf{T}^k(\mathbf{Z})) - \hat{\mathbf{x}}^k(\mathbf{T}^k(\mathbf{Z}) + h\boldsymbol{\xi}^k) \right) \middle| \bar{\mathbf{Z}}^k \right] \quad (26)$$

for a fixed k . We first note

$$\mathbf{T}^k(\mathbf{Z}) = \mathbf{e}^k \circ (\mathbf{L}\mathbf{Z} + \mathbf{l}) = \mathbf{L}^k \mathbf{Z}^k + \bar{\mathbf{L}}^k \bar{\mathbf{Z}}^k + \mathbf{l}^k = \mathbf{L}^k \boldsymbol{\xi}^k + \mathbf{L}^k \boldsymbol{\mu}^k + \bar{\mathbf{L}}^k \bar{\mathbf{Z}}^k + \mathbf{l}^k$$

where $\mathbf{e}^k$ is a vector where component j is 1 if $j \in S_k$ and 0 otherwise. The matrix $\mathbf{L}^k \in \mathbb{R}^{|S_k| \times |S_k|}$ is the square submatrix induced by the rows and columns in S_k . (Recall, this matrix is invertible by Assumption 6.3.) The matrix $\bar{\mathbf{L}}^k \in \mathbb{R}^{|S_k| \times (n-|S_k|)}$ is the matrix with rows corresponding S_k and columns corresponding to S_k^c . We then fix a k and $\bar{\mathbf{Z}}^k$, and temporarily suppress k in the notation. We define $\mathbf{y}_0 = \mathbf{L}\boldsymbol{\xi}$, $\mathbf{y}_h = \mathbf{L}\boldsymbol{\xi} + h\boldsymbol{\xi}$ and $g(\mathbf{y}) = \hat{\mathbf{x}}(\mathbf{y} + \mathbf{L}\boldsymbol{\mu} + \bar{\mathbf{L}}\bar{\mathbf{Z}} + \mathbf{l}) \in \mathbb{R}^{|S_k|}$. Then, the conditional expectation Eq. (26) is

$$\begin{aligned} & \left| \mathbb{E} \left[(\mathbf{L}^{-1} \mathbf{y}_0)^\top g(\mathbf{y}_0) \right] - \mathbb{E} \left[((\mathbf{L} + h\mathbf{I})^{-1} \mathbf{y}_h)^\top g(\mathbf{y}_h) \right] \right| \\ & \leq \underbrace{\left| \mathbb{E} \left[((\mathbf{L}^{-1} - (\mathbf{L} + h\mathbf{I})^{-1}) \mathbf{y}_h)^\top g(\mathbf{y}_h) \right] \right|}_{\text{Term (i)}} + \underbrace{\left| \mathbb{E} \left[(\mathbf{L}^{-1} \mathbf{y}_0)^\top g(\mathbf{y}_0) \right] - \mathbb{E} \left[(\mathbf{L}^{-1} \mathbf{y}_h)^\top g(\mathbf{y}_h) \right] \right|}_{\text{Term (ii)}} \end{aligned}$$

Bounding Term (i)

Since $\|g(\mathbf{y})\|_\infty \leq 1$, we have that

$$\begin{aligned} \left| \left[\left((\mathbf{L}^{-1} - (\mathbf{L} + h\mathbf{I})^{-1}) \mathbf{y}_h \right)^\top g(\mathbf{y}_h) \right] \right| &\leq \| (\mathbf{L}^{-1} - (\mathbf{L} + h\mathbf{I})^{-1}) \mathbf{y}_h \|_1 = h \| \mathbf{L}^{-1} \boldsymbol{\xi} \|_1 \\ &\leq h \sqrt{|S_k|} \| \mathbf{L}^{-1} \boldsymbol{\xi} \|_2 \leq \frac{h \sqrt{|S_k|}}{\sigma_{\min}} \| \boldsymbol{\xi} \|_2 \end{aligned}$$

where last inequality holds by definition of the operator norm on the matrix $\mathbf{L}^{-1}$.

Hence, we have

$$\text{Term (i)} = \left| \mathbb{E} \left[\left((\mathbf{L}^{-1} - (\mathbf{L} + h\mathbf{I})^{-1}) \mathbf{y}_h \right)^\top g(\mathbf{y}_h) \right] \right| \leq \frac{h \sqrt{|S_k|}}{\sigma_{\min}} \mathbb{E} [\| \boldsymbol{\xi} \|_2] \leq \frac{h |S_k|}{\sigma_{\min} \sqrt{\nu_{\min}}}.$$

Bounding Term (ii)

To bound the second term, we would like to appeal to a total variation argument, however the terms in the expectation are not bounded. We thus consider truncating the terms at U (a constant we will determine later).

$$\begin{aligned} &\left| \mathbb{E} \left[(\mathbf{L}^{-1} \mathbf{y}_0)^\top g(\mathbf{y}_0) \right] - \mathbb{E} \left[(\mathbf{L}^{-1} \mathbf{y}_h)^\top g(\mathbf{y}_h) \right] \right| \\ &\leq \left| \mathbb{E} \left[(\mathbf{L}^{-1} \mathbf{y}_h)^\top g(\mathbf{y}_h) \mathbb{I} \{ \| \mathbf{L}^{-1} \mathbf{y}_h \|_1 > U \} \right] \right| \end{aligned} \quad (27a)$$

$$+ \left| \mathbb{E} \left[(\mathbf{L}^{-1} \mathbf{y}_0)^\top g(\mathbf{y}_0) \mathbb{I} \{ \| \mathbf{L}^{-1} \mathbf{y}_0 \|_1 > U \} \right] \right| \quad (27b)$$

$$+ \left| \mathbb{E} \left[(\mathbf{L}^{-1} \mathbf{y}_0)^\top g(\mathbf{y}_0) \mathbb{I} \{ \| \mathbf{L}^{-1} \mathbf{y}_0 \|_1 \leq U \} \right] - \mathbb{E} \left[(\mathbf{L}^{-1} \mathbf{y}_h)^\top g(\mathbf{y}_h) \mathbb{I} \{ \| \mathbf{L}^{-1} \mathbf{y}_h \|_1 \leq U \} \right] \right| \quad (27c)$$

We first bound Eq. (27a). By Holder's inequality, we have

$$\left| \mathbb{E} \left[(\mathbf{L}^{-1} \mathbf{y}_h)^\top g(\mathbf{y}_h) \mathbb{I} \{ \| \mathbf{L}^{-1} \mathbf{y}_h \|_1 > U \} \right] \right| \leq \mathbb{E} \left[\left((\mathbf{L}^{-1} \mathbf{y}_h)^\top g(\mathbf{y}_h) \right)^2 \right]^{1/2} \mathbb{P} \{ \| \mathbf{L}^{-1} \mathbf{y}_h \|_1 > U \}^{1/2}. \quad (28)$$

To bound Eq. (28), we first bound the term inside the expectation. Applying Holder's inequality again shows

$$\left((\mathbf{L}^{-1} \mathbf{y}_h)^\top g(\mathbf{y}_h) \right)^2 \leq \| \mathbf{L}^{-1} \mathbf{y}_h \|_1^2 \| g(\mathbf{y}_h) \|_\infty^2 \leq \| \mathbf{L}^{-1} \mathbf{y}_h \|_1^2,$$

where the second holds since $\|g(\mathbf{y}_h)\|_\infty \leq 1$. For $0 \leq h \leq 1$ and $\sigma_{\min} \leq 1$, the last term can be further bounded as follows,

$$\begin{aligned} \| \mathbf{L}^{-1} \mathbf{y}_h \|_1^2 &= \| \mathbf{L}^{-1} (\mathbf{L} + h\mathbf{I}) \boldsymbol{\xi} \|_1^2 \\ &\leq 2 \| \boldsymbol{\xi} \|_1^2 + 2h^2 \| \mathbf{L}^{-1} \boldsymbol{\xi} \|_1^2 && (\text{Triangle Inequality and } (a+b)^2 \leq 2a^2 + 2b^2) \\ &\leq 2 |S_k| \| \boldsymbol{\xi} \|_2^2 + 2 |S_k| h^2 \| \mathbf{L}^{-1} \boldsymbol{\xi} \|_2^2 \\ &\leq 2 |S_k| \| \boldsymbol{\xi} \|_2^2 + \frac{2 |S_k| h^2}{\sigma_{\min}^2} \| \boldsymbol{\xi} \|_2^2 \\ &\leq \frac{4 |S_k|}{\sigma_{\min}^2} \| \boldsymbol{\xi} \|_2^2, && (\text{since } \sigma_{\min} \leq 1, h \leq 1). \end{aligned}$$

Applying this bound to the the probability term as well, we have

$$\mathbb{P} \left\{ \|\mathbf{L}^{-1} \mathbf{y}_h\|_1 > U \right\}^{1/2} \leq \mathbb{P} \left\{ \frac{2\sqrt{|S_k|}}{\sigma_{\min}} \|\boldsymbol{\xi}\|_2 > U \right\}^{1/2} = \mathbb{P} \left\{ \|\boldsymbol{\xi}\|_2 > \frac{\sigma_{\min} U}{2\sqrt{|S_k|}} \right\}^{1/2}.$$

By Gupta and Rusmevichientong (2021, Lemma A.1), $\|\boldsymbol{\xi}\|_2$ is subGaussian with variance proxy at most a constant times $\kappa^2 |S_k|$. Thus,

$$\mathbb{P} \left\{ \|\boldsymbol{\xi}\|_2 > \frac{\sigma_{\min} U}{2\sqrt{|S_k|}} \right\} \leq 2 \exp \left(-C_2 \frac{\sigma_{\min}^2 U^2}{\kappa^2 |S_k|^2} \right),$$

for some constant C_2 .

Plugging in our bounds to Eq. (28), we have

$$\text{Eq. (27a)} \leq \mathbb{E} \left[\frac{4|S_k|}{\sigma_{\min}^2} \|\boldsymbol{\xi}\|_2^2 \right]^{1/2} \mathbb{P} \left\{ \|\boldsymbol{\xi}\|_2 > \frac{\sigma_{\min} U}{2\sqrt{|S_k|}} \right\}^{1/2} \leq \frac{C_3 |S_k|}{\sigma_{\min} \sqrt{\nu_{\min}}} \exp \left(\frac{-\sigma_{\min}^2 U^2}{C_3 \kappa^2 |S_k|^2} \right),$$

for some constant C_3 .

To bound Eq. (27b), we can repeat the same argument but take $h = 0$ throughout. Consequently,

$$\text{Eq. (27b)} \leq \frac{C_3 |S_k|}{\sigma_{\min} \sqrt{\nu_{\min}}} \exp \left(\frac{-\sigma_{\min}^2 U^2}{C_3 \kappa^2 |S_k|^2} \right).$$

Combining the two bounds, we have for $\sigma_{\min} \leq 1$ and $0 \leq h \leq 1$,

$$\text{Eq. (27a)} + \text{Eq. (27b)} \leq \frac{2C_3 |S_k|}{\sigma_{\min} \sqrt{\nu_{\min}}} \exp \left(\frac{-\sigma_{\min}^2 U^2}{C_3 \kappa^2 |S_k|^2} \right).$$

Finally, we bound Eq. (27c) using a Total Variation argument. We first note the function $\mathbf{y} \mapsto (\mathbf{L}^{-1} \mathbf{y})^\top g(\mathbf{y}) \mathbb{I} \{ \|\mathbf{L}^{-1} \mathbf{y}\|_1 \leq U \}$ is bounded by U for all $\mathbf{y}$ as

$$\left| (\mathbf{L}^{-1} \mathbf{y})^\top g(\mathbf{y}) \mathbb{I} \{ \|\mathbf{L}^{-1} \mathbf{y}_h\|_1 \leq U \} \right| \leq \mathbb{I} \{ \|\mathbf{L}^{-1} \mathbf{y}\|_1 \leq U \} \|\mathbf{L}^{-1} \mathbf{y}\|_1 \|g(\mathbf{y})\|_\infty \leq U.$$

Thus, we have

$$\text{Eq. (27c)} \leq U \cdot \text{TV}(\mathbf{y}_0, \mathbf{y}_h) = U \cdot \text{TV}(\mathbf{L}\boldsymbol{\xi}, (\mathbf{L} + h\mathbf{I})\boldsymbol{\xi}),$$

where $\text{TV}(\cdot, \cdot)$ denotes total variation distance. Our goal is to apply Lemma F.3.

To that end, we verify its assumptions. We reintroduce k in the notation for clarity. By Assumption 6.3, $\mathbf{L}^k$ is invertible and $\sigma_{\min}(\mathbf{L}^k) \geq \sigma_{\min} > 0$. Furthermore, $\mathbb{E}[\|\boldsymbol{\xi}^k\|^2] \leq |S_k|/\nu_{\min}$. Thus, applying Lemma F.3 and simplifying yields

$$\text{Eq. (27c)} \leq \frac{Uh|S_k|}{\sigma_{\min}} \left(1 + \sqrt{\frac{\mathbb{E}[\|\nabla \log p(\boldsymbol{\xi}^k)\|^2]}{|S_k| \nu_{\min}}} \right).$$

Putting it all together in Eq. (27), we have

$$\text{Term (ii)} \leq C_4 U h |S_k| \left(1 + \sqrt{\frac{\mathbb{E}[\|\nabla \log p(\boldsymbol{\xi}^k)\|^2]}{|S_k| \nu_{\min}}} \right) + C_4 |S_k| \exp \left(\frac{-U^2}{C_4^2 |S_k|^2} \right).$$

for some constant C_4 (depending on $\nu_{\min}$, $\nu_{\max}$, $\sigma_{\min}$, κ). Choosing $U = C_4 |S_k| \sqrt{\log h^{-1}}$ and collecting constants shows

$$\begin{aligned} \text{Term (ii)} &\leq C_5 \sqrt{\log(1/h)} |S_k|^2 h \left(1 + \sqrt{\frac{\mathbb{E} [\|\nabla \log p(\boldsymbol{\xi}^k)\|^2]}{|S_k| \nu_{\min}}} \right) + C_5 |S_k| h, \\ &\leq C_6 h \sqrt{\log(1/h)} |S_k|^2 \left(1 + \sqrt{\frac{\mathbb{E} [\|\nabla \log p(\boldsymbol{\xi}^k)\|^2]}{|S_k| \nu_{\min}}} \right) \end{aligned}$$

for some C_5, C_6 (depending on $\nu_{\min}$, $\nu_{\max}$, $\sigma_{\min}$, κ).

Combining Term (i) and Term (ii) and summing across k , we have

$$\begin{aligned} \text{Term (i)} + \text{Term (ii)} &\leq C_7 h \sum_{k=1}^K |S_k| + C_7 h \sqrt{\log\left(\frac{1}{h}\right)} \sum_{k=1}^K |S_k|^2 \left(1 + \sqrt{\frac{\mathbb{E} [\|\nabla \log p(\boldsymbol{\xi}^k)\|^2]}{|S_k| \nu_{\min}}} \right) \\ &\leq C_8 h \sqrt{\log\left(\frac{1}{h}\right)} \sum_{k=1}^K |S_k|^2 \sqrt{\frac{\mathbb{E} [\|\nabla \log p(\boldsymbol{\xi}^k)\|^2]}{|S_k|}}. \end{aligned}$$

Focusing on the sum,

$$\begin{aligned} \sum_{k=1}^K |S_k|^2 \sqrt{\frac{\mathbb{E} [\|\nabla \log p(\boldsymbol{\xi}^k)\|^2]}{|S_k|}} &= \sum_{k=1}^K |S_k|^{3/2} \sqrt{\mathbb{E} [\|\nabla \log p(\boldsymbol{\xi}^k)\|^2]} \\ &\leq \sqrt{\sum_{k=1}^K |S_k|^3 \cdot \sum_{k=1}^K \mathbb{E} [\|\nabla \log p(\boldsymbol{\xi}^k)\|^2]} \\ &\leq S_{\max} \sqrt{n \mathbb{E} [\|\nabla \log p(\boldsymbol{\xi})\|_2^2]}, \end{aligned}$$

where we've used the independence of the components of $\boldsymbol{\xi}$ to rewrite the sum of componentwise informations as the total information.

Substituting above shows

$$\text{Term (i)} + \text{Term (ii)} \leq C_8 h \sqrt{\log\left(\frac{1}{h}\right)} S_{\max} \sqrt{n \mathbb{E} [\|\nabla \log p(\boldsymbol{\xi})\|_2^2]}.$$

for a constant C_6 . Invoking Assumption 6.4 completes the proof. $\square$

F.4. Bound on Eq. (25b)

As described in the main body, our approach will be to use a coloring of $G(V, E)$ to group subproblems by color, and then take a union bound over colors. Hence, we first prove Lemma 6.5 which establishes components of the same color are independent.

Proof of Lemma 6.5 By contradiction, suppose these subproblems were dependent. Then, there must exist an $i \in \mathcal{S}_k$ and a $j \in \mathcal{S}_{\bar{k}}$ such that $T_i(\mathbf{Z}, \bar{\boldsymbol{\theta}})$ and $T_j(\mathbf{Z}, \bar{\boldsymbol{\theta}})$ both depend on some component ξ_l , which suggests $e(k, \bar{k})$ is present, and hence k and $\bar{k}$ cannot be the same color. This is a contradiction. $\square$

Lemma F.5 (Convergence of Solution Stability to Expectation) *Suppose Assumption 3.2 holds. Then, there exists a constant C such that with probability at least $1 - \epsilon$,*

$$\begin{aligned} & \sup_{\substack{\bar{\theta} \in \bar{\Theta} \\ h' \in \{0, h\}}} \left| \boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h' \boldsymbol{\xi}) - \mathbb{E} [\boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h' \boldsymbol{\xi})] \right| \\ & \leq C \kappa S_{\max} \sqrt{\chi(G) n \log(\mathcal{X}_{\max})} \log(\chi(G)) \log(1/\epsilon) \\ & \leq C S_{\max}^{3/2} \sqrt{n T_{\max} \log \mathcal{X}_{\max}} \cdot \log \left(\frac{S_{\max} T_{\max}}{\epsilon} \right) \end{aligned}$$

Proof for Lemma F.5. Let $R_1, \dots, R_{\chi(G)}$ be the partition of V corresponding to the different colors. We first decompose Eq. (25b) into the sets determined by the vertex coloring as follows,

$$\begin{aligned} & \sup_{\substack{\bar{\theta} \in \bar{\Theta} \\ h' \in \{0, h\}}} \left| \boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h' \boldsymbol{\xi}) - \mathbb{E} [\boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h' \boldsymbol{\xi})] \right| \\ & \leq \sum_{l=1}^{\chi(G)} \sup_{\substack{\bar{\theta} \in \bar{\Theta} \\ h' \in \{0, h\}}} \sum_{k \in R_l} (\boldsymbol{\xi}^k)^\top \hat{\mathbf{x}}^k(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h' \boldsymbol{\xi}) - \mathbb{E} [(\boldsymbol{\xi}^k)^\top \hat{\mathbf{x}}^k(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h' \boldsymbol{\xi})]. \end{aligned}$$

We bound the l^{th} term of the outer summation using Theorem A.1 of Gupta, Huang, and Rusmevichientong (2022a) since $f_k(\mathbf{Z}, \bar{\theta}, h') = (\boldsymbol{\xi}^k)^\top \hat{\mathbf{x}}^k(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h' \boldsymbol{\xi})$ for $k \in R_l$ are independent. Fixing an l , we take

$$\mathbf{F}(\mathbf{Z}) = (\|\boldsymbol{\xi}^k\|_1)_{k \in R_l} \in \mathbb{R}^{|R_l|}$$

We can bound the Orlicz norm as follows,

$$\|\|\mathbf{F}(\mathbf{Z})\|_2\|_\Psi \leq \left\| \sqrt{S_{\max}} \|(\boldsymbol{\xi}^k)_{k \in R_l}\|_2 \right\|_\Psi \leq \sqrt{S_{\max}} \kappa \sqrt{2 S_{\max} |R_l|} = S_{\max} \kappa \sqrt{2 |R_l|}$$

where the first inequality holds by noting $\|\boldsymbol{\xi}^k\|_1 \leq \sqrt{S_{\max}} \|\boldsymbol{\xi}^k\|_2$ and the second inequality holds by Gupta and Rusmevichientong (2021, Lemma A.1).

Note that

$$\left| \left\{ ((\boldsymbol{\xi}^k)^\top \hat{\mathbf{x}}^k(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h' \boldsymbol{\xi}))_{k \in R_l} : \bar{\theta} \in \bar{\Theta}, h' \in \{0, h\} \right\} \right| \leq 2 \mathcal{X}_{\max}.$$

Now apply Gupta and Rusmevichientong (2021, Theorem A.1) to show that there exists an absolute constant C such that probability at least $1 - \epsilon$,

$$\begin{aligned} & \sup_{\substack{\bar{\theta} \in \bar{\Theta} \\ h' \in \{0, h\}}} \left| \sum_{k \in R_l} (\boldsymbol{\xi}^k)^\top \hat{\mathbf{x}}^k(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h' \boldsymbol{\xi}) - \mathbb{E} [(\boldsymbol{\xi}^k)^\top \hat{\mathbf{x}}^k(\mathbf{T}(\mathbf{Z}, \bar{\theta}) + h' \boldsymbol{\xi})] \right| \\ & \leq C S_{\max} \kappa \sqrt{|R_l| \log(\mathcal{X}_{\max})} \log \left(\frac{1}{\epsilon} \right). \end{aligned}$$

To complete the proof, we take a union bound over l allowing us to show that with probability $1 - \epsilon$

$$\begin{aligned}
& \sup_{\substack{\bar{\boldsymbol{\theta}} \in \bar{\Theta} \\ h' \in \{0, h\}}} |\boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h' \boldsymbol{\xi}) - \mathbb{E} [\boldsymbol{\xi}^\top \hat{\mathbf{x}}(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + h' \boldsymbol{\xi})]| \\
& \leq \sum_{l=1}^{\chi(G)} C S_{\max} \kappa \sqrt{|R_l| \log(\mathcal{X}_{\max})} \log \left(\frac{\chi(G)}{\epsilon} \right). \\
& \leq C \chi(G) S_{\max} \kappa \sqrt{\frac{K}{\chi(G)} \log(\mathcal{X}_{\max})} \log \left(\frac{\chi(G)}{\epsilon} \right) \\
& \leq C \kappa S_{\max} \sqrt{\chi(G) n \log(\mathcal{X}_{\max})} \log \left(\frac{\chi(G)}{\epsilon} \right)
\end{aligned}$$

where the first inequality is the union bound, and the second inequality uses $\sum_{l=1}^{\chi(G)} |R_l| = K$ and Jensen's inequality. Collecting constants proves the first part of the lemma.

For the second, by Brook's Theorem, $\chi(G)$ is at most the maximal degree of $G(V, E)$ plus one. Let $T_{\max} = \max_j \sum_{i=1}^n \sup_{\bar{\boldsymbol{\theta}} \in \bar{\Theta}} \mathbb{I} \{ \mathbf{L}_{ij}(\bar{\boldsymbol{\theta}}) \neq 0 \}$. Then we claim that the maximal degree of $G(V, E)$ is at most $S_{\max} T_{\max}$. To see this, consider S_1 . For each $k \in S_1$, $T_k(\mathbf{Z}, \bar{\boldsymbol{\theta}})$ can depend on at most $T_{\max}$ different components of $\boldsymbol{\xi}$. In a worst-case, these components each belong to different subproblems, creating at most $T_{\max}$ outgoing edges from node 1. This is true for each $k \in S_1$, so S_1 has an outdegree of at most $|S_1| T_{\max}$. This is true for each subproblem, yielding the upperbound $S_{\max} T_{\max}$. Substituting above collecting constants yields the second part of the lemma. $\square$

F.5. Proof for Block Decoupled Estimation Error, Theorem 6.7

We apply Theorem 4.2 and use Lemma 6.6 to bound the solution stability term. We complete the bound by obtaining C_T for the affine plug-in class. We see

$$\begin{aligned}
\|\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) - \mathbf{T}(\mathbf{Y}, \bar{\boldsymbol{\theta}})\|_1 &= \|\mathbf{L}(\bar{\boldsymbol{\theta}})(\mathbf{Z} - \mathbf{Y})\|_1 \\
&\leq \sum_{j=1}^n \|\mathbf{L}_j(\bar{\boldsymbol{\theta}})(Z_j - Y_j)\|_1 = \left(\max_j \|\mathbf{L}_j(\bar{\boldsymbol{\theta}})\|_1 \right) \|\mathbf{Z} - \mathbf{Y}\|_1 \leq T_{\max} L_{\max} \|\mathbf{Z} - \mathbf{Y}\|_1,
\end{aligned}$$

so $C_T = T_{\max} L_{\max}$.

Appendix G: Weakly-Coupled Policy Proofs

Proof for Theorem 6.12 We can bound the estimation error for weakly-coupled problems as follows,

$$\begin{aligned}
& \sup_{\boldsymbol{\theta} \in \Theta} |\boldsymbol{\xi}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - D^{WC}(\mathbf{Z}, \boldsymbol{\theta}, h)| \\
& \leq \sup_{\boldsymbol{\theta} \in \Theta} |\boldsymbol{\xi}^\top (\mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - \tilde{\mathbf{x}}(\mathbf{Z}, \boldsymbol{\lambda}(\mathbf{Z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta})), \mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta}), \bar{\boldsymbol{\theta}}))|
\end{aligned}$$

$$\begin{aligned}
& + \sup_{\theta \in \Theta} \left| \xi^\top \tilde{x}(\mathbf{Z}, \lambda(\mathbf{Z}, \bar{\theta}, x^0(\mathbf{Z}, \theta)), x^0(\mathbf{Z}, \theta), \bar{\theta}) - \tilde{D}(\mathbf{Z}, \lambda(\mathbf{Z}, \bar{\theta}, x^0(\mathbf{Z}, \theta)), x^0(\mathbf{Z}, \theta), \bar{\theta}, h) \right| \\
& \leq \underbrace{\sup_{\theta \in \Theta} \|\xi\|_\infty \|x(\mathbf{Z}, \theta) - \tilde{x}(\mathbf{Z}, \lambda(\mathbf{Z}, \bar{\theta}, x^0(\mathbf{Z}, \theta)), x^0(\mathbf{Z}, \theta), \bar{\theta})\|_1}_{(a)} \\
& \quad + \underbrace{\sup_{\theta \in \Theta, x^0 \in \mathcal{X}^0} \left| \xi^\top \tilde{x}(\mathbf{Z}, \lambda(\mathbf{Z}, \bar{\theta}, x^0), x^0, \bar{\theta}) - \tilde{D}(\mathbf{Z}, \lambda(\mathbf{Z}, \bar{\theta}, x^0), x^0, \bar{\theta}, h) \right|}_{(b)}
\end{aligned}$$

The first inequality adds and subtracts our approximate policy. The last inequality gives us the key terms we must bound.

To bound (a), we apply Lemma G.1, showing

$$\sup_{\theta \in \Theta} \|\xi\|_\infty \|x(\mathbf{Z}, \theta) - \tilde{x}(\mathbf{Z}, \lambda^0(\mathbf{Z}, \theta, x^0(\mathbf{Z}, \theta)), x^0(\mathbf{Z}, \theta), \bar{\theta})\|_1 \leq \|\xi\|_\infty \cdot m \cdot S_{\max}$$

Since ξ is a vector of independent sub-Gaussian random variables, with probability at least $1 - \epsilon$,

$$\|\xi\|_\infty \leq C_1 \sqrt{\frac{\log n}{\nu_{\min}}} \cdot \log\left(\frac{1}{\epsilon}\right)$$

where C_1 is a universal constant (Wainwright, 2019). Thus, with probability at least $1 - \epsilon$,

$$(a) \leq C_1 \cdot m \cdot S_{\max} \sqrt{\frac{\log n}{\nu_{\min}}} \cdot \log\left(\frac{1}{\epsilon}\right).$$

To bound (b), we first observe by Lemma G.2 $\|\lambda^0(\mathbf{Z}, \bar{\theta}, x^0)\|_1 \leq \frac{2}{s} \|\mathbf{T}(\mathbf{Z}, \bar{\theta})\|_1$, for every $\bar{\theta}, x^0$. Moreover,

$$\|\mathbf{T}(\mathbf{Z}, \bar{\theta})\|_1 \leq \|\mathbf{L}(\bar{\theta})\|_1 \|\mathbf{Z}\|_1 + \|\mathbf{l}(\bar{\theta})\|_1 \leq \|\mu\|_1 + \|\xi\|_1 + 1 \leq C_\mu(n+1) + n \cdot \|\xi\|_\infty.$$

Plugging in our bound on $\|\xi\|_\infty$, we see with probability at least $1 - \epsilon$

$$\|\mathbf{T}(\mathbf{Z}, \bar{\theta})\|_1 \leq 2C_1 C_\mu n \sqrt{\frac{\log n}{\nu_{\min}}} \cdot \log\left(\frac{1}{\epsilon}\right).$$

Letting $\epsilon = e^{-n}$, we see that for $\lambda_{\max} = 4C_1 C_\mu \bar{s}^{-1} \nu_{\min}^{-1/2} n^2$ that $\mathbb{P}\{\|\lambda(\mathbf{Z}, \theta, x^0(\mathbf{Z}, \theta, x^0))\|_1 \geq \lambda_{\max}\} < e^{-n}$.

Let $\Lambda = \{\lambda \in \mathbb{R}_+^m : \|\lambda\|_1 \leq \lambda_{\max}\}$. Then, when $\lambda(\mathbf{Z}, \theta, x^0) \in \Lambda$,

$$\begin{aligned}
(b) &= \sup_{\theta \in \Theta, x^0 \in \mathcal{X}^0} \left| \xi^\top \tilde{x}(\mathbf{Z}, \lambda^0(\mathbf{Z}, \theta, x^0), x^0, \theta) - \tilde{D}(\mathbf{Z}, \lambda^0(\mathbf{Z}, \theta, x^0), x^0, \theta, h) \right| \\
&\leq \sup_{x^0 \in \mathcal{X}^0} \sup_{\substack{\theta \in \Theta \\ \lambda \in \Lambda}} \left| \xi^\top \tilde{x}(\mathbf{Z}, \lambda, x_0, \theta) - \tilde{D}(\mathbf{Z}, \lambda, x_0, \bar{\theta}, h) \right|
\end{aligned}$$

We complete the bound on (b) by first fixing $\mathbf{x}^0$ in the last line and then applying Corollary 6.8 to the block decoupled setting induced by the lifted affine policy class to show

$$\begin{aligned} & \sup_{(\bar{\boldsymbol{\theta}}, \boldsymbol{\lambda}) \in \tilde{\Theta}} \left| \boldsymbol{\xi}^\top \tilde{\mathbf{x}}(\mathbf{Z}, \boldsymbol{\lambda}, \mathbf{x}_0, \boldsymbol{\theta}) - \tilde{D}(\mathbf{Z}, \boldsymbol{\lambda}, \mathbf{x}_0, \bar{\boldsymbol{\theta}}, h) \right| \\ & \leq C_2 S_{\max}^{3/4} n^{3/4} \log n \sqrt{T_{\max} \log \mathcal{X}_{\max} \log |\tilde{\Theta}_0(n^{-1/2})|} \cdot \log \left(\frac{S_{\max} T_{\max}}{\epsilon} \right), \end{aligned}$$

where C_2 is the constant from the corollary and $(\bar{\boldsymbol{\theta}}, \boldsymbol{\lambda}) \in \tilde{\Theta} \equiv \Theta \times \Lambda$. Note that the new plug-in class $\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \boldsymbol{\lambda}^\top \mathbf{A}^0(\mathbf{x}^0)$ indexed by $\tilde{\Theta}$ is still an affine policy class and satisfies Assumption 2.3 with the same Lipschitz constant as the original plug-in class.

We can further bound $|\tilde{\Theta}_0(n^{-1/2})|$ with respect to the covering $\Theta_0(\epsilon)$. Letting $\Lambda_0(\epsilon)$ be the ϵ -covering of Λ_0 , we see

$$|\tilde{\Theta}_0(2\epsilon)| \stackrel{(i)}{\leq} |\Theta_0(\epsilon)| \cdot \left| \Lambda_0 \left(\frac{\epsilon}{\|\mathbf{A}^0\|_\infty n} \right) \right| \stackrel{(ii)}{\leq} |\Theta_0(\epsilon)| \cdot \left(\frac{3\lambda_{\max} \|\mathbf{A}^0\|_\infty n}{\epsilon} \right)^m.$$

The inequality (i) follows because $\Theta_0(\epsilon) \times \Lambda_0 \left(\frac{\epsilon}{\|\mathbf{A}^0\|_\infty n} \right)$ is a valid covering that satisfies Assumption 2.5 since for each $(\bar{\boldsymbol{\theta}}, \boldsymbol{\lambda}) \in \tilde{\Theta}$ there exists $(\bar{\boldsymbol{\theta}}_0, \boldsymbol{\lambda}_0) \in \tilde{\Theta}_0(n^{-1/2})$ we have

$$\begin{aligned} \|\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \boldsymbol{\lambda}^\top \mathbf{A}^0(\mathbf{x}^0) - (\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}_0) + \boldsymbol{\lambda}_0^\top \mathbf{A}^0(\mathbf{x}^0))\|_1 & \leq \|\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) - \mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}_0)\|_1 + \|(\boldsymbol{\lambda} - \boldsymbol{\lambda}_0)^\top \mathbf{A}^0(\mathbf{x}^0)\|_1 \\ & \leq C(\|\mathbf{Z}\|_1 + 1)\epsilon + \sum_{j=1}^n \left| (\boldsymbol{\lambda} - \boldsymbol{\lambda}_0)^\top \mathbf{A}_j^0(\mathbf{x}^0) \right| \\ & \leq C(\|\mathbf{Z}\|_1 + 1)\epsilon + n \|\boldsymbol{\lambda} - \boldsymbol{\lambda}_0\|_1 \|\mathbf{A}^0\|_\infty \\ & \leq C(\|\mathbf{Z}\|_1 + 1)\epsilon + \epsilon \\ & \leq 2C(\|\mathbf{Z}\|_1 + 1)\epsilon. \end{aligned}$$

Inequality (ii) applies Pollard (1990, Lemma 4.1).

We can further simplify our bound for $n > 2$ by

$$\begin{aligned} \log |\tilde{\Theta}_0(2n^{-1/2})| & \leq \log |\Theta_0(n^{-1/2})| + \log (3\lambda_{\max} \|\mathbf{A}^0\|_\infty n^{1/2})^m \\ & \leq (\log |\Theta_0(n^{-1/2})|) m \log (3\lambda_{\max} \|\mathbf{A}^0\|_\infty n^{1/2}) \\ & \leq m (\log |\Theta_0(n^{-1/2})|) (\log (3\lambda_{\max} \|\mathbf{A}^0\|_\infty) + \log n^{1/2}) \\ & \leq m \log n \cdot (\log |\Theta_0(n^{-1/2})|) (\log (3\lambda_{\max} \|\mathbf{A}^0\|_\infty) + 1). \end{aligned}$$

Redefining $2n^{-1/2} \rightarrow n^{-1/2}$ shows

$$\log |\tilde{\Theta}_0(n^{-1/2})| \leq C_3 m \log n \cdot (\log |\Theta_0(n^{-1/2}/2)|)$$

for some constant C_3 depending on $\log(\lambda_{\max})$ and $\log(\|\mathbf{A}^0\|_\infty)$.

Taking a union bound over $\mathbf{x}^0 \in \mathcal{X}^0$, and collecting terms, we have

$$\begin{aligned} (b) &\leq \sup_{\mathbf{x}^0 \in \mathcal{X}^0} \sup_{\substack{\boldsymbol{\theta} \in \Theta \\ \boldsymbol{\lambda} \in \Lambda}} |\boldsymbol{\xi}^\top \tilde{\mathbf{x}}(\mathbf{Z}, \boldsymbol{\lambda}, \mathbf{x}_0, \boldsymbol{\theta}) - \tilde{D}(\mathbf{Z}, \boldsymbol{\lambda}, \mathbf{x}_0, \bar{\boldsymbol{\theta}}, h)| \\ &\leq C_4 S_{\max}^{3/4} n^{3/4} \log^{3/2} n \sqrt{m T_{\max} \log \mathcal{X}_{\max} \log |\Theta_0| (n^{-1/2}/2)} \cdot \log \left(\frac{S_{\max} T_{\max} |\mathcal{X}^0|}{\epsilon} \right) \end{aligned}$$

where C_4 depends on $\log \lambda_{\max}$, $\log \|\mathbf{A}^0\|_\infty$, $\nu_{\min}$, $\nu_{\max}$, $\sigma_{\min}$, $\|\boldsymbol{\mu}\|_\infty$, $L_{\max}$.

Combining bounds on (a) and (b) completes the proof. $\square$

Lemma G.1 (Error Approximating Weakly-Coupled by Constraints Policies) *Assume Assumption 6.10 holds. Then,*

$$\|\mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - \tilde{\mathbf{x}}(\mathbf{Z}, \boldsymbol{\lambda}(\mathbf{Z}, \boldsymbol{\theta}, \mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta})), \mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta}), \boldsymbol{\theta})\|_1 \leq m \cdot S_{\max}, \quad a.s.$$

Proof for Lemma G.1 To streamline the proof, we fix a $\boldsymbol{\theta} \in \Theta$ and $\mathbf{Z}$, dropping them from the notation. We let $\bar{\mathbf{x}}$ and $\bar{\boldsymbol{\lambda}}$ represent the optimal primal solution $\mathbf{x}(\mathbf{Z}, \boldsymbol{\theta})$ and optimal dual solution $\boldsymbol{\lambda}(\mathbf{Z}, \boldsymbol{\theta}, \mathbf{x}^0(\mathbf{Z}, \boldsymbol{\theta}))$, respectively.

Our goal is to bound

$$\|\bar{\mathbf{x}} - \tilde{\mathbf{x}}(\bar{\boldsymbol{\lambda}}, \bar{\mathbf{x}}^0)\|_1 = \sum_{k=0}^K \|\bar{\mathbf{x}}^k - \tilde{\mathbf{x}}^k(\bar{\boldsymbol{\lambda}}, \bar{\mathbf{x}}^0)\|_1.$$

Focusing on the k^{th} term, recall $\tilde{\mathbf{x}}^k(\bar{\boldsymbol{\lambda}}, \bar{\mathbf{x}}^0)$ solves the optimization problem,

$$\min_{\mathbf{x}^k \geq \mathbf{0}} \sum_{j \in S_k} (T_j + \bar{\boldsymbol{\lambda}}^\top \mathbf{A}_j^0(\bar{\mathbf{x}}^0)) x_j, \quad \text{s.t.} \quad \mathbf{x}^k \in \mathcal{X}^k(\bar{\mathbf{x}}^0). \quad (29)$$

We claim that if Problem (29) has a unique solution, then $\bar{\mathbf{x}}^k = \tilde{\mathbf{x}}^k(\bar{\boldsymbol{\lambda}}, \bar{\mathbf{x}}^0)$. To prove the claim, we show first show $\bar{\mathbf{x}}^k$ is the optimal solution of Problem (29). Let

$$g(\boldsymbol{\lambda}) = -\langle \mathbf{b}(\bar{\mathbf{x}}^0), \boldsymbol{\lambda} \rangle + \sum_{k=1}^K \min_{\mathbf{x}^k \in \mathcal{X}^k} \langle \mathbf{T}^k + \boldsymbol{\lambda}^\top \mathbf{A}_{S_k}^0(\bar{\mathbf{x}}^0), \mathbf{x}^k \rangle$$

be the dual problem objective solved by $\bar{\boldsymbol{\lambda}}$. Using strong duality of Problem (16) with a fixed $\mathbf{x}^0$, we can show

$$\begin{aligned} \sum_{k=1}^K \langle \mathbf{T}^k, \bar{\mathbf{x}}^k \rangle &= g(\bar{\boldsymbol{\lambda}}), \text{ by strong duality,} \\ &= -\langle \mathbf{b}(\bar{\mathbf{x}}^0), \bar{\boldsymbol{\lambda}} \rangle + \sum_{k=1}^K \min_{\mathbf{x}^k \in \mathcal{X}^k} \langle \mathbf{T}^k + \bar{\boldsymbol{\lambda}}^\top \mathbf{A}_{S_k}^0(\bar{\mathbf{x}}^0), \mathbf{x}^k \rangle \\ &\stackrel{(*)}{\leq} -\langle \mathbf{b}(\bar{\mathbf{x}}^0), \bar{\boldsymbol{\lambda}} \rangle + \sum_{k=1}^K \langle \mathbf{T}^k + \bar{\boldsymbol{\lambda}}^\top \mathbf{A}_{S_k}^0(\bar{\mathbf{x}}^0), \bar{\mathbf{x}}^k \rangle, \text{ since } \bar{\mathbf{x}}^k \in \mathcal{X}^k(\bar{\mathbf{x}}^0), \end{aligned}$$

$$\begin{aligned}
&= \underbrace{\langle \mathbf{A}^0(\bar{\mathbf{x}}^0)\bar{\mathbf{x}} - \mathbf{b}(\bar{\mathbf{x}}^0), \bar{\boldsymbol{\lambda}} \rangle}_{\leq 0, \text{ since } \bar{\mathbf{x}} \in \mathcal{X}^{WC} \text{ and } \bar{\boldsymbol{\lambda}} \geq \mathbf{0}} + \sum_{k=1}^K \langle \mathbf{T}^k, \bar{\mathbf{x}}^k \rangle \\
&\leq \sum_{k=1}^K \langle \mathbf{T}^k, \bar{\mathbf{x}}^k \rangle.
\end{aligned}$$

Here we have shown that equality holds throughout. Since inequality $(*)$ is an equality, we see that $\bar{\mathbf{x}}^k$ minimizes Problem (29) for each k . Since we claim Problem (16) has a unique solution, we have $\bar{\mathbf{x}}^k = \tilde{\mathbf{x}}^k(\bar{\boldsymbol{\lambda}}, \bar{\mathbf{x}}^0)$.

Thus, to bound $\|\bar{\mathbf{x}} - \tilde{\mathbf{x}}(\bar{\boldsymbol{\lambda}}, \bar{\mathbf{x}}^0)\|_1$, we count the number of sub-problems that do not have unique solutions. Let $\mathcal{K}$ be the set of k where $\tilde{\mathbf{x}}^k(\bar{\boldsymbol{\lambda}}, \bar{\mathbf{x}}^0)$ is not a unique solution. We see $k \in \mathcal{K}$ if:

1. The subproblem cost vector is $\mathbf{0}$, or formally,

$$\begin{pmatrix} T_j \\ \mathbf{A}_j^0 \\ 0 \end{pmatrix}^\top \begin{pmatrix} 1 \\ \boldsymbol{\lambda} \\ 0 \end{pmatrix} = 0, \quad \forall j \in S_k$$

2. The subproblem cost vector points in a direction corresponding to a unit vector in $\mathcal{D}^k(\bar{\mathbf{x}}^0)$, or formally, there exists $\mathbf{d} \in \mathcal{D}^k(\bar{\mathbf{x}}^0)$ and $\alpha^k > 0$ such that

$$\begin{pmatrix} T_j \\ \mathbf{A}_j^0 \\ -d_j \end{pmatrix}^\top \begin{pmatrix} 1 \\ \boldsymbol{\lambda} \\ \alpha^k \end{pmatrix} = 0, \quad \forall j \in S_k$$

We let $\mathcal{K}_1$ define the set of k that satisfy the first condition.

Combining all the equalities across $k \in \mathcal{K}, j \in S_k$, we see they each represent a linear relation among the vectors of $\mathcal{F}$ in Assumption 6.10. By the assumption, the maximal number of equalities we can have among the vectors is at most the number of variables:

$$\underbrace{m}_{\text{of } \boldsymbol{\lambda}} + \underbrace{|\mathcal{K}| - |\mathcal{K}_1|}_{\text{of } \alpha^k \text{ for } k \in \mathcal{K} \setminus \mathcal{K}_1} + 1.$$

Hence

$$m + |\mathcal{K}| - |\mathcal{K}_1| + 1 \geq \sum_{k \in \mathcal{K}} |S_k| = \sum_{k \in \mathcal{K} \setminus \mathcal{K}_1} |S_k| + \sum_{k \in \mathcal{K}_1} |S_k|.$$

Next, we claim that for any $k \in \mathcal{K} \setminus \mathcal{K}_1$, we must have $|S_k| \geq 2$. Indeed, if $|S_k| = 1$, then $\mathcal{X}^k$ is an interval, and the only way to have multiple solutions is if the subproblem cost vector is $\mathbf{0}$, i.e., $k \in \mathcal{K}_1$, a contradiction. As a result, we can further lower bound this last quantity by

$$m + |\mathcal{K}| - |\mathcal{K}_1| + 1 \geq 2(|\mathcal{K}| - |\mathcal{K}_1|) + |\mathcal{K}_1|.$$

Rearranging, proves $|\mathcal{K}| \leq m + 1$.

Finally, for $k \in \mathcal{K}$, $\|\bar{\mathbf{x}}^k - \tilde{\mathbf{x}}^k(\bar{\boldsymbol{\lambda}}, \bar{\mathbf{x}}^0)\|_1 \leq |S_k|$. Hence, $\|\bar{\mathbf{x}} - \tilde{\mathbf{x}}(\bar{\boldsymbol{\lambda}}, \bar{\mathbf{x}}^0)\|_1 \leq m \cdot S_{\max}$. This completes the proof. $\square$

Lemma G.2 (Dual Variables Bounded by Plug-in) *Under Assumption 6.11, we see for all $\boldsymbol{\theta} \in \{0\} \times \bar{\Theta}$, $\mathbf{z} \in \mathbb{R}^n$, and $\mathbf{x}^0 \in \mathcal{X}^0$ that*

$$\|\boldsymbol{\lambda}(\mathbf{z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0)\|_1 \leq \frac{2}{\bar{s}} \|\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})\|_1.$$

Proof of Lemma G.2 Fix $\mathbf{z}, \boldsymbol{\theta}, \mathbf{x}^0$. To reduce notation, let $\tilde{\mathbf{x}} = \tilde{\mathbf{x}}(\mathbf{z}, \boldsymbol{\lambda}(\mathbf{z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0), \mathbf{x}^0, \bar{\boldsymbol{\theta}})$ and

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})^\top \mathbf{x} + \boldsymbol{\lambda}^\top (\mathbf{A}^0(\mathbf{x}^0)\mathbf{x} - \mathbf{b}^0(\mathbf{x}^0)),$$

By optimality,

$$\mathcal{L}(\tilde{\mathbf{x}}, \boldsymbol{\lambda}(\mathbf{z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0)) \geq \mathcal{L}(\tilde{\mathbf{x}}, \mathbf{0}) \geq -\|\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})\|_1$$

where the last inequality holds since decision variables are bounded between 0 and 1.

Since $\boldsymbol{\lambda}(\mathbf{z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0) \geq 0$, we see $\|\boldsymbol{\lambda}(\mathbf{z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0)\|_1 = \mathbf{e}^\top \boldsymbol{\lambda}(\mathbf{z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0)$. Thus,

$$\begin{aligned} \|\boldsymbol{\lambda}(\mathbf{z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0)\|_1 &\leq \max_{\boldsymbol{\lambda} \geq 0} \mathbf{e}^\top \boldsymbol{\lambda} \\ \text{s.t. } \min_{\mathbf{x} \in \tilde{\mathcal{X}}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) &\geq -\|\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})\|_1, \end{aligned}$$

where $\tilde{\mathcal{X}} = \{\mathbf{x} : \mathbf{x}^k \in \mathcal{X}^k(\mathbf{x}^0), \forall k = 1, \dots, K\}$. We upper bound the optimization problem by relaxing our one constraint with penalty $1/\bar{s} > 0$ to show,

$$\begin{aligned} \|\boldsymbol{\lambda}(\mathbf{z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0)\|_1 &\leq \max_{\boldsymbol{\lambda} \geq 0} \mathbf{e}^\top \boldsymbol{\lambda} + \frac{1}{\bar{s}} \left(\min_{\mathbf{x} \in \tilde{\mathcal{X}}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) + \|\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})\|_1 \right) \\ &= \max_{\boldsymbol{\lambda} \geq 0} \mathbf{e}^\top \boldsymbol{\lambda} + \frac{1}{\bar{s}} \left(\min_{\mathbf{x} \in \tilde{\mathcal{X}}} \mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})^\top \mathbf{x} + \boldsymbol{\lambda}^\top (\mathbf{A}^0(\mathbf{x}^0)\mathbf{x} - \mathbf{b}^0(\mathbf{x}^0)) + \|\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})\|_1 \right) \\ &\leq \max_{\boldsymbol{\lambda} \geq 0} \mathbf{e}^\top \boldsymbol{\lambda} + \frac{1}{\bar{s}} (\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})^\top \bar{\mathbf{x}} + \boldsymbol{\lambda}^\top (\mathbf{A}^0(\mathbf{x}^0)\bar{\mathbf{x}} - \mathbf{b}^0(\mathbf{x}^0)) + \|\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})\|_1), \\ &= \max_{\boldsymbol{\lambda} \geq 0} \frac{1}{\bar{s}} (\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})^\top \bar{\mathbf{x}} + \boldsymbol{\lambda}^\top (\mathbf{A}^0(\mathbf{x}^0)\bar{\mathbf{x}} + \bar{s}\mathbf{e} - \mathbf{b}^0(\mathbf{x}^0)) + \|\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})\|_1) \end{aligned}$$

where $\bar{\mathbf{x}} \in \mathcal{X}^{WC}$ satisfies our strict feasibility assumption (Assumption 6.11) for $\mathbf{x}^0$. The first inequality is our upper-bound from relaxing the one constraint. The first equality expands $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$. The second inequality holds since $\bar{\mathbf{x}} \in \mathcal{X}^{WC} \subseteq \tilde{\mathcal{X}}$. The second equality collects terms.

Since $\mathbf{A}^0(\mathbf{x}^0)\bar{\mathbf{x}} + \bar{s}\mathbf{e} - \mathbf{b}^0(\mathbf{x}^0) \leq \mathbf{0}$ by the strict feasibility assumption,

$$\|\boldsymbol{\lambda}(\mathbf{z}, \bar{\boldsymbol{\theta}}, \mathbf{x}^0)\|_1 \leq \frac{1}{\bar{s}} (\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})^\top \bar{\mathbf{x}} + \|\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})\|_1) \leq \frac{2}{\bar{s}} \|\mathbf{T}(\mathbf{z}, \bar{\boldsymbol{\theta}})\|_1$$

Note the bound holds for any choice of $\mathbf{z}, \boldsymbol{\theta}, \mathbf{x}^0$. This completes the proof. $\square$

Appendix H: Details for Numerics and Examples

H.1. Implementation Details for Numerics

We describe the implementation of the one-shot VGC, multi-shot VGC, and Stein correction for the speed hump case study. For the lifted policy classes, we let $\boldsymbol{\lambda}(\mathbf{Z}, \bar{\boldsymbol{\theta}})$ be the optimal dual variable corresponding to the budget constraint for the cost vector $\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}})$.

H.1.1. One-Shot VGC Implementation The primary challenge of implementing the one-shot VGC is choosing a distribution for δ_h for each policy and evaluating the expectation

$$\mathbb{E} [V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \delta_h, \rho) | \mathbf{Z}].$$

In Section 9, for a fixed $\mathbf{L}$, we choose $\delta_h \sim \mathcal{N}(\mathbf{0}, \hat{\boldsymbol{\Sigma}}^h)$ and evaluate the expectation by Monte Carlo simulation with 25 samples of δ_h . To construct $\hat{\boldsymbol{\Sigma}}^h$, consider the matrix $\boldsymbol{\Sigma}^h = h(\mathbf{L}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\mathbf{L}^\top) + h^2\boldsymbol{\Sigma}$. If $\boldsymbol{\Sigma}^h$ is positive semi-definite, then $\hat{\boldsymbol{\Sigma}}^h = \boldsymbol{\Sigma}^h$. If $\boldsymbol{\Sigma}^h$ is not positive semi-definite, then we form the eigenvalue decomposition $\boldsymbol{\Sigma}^h = \mathbf{Q}^h \boldsymbol{\Lambda}^h \mathbf{Q}^{h\top}$ and construct

$$\hat{\boldsymbol{\Sigma}}^h = \rho \mathbf{Q}^h \boldsymbol{\Lambda}_+^h \mathbf{Q}^{h\top}, \text{ where } \boldsymbol{\Lambda}_+^h = \text{diag} \{ \max(0, \Lambda_{11}^h), \dots, \max(0, \Lambda_{nn}^h) \} \text{ and } \rho = \frac{\text{tr}(\boldsymbol{\Lambda}^h)}{\text{tr}(\boldsymbol{\Lambda}_+^h)}.$$

This method of “correcting” a non-positive definite matrix was also used in Chan and Wood, 1997 when simulating stationary Gaussian random fields.

To estimate the conditional expectation, we simulate $\delta_h^i \sim \mathcal{N}(\mathbf{0}, \hat{\boldsymbol{\Sigma}}^h)$ for $i = 1, \dots, S$. The resulting one-shot VGC out-of-sample performance estimate is

$$\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - \frac{\left(\frac{1}{S} \sum_{i=1}^S V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \delta_h^i, \rho) \right) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), \rho)}{h}.$$

In our experiments, we choose $S = 25$.

H.1.2. Multi-Shot VGC Implementation For the multi-shot VGC, we let $\delta_j^h \sim \mathcal{N}(0, (h^2 + 2h)/\nu_j)$ and define $\bar{\delta}_h(\bar{\boldsymbol{\theta}}), a_j(\bar{\boldsymbol{\theta}}) \sim L_{jj}(\bar{\boldsymbol{\theta}})\delta_j^h e_j, L_{jj}(\bar{\boldsymbol{\theta}})$ with probability $1/n$ for every j . We simulate $\bar{\delta}_h^i(\bar{\boldsymbol{\theta}}), a_j^i(\bar{\boldsymbol{\theta}})$ from the distribution. The resulting multi-shot VGC out-of-sample performance estimate is

$$\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - \frac{1}{S} \sum_{i=1}^S \frac{V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \bar{\delta}_h^i(\bar{\boldsymbol{\theta}}), \rho) - V(\mathbf{T}(\mathbf{Z}, \bar{\boldsymbol{\theta}}), \rho)}{a_j^i(\bar{\boldsymbol{\theta}})h/n}.$$

In our experiments, we choose $S = n$.

H.1.3. Stein Correction Implementation We implement the Stein correction on the lifted plug-in policy which was also proposed in Gupta and Rusmevichientong, 2021. Since the lifted policy fully decouples the optimization problem, the decisions correspond to indicator variables. The Stein correction out-of-sample performance estimate is calculated by

$$\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - \sum_{j=1}^n \frac{1}{2h\sqrt{\nu_j}} \left(\mathbb{I} \left\{ T_j \left(\mathbf{Z} + \frac{h}{\sqrt{\nu_j}}, \bar{\boldsymbol{\theta}} \right) - \lambda(\mathbf{Z}, \bar{\boldsymbol{\theta}}) \geq 0 \right\} - \mathbb{I} \left\{ T_j \left(\mathbf{Z} - \frac{h}{\sqrt{\nu_j}}, \bar{\boldsymbol{\theta}} \right) - \lambda(\mathbf{Z}, \bar{\boldsymbol{\theta}}) \geq 0 \right\} \right)$$

H.1.4. Lifted One-Shot VGC To define the one-shot VGC on the lifted plug-in policy, we first let $\tilde{V}(t) = t \cdot \mathbb{I}\{t \geq 0\}$ which represents the optimal plug-in objective value with plug-in t for one element of the fully decoupled optimization problem. Again, we let $\delta_h^i \sim \mathcal{N}(\mathbf{0}, \hat{\boldsymbol{\Sigma}}^h)$ for $i = 1, \dots, S$ and $S = 25$. The one-shot VGC out-of-sample performance estimate is

$$\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - \frac{1}{S} \sum_{i=1}^S \frac{1}{h} \left(\tilde{V}(T_j(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + \delta_j^{h,i} - \lambda(\mathbf{Z}, \bar{\boldsymbol{\theta}})) - \tilde{V}(T_j(\mathbf{Z}, \bar{\boldsymbol{\theta}}) - \lambda(\mathbf{Z}, \bar{\boldsymbol{\theta}})) \right)$$

H.1.5. Lifted Multi-Shot VGC Similar to the multi-shot VGC on the original policy, we have $\delta_j^{h,i} \sim \mathcal{N}(0, (h^2 + 2h)/\nu_j)$ for $i = 1, \dots, S$ and $j = 1, \dots, n$. The out-of-sample estimate is then computed as

$$\mathbf{Z}^\top \mathbf{x}(\mathbf{Z}, \boldsymbol{\theta}) - \frac{1}{S} \sum_{i=1}^S \frac{n}{L_{jj}(\bar{\boldsymbol{\theta}})h} \left(\tilde{V}(T_j(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + L_{jj}(\bar{\boldsymbol{\theta}})\delta_j^{h,i} - \lambda(\mathbf{Z}, \bar{\boldsymbol{\theta}})) - \tilde{V}(T_j(\mathbf{Z}, \bar{\boldsymbol{\theta}}) - \lambda(\mathbf{Z}, \bar{\boldsymbol{\theta}})) \right)$$

for $S = 25$. Note that $T_j(\mathbf{Z}, \bar{\boldsymbol{\theta}}) + L_{jj}(\bar{\boldsymbol{\theta}})\delta_j^{h,i} = T_j(\mathbf{Z} + \delta_j^{h,i}, \bar{\boldsymbol{\theta}})$ since we are only looking at the j^{th} element.

H.2. Lepski's Method Implementation

Our Lepski's method implementation selects the bandwidth h_j where

$$\hat{j} = \max \left\{ j : |D(\mathbf{Z}, \boldsymbol{\theta}, h_j) - D(\mathbf{Z}, \boldsymbol{\theta}, h_i)| \leq CNF(j) + (\sqrt{6} - 1)CNF(i), \forall i \leq j \right\},$$

for each choice of $\boldsymbol{\theta}$. In Section 7, we suggested the theoretically rigorous choice of $CNF(j) \equiv 2C_{T_{h_j}}^{\kappa} \sqrt{n \log(\frac{2}{\epsilon})}$ motivated by Theorem 3.3. However, the one-shot VGC for fully decoupled problems is essentially a sum of weakly-dependent random variables. Thus, by the Central Limit Theorem, we might expect it to have Gaussian tails. Similarly, for weakly-coupled problems, Donsker-type theorems for the process given by the VGC correction as a function of the lifting dual variables also suggest that it may have Gaussian tails.

Thus, in our implementation, we take the heuristic choice

$$CNF(j) = 1.64 \cdot \sqrt{\text{Var}(D(\mathbf{Z}, \boldsymbol{\theta}, h_j))},$$

where $\text{Var}(D(\mathbf{Z}, \boldsymbol{\theta}, h_j))$ is the variance of the one-shot VGC and 1.64 corresponds to the z-score for a two-tailed test at 90% significance. In other words, the CNF can be interpreted as the half-width or margin of error of a 90% confidence interval for $D(\mathbf{Z}, \boldsymbol{\theta}, h_j)$. Practically, we find this choice of CNF performs better as it is less conservative, than the tail-bound quantity from Theorem 3.3.

To estimate $\text{Var}(D(\mathbf{Z}, \boldsymbol{\theta}, h_j))$, we utilize parametric bootstrap to generate S samples $\mathbf{Z}^{(1)}, \dots, \mathbf{Z}^{(S)}$ and calculate the sample variance with the samples $D(\mathbf{Z}^{(1)}, \boldsymbol{\theta}, h_j), \dots, D(\mathbf{Z}^{(S)}, \boldsymbol{\theta}, h_j)$. In the case where $\mathbf{Z}$ is Gaussian, we let $\mathbf{Z}^{(i)} \sim \mathcal{N}(\mathbf{Z}, \boldsymbol{\Sigma})$ for $i = 1, \dots, S$. In the numerical case study in Section 9, we let $Z_j^{(i)} = \lambda_j \hat{I}_j^{(i)} / (\gamma N_j)$ where $\hat{I}_j^{(i)} \sim \text{Binomial}(\gamma N_j, \hat{p}_j)$ and $\hat{p}_j = \hat{I}_j / (\gamma N_j)$ for $i = 1, \dots, S; j = 1, \dots, n$.

H.3. Additional Results

Additional Figures from Section 9.1. In Fig. EC.1, we include a larger version of Fig. 3 that includes the VGC variants with the lifted policy for the reader's convenience.



Figure EC.1 Performance Results. We compare the estimated expected out-of-sample performance of our method to various benchmarks and various over 100 trials. The error bars are 95% confidence intervals. The experiments vary the amount of data available but keep the number of decisions fixed.

Additional Results for Section 8.2. We also consider a synthetic experiment that compares the bandwidth h selected by Lepski's method against the rate optimal h . Similar to Section 8.2, we consider the full information optimization problem,

$$\max_{\mathbf{x} \in \mathcal{X}} \sum_{k=1}^K \sum_{j=1}^3 \mu_{kj} x_{kj}, \quad \text{where } \mathcal{X} = \left\{ \mathbf{x} : \sum_{j=1}^B x_{kj} \leq 1 \forall k; x_{kj} \in \{0, 1\} \forall k, j \right\}.$$

We observe $Z_{kj} \sim \mathcal{N}(\mu_{kj}, 1/(\alpha \nu_{kj}))$ with scaling term $\alpha \in \{0.125, 0.25, 0.5, 1, 2, 4\}$ and

$$(\mu_{kj}, \nu_{kj}) = \begin{cases} (0, 2) & \text{for } j = 1, \\ (0.5, 10) & \text{for } j = 2, \\ (1, 6) & \text{for } j = 3. \end{cases}$$

We compare Lepski's method against the rate optimal choice of $h = n^{-1/4}$ and the best h across 100 trials for each scale term α . The results show Lepski's method is comparable or better than the rate optimal h , especially when the noise level is low.

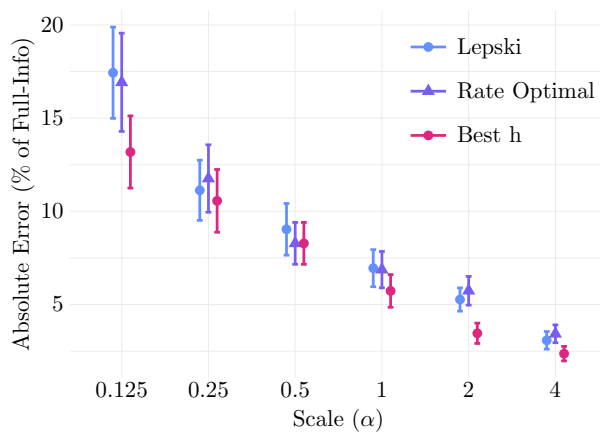


Figure EC.2 **Lepski’s Method on Synthetic Problem.** The figure plots the absolute error scaled by the full information objective between the in-sample optimism and the VGC estimate across different methods of choosing the bandwidth parameter h for $K = 160$. The error bars are 95% confidence intervals computed over 100 trials.